

Lecture notes in Computer Algebra

A first introduction to Computational Algebraic Geometry and Commutative Algebra

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Introduction

WHAT IS THIS DOCUMENT?

This set of notes is designed to supply the course material for the course MA522 at North Carolina State University. This document indicates *all* the contents of the course. You are strongly advised to follow the contents and instructions of these notes.

We aim to provide a first introduction to *Computer Algebra*. Computer algebra is a vast area that borrows and interconnects ideas from several fields of mathematics and computer science. However, we will focus on computations involving the solutions to systems of polynomial equations; these solutions are called *algebraic varieties*. Therefore, this course can be seen as a first introduction to *computational ideas in Algebraic Geometry and Commutative Algebra*. Our main tool will be the notion of *Gröbner bases* and we will see how they reduce complicated problems to questions relating to *monomial ideals*. As a consequence, we will spend some time studying combinatorial properties of monomial ideals.

The following textbooks will be our primary resources:

- [Ideals, Varieties, and Algorithms: An Introduction to Computational Algebraic Geometry and Commutative Algebra](#) by David Cox, John Little, and Donal O’Shea ([3]).
- [Monomial ideals](#) by Jürgen Herzog and Takayuki Hibi ([4]).
- [Using Algebraic Geometry](#) by David Cox, John Little, and Donal O’Shea ([2]).
- [Combinatorial commutative algebra](#) by Ezra Miller and Bernd Sturmfels ([8]).
- [A course in commutative algebra](#) by Gregor Kemper ([5]).

Throughout this document we use the following conventions:

- $\mathbb{N} = \{0, 1, 2, \dots\}$ is the set of nonnegative integers.
- $\mathbb{Z}_+ = \{1, 2, \dots\}$ is the set of positive integers.
- $\mathbb{k}$ is a field.
- $S = \mathbb{k}[x_1, \dots, x_n]$ is the polynomial ring in n variables $x_1, \dots, x_n$ over the field $\mathbb{k}$.
- $[k] = \{1, \dots, k\}$ for any $k \geq 1$.

CHAPTER 1

Polynomial rings, ideals and varieties

This chapter introduces the basic objects we shall study: *polynomial rings, ideals and varieties*. In this chapter we essentially follow [3, Chapter 1].

1.1. Polynomial rings

Let $\mathbb{k}$ be a field and $S = \mathbb{k}[x_1, \dots, x_n]$ be a polynomial ring. In most of our situations, the field $\mathbb{k}$ will be:

- (i) $\mathbb{k} = \mathbb{Q}$ the field of rational numbers (a good field for computers to work with).
- (ii) $\mathbb{k} = \mathbb{R}$ the field of real numbers (the field of *real life*).
- (iii) $\mathbb{k} = \mathbb{C}$ the field of complex numbers (our preferred algebraically closed field).

The underlying set of the ring S has the structure of a $\mathbb{k}$ -vector space. The basis elements are the *monomials*.

DEFINITION 1.1.1. A *monomial* in S is a product of the form

$$x_1^{\alpha_1} \cdots x_n^{\alpha_n}$$

where all of the exponents $\alpha_1, \dots, \alpha_n \in \mathbb{N}$ are nonnegative integers. We shall abbreviate $x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$ with $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}^n$. The (total) *degree* of this monomial is given by the sum $\deg(x^\alpha) := |\alpha| := \alpha_1 + \cdots + \alpha_n$.

Then the elements in the polynomial ring S are the polynomials.

DEFINITION 1.1.2. A *polynomial* $f \in S$ is a finite linear combination (with coefficients in $\mathbb{k}$) of monomials. We will write a polynomial f in the form

$$f = \sum_{\alpha} c_{\alpha} x^{\alpha}, \quad c_{\alpha} \in \mathbb{k}$$

where the sum is over a finite number of n -tuples $\alpha = (\alpha_1, \dots, \alpha_n)$.

DEFINITION 1.1.3. Let $f = \sum_{\alpha} c_{\alpha} x^{\alpha}$ be a polynomial in S .

- (i) We call c_{α} the *coefficient* of the monomial x^{α} .
- (ii) If $c_{\alpha} \neq 0$, then we call $c_{\alpha} x^{\alpha}$ a *term* of f .
- (iii) The (total) *degree* of $f \neq 0$, denoted $\deg(f)$, is the maximum $|\alpha|$ such that the coefficient c_{α} is nonzero. The (total) degree of the zero polynomial is undefined.

(iv) If all the terms of $f \neq 0$ have the same degree, then we say that f is a *homogeneous* polynomial.

When the number of variables is small, we shall use the notation $S = \mathbb{k}[x]$, $S = \mathbb{k}[x, y]$ or $S = \mathbb{k}[x, y, z]$.

EXAMPLE 1.1.4. (i) $2x^2 + 17xy^3z^2 + 5$ is a polynomial in $S = \mathbb{Q}[x, y, z]$ of degree 6 and it has 3 terms. But it is not homogeneous.

(ii) The polynomial $2x^6 + 17xy^3z^2 + 5z^6$ is homogeneous of degree 6 in $S = \mathbb{Q}[x, y, z]$.

The polynomial ring has the following direct sum decomposition

$$S = \bigoplus_{i \geq 0} S_i \quad \text{with} \quad S_i := \bigoplus_{|\alpha|=i} \mathbb{k} \cdot x^\alpha.$$

In other words, S_i is the vector space spanned by the monomials of degree i . With our conventions above, notice that the nonzero elements of S_i are the homogeneous polynomials of degree i . With this decomposition by total degree, we say that S is a *graded ring*.

We are interested in the following object, which is the geometrical counterpart of the polynomial ring $S = \mathbb{k}[x_1, \dots, x_n]$.

DEFINITION 1.1.5. We define the n -dimensional *affine space* over $\mathbb{k}$ to be the set

$$\mathbb{A}_{\mathbb{k}}^n := \{(a_1, \dots, a_n) \mid a_1, \dots, a_n \in \mathbb{k}\}.$$

As a set $\mathbb{A}_{\mathbb{k}}^n$ is just the n -dimensional $\mathbb{k}$ -vector space $\mathbb{k}^n$. But we write $\mathbb{A}_{\mathbb{k}}^n$ to stress that we will be employing the *Zariski topology* (which will be introduced later in [Section 1.3](#)).

Now our discussion begs the following question: *what is the relation between the polynomial ring $S = \mathbb{k}[x_1, \dots, x_n]$ and the affine space $\mathbb{A}_{\mathbb{k}}^n$?* Our answer will come from the fact that a polynomial $f \in S$ naturally gives a function

$$f : \mathbb{k}^n \rightarrow \mathbb{k}, \quad a = (a_1, \dots, a_n) \in \mathbb{k}^n \mapsto f(a_1, \dots, a_n) \in \mathbb{k},$$

by evaluating the polynomial f . We can describe *algebraic geometry* as the area of mathematics that studies the set of zeroes of polynomial equations.

The following example shows that some care should be taken when working over a finite field. We can have a nonzero polynomial that yields the zero function.

EXAMPLE 1.1.6. Let $p > 1$ be a prime number, $\mathbb{k} = \mathbb{Z}/p\mathbb{Z}$ be the field of p elements and $S = \mathbb{k}[x]$. Consider the *nonzero polynomial* $f(x) = x^p - x \in S$. By [Fermat's little theorem](#), we have $a^p \equiv a \pmod{p}$ for all $a \in \mathbb{Z}$. Therefore the function $f : \mathbb{k} \rightarrow \mathbb{k}$, $a \in \mathbb{k} \mapsto f(a) = a^p - a$ obtained by evaluating f is the *zero function*.

The following proposition shows that this phenomenon does not appear when the field $\mathbb{k}$ is infinite.

PROPOSITION 1.1.7. *Suppose $\mathbb{k}$ is an infinite field and $f \in S = \mathbb{k}[x_1, \dots, x_n]$ is a polynomial. Then $f = 0$ in S if and only if $f : \mathbb{k}^n \rightarrow \mathbb{k}$ is the zero function.*

Before proving the proposition, we need the following basic result from linear algebra.

LEMMA 1.1.8 (Vandermonde determinant). *Let $m \geq 1$ and let $v_1, \dots, v_m \in \mathbb{k}$. Consider the following matrix*

$$V = V(v_1, \dots, v_m) := \begin{pmatrix} 1 & v_1 & v_1^2 & \cdots & v_1^{m-1} \\ 1 & v_2 & v_2^2 & \cdots & v_2^{m-1} \\ 1 & v_3 & v_3^2 & \cdots & v_3^{m-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & v_m & v_m^2 & \cdots & v_m^{m-1} \end{pmatrix}.$$

Then $\det(V) = \prod_{1 \leq i < j \leq m} (v_j - v_i)$.

PROOF. We proceed by induction on m . For $m = 1$, the formula is clear. Assume that $m \geq 2$ and that the result holds for matrices of size $m - 1$. For $j = m, m - 1, \dots, 2$, subtract v_1 times the $(j - 1)$ -st column from the j -th column. These operations do not change the determinant, and they yield the matrix

$$V' = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 1 & v_2 - v_1 & v_2^2 - v_1 v_2 & \cdots & v_2^{m-1} - v_1 v_2^{m-2} \\ 1 & v_3 - v_1 & v_3^2 - v_1 v_3 & \cdots & v_3^{m-1} - v_1 v_3^{m-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & v_m - v_1 & v_m^2 - v_1 v_m & \cdots & v_m^{m-1} - v_1 v_m^{m-2} \end{pmatrix}.$$

Let V'' be the $(m - 1) \times (m - 1)$ matrix obtained by deleting the first row and the first column of V' . Expanding along the first row of V' and factoring $v_i - v_1$ from the row of V'' corresponding to v_i , for $2 \leq i \leq m$, gives the first equality below. Hence, by the inductive hypothesis, we obtain

$$\det(V) = \prod_{i=2}^m (v_i - v_1) \cdot \det(V(v_2, \dots, v_m)) = \prod_{i=2}^m (v_i - v_1) \cdot \prod_{2 \leq i < j \leq m} (v_j - v_i) = \prod_{1 \leq i < j \leq m} (v_j - v_i),$$

as required. $\square$

COROLLARY 1.1.9. *Let $f = c_d x^d + c_{d-1} x^{d-1} + \cdots + c_0 \in \mathbb{k}[x]$ be a polynomial of degree d . Then f can have at most d distinct roots.*

PROOF. By contradiction, suppose that f has $d + 1$ distinct roots; say $v_1, v_2, \dots, v_{d+1} \in \mathbb{k}$ with $f(v_i) = 0$ and $v_i \neq v_j$ for $i \neq j$. Consider the Vandermonde matrix $V = V(v_1, \dots, v_{d+1})$. We can

form the following linear system

$$\begin{pmatrix} 1 & v_1 & v_1^2 & \cdots & v_1^d \\ 1 & v_2 & v_2^2 & \cdots & v_2^d \\ 1 & v_3 & v_3^2 & \cdots & v_3^d \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & v_{d+1} & v_{d+1}^2 & \cdots & v_{d+1}^d \end{pmatrix} \cdot \begin{pmatrix} c_0 \\ \vdots \\ c_{d-1} \\ c_d \end{pmatrix} = \begin{pmatrix} f(v_1) \\ \vdots \\ f(v_d) \\ f(v_{d+1}) \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 0 \end{pmatrix}.$$

From [Lemma 1.1.8](#) and the fact that all the v_i are distinct, it follows that

$$\det(V) = \prod_{1 \leq i < j \leq d+1} (v_j - v_i) \neq 0.$$

Therefore, since V is an invertible matrix, we have $c_0 = \cdots = c_d = 0$. This means that $f = 0 \in \mathbb{k}[x]$ is the zero polynomial, a contradiction. $\square$

Now we are ready for the proof of [Proposition 1.1.7](#).

PROOF OF PROPOSITION 1.1.7. If $f = 0$ is the zero polynomial in S , then it is clear that $f : \mathbb{k}^n \rightarrow \mathbb{k}$ is the zero function. Hence we only need to show the reverse implication.

Suppose that $f : \mathbb{k}^n \rightarrow \mathbb{k}$ is the zero function. We proceed by induction on the number of variables n . If $n = 1$, then [Corollary 1.1.9](#) yields that $f = 0$ in S (because we are assuming $\mathbb{k}$ is an infinite field). Thus assume $n \geq 2$. We can write

$$f(x_1, \dots, x_n) = \sum_{i=0}^e g_i(x_1, \dots, x_{n-1}) x_n^i,$$

where each $g_i(x_1, \dots, x_{n-1})$ is a polynomial in $\mathbb{k}[x_1, \dots, x_{n-1}]$. Let $\mathbf{a} = (a_1, \dots, a_{n-1}) \in \mathbb{k}^{n-1}$ be any tuple and consider the polynomial

$$f_{\mathbf{a}}(x_n) = \sum_{i=0}^e g_i(a_1, \dots, a_{n-1}) x_n^i \in \mathbb{k}[x_n].$$

For any $v \in \mathbb{k}$, by assumption $f_{\mathbf{a}}(v) = f(a_1, \dots, a_{n-1}, v) = 0$, and so [Corollary 1.1.9](#) implies that $f_{\mathbf{a}}(x_n) = 0$ in $\mathbb{k}[x_n]$. Hence we have shown that $g_i(a_1, \dots, a_{n-1}) = 0$ for any $(a_1, \dots, a_{n-1}) \in \mathbb{k}^{n-1}$. By the induction hypothesis, we get $g_i(x_1, \dots, x_{n-1}) = 0$ in $\mathbb{k}[x_1, \dots, x_{n-1}]$. Finally, it follows that $f = 0$ in S . $\square$

COROLLARY 1.1.10. Suppose $\mathbb{k}$ is an infinite field and let $f, g \in S$ be two polynomials. Then $f = g$ in S if and only if $f : \mathbb{k}^n \rightarrow \mathbb{k}$ and $g : \mathbb{k}^n \rightarrow \mathbb{k}$ are the same function.

PROOF. Apply [Proposition 1.1.7](#) to the polynomial $f - g \in S$. $\square$

Below is the basic geometrical object we shall study.

DEFINITION 1.1.11. Let $f_1, \dots, f_s \in S$ be polynomials. The *affine variety* defined by $f_1, \dots, f_s$ is given by

$$V(f_1, \dots, f_s) := \{ (a_1, \dots, a_n) \in \mathbb{k}^n \mid f_i(a_1, \dots, a_n) = 0 \text{ for all } 1 \leq i \leq s \} \subset \mathbb{A}_{\mathbb{k}}^n.$$

To close this section, we recall the following result.

THEOREM 1.1.12 (Fundamental theorem of algebra). *Every nonconstant polynomial $f \in \mathbb{C}[x]$ has a root in $\mathbb{C}$ (i.e., $\mathbb{C}$ is algebraically closed).*

As a consequence, the variety $V(f)$ defined by a nonconstant polynomial $f \in \mathbb{C}[x]$ with complex coefficients is nonempty. Notice that the above theorem does not hold for the real numbers. For instance, the polynomial $x^2 + 1 \in \mathbb{R}[x]$ does not have any root in $\mathbb{R}$.

1.2. Ideals

We now introduce the basic algebraic objects that we study.

DEFINITION 1.2.1. A subset $I \subset S = \mathbb{k}[x_1, \dots, x_n]$ is an *ideal* if it satisfies:

- (i) $0 \in I$.
- (ii) If $f, g \in I$, then $f + g \in I$.
- (iii) If $f \in I$ and $h \in S$, then $hf \in I$.

We say that I is a *proper ideal* if $I \subsetneq S$ and that I is the *unit ideal* if $I = S$. Notice that $I = S$ is the unit ideal if and only if $1 \in I$. A natural way to present an ideal is by utilizing *generators*:

DEFINITION 1.2.2. Let $f_1, \dots, f_s \in S$ be polynomials. Then we write

$$(f_1, \dots, f_s) := \left\{ \sum_{i=1}^s h_i f_i \mid h_1, \dots, h_s \in S \right\}.$$

More generally, given any (possibly infinite) set $A \subset S$ of polynomials, we also write

$$(A) := \left\{ \sum_{i=1}^s h_i f_i \mid s \geq 0, f_1, \dots, f_s \in A \text{ and } h_1, \dots, h_s \in S \right\},$$

where the sum for $s = 0$ is understood to be 0.

If we are given two ideals $I \subset S$ and $J \subset S$, then we have the following basic operations:

- (1) $I + J := \{f + g \mid f \in I, g \in J\}$ (*sum*).
- (2) $I \cap J := \{f \in S \mid f \in I \text{ and } f \in J\}$ (*intersection*).
- (3) $IJ := \{f_1 g_1 + \dots + f_s g_s \mid s \geq 1, f_1, \dots, f_s \in I \text{ and } g_1, \dots, g_s \in J\}$ (*product*).

Notice that by definition we always have $IJ \subset I \cap J$.

The lemma below tells us that ideals behave well under these operations.

LEMMA 1.2.3. (i) *The sum $I + J$ of two ideals $I, J \subset S$ is an ideal.*

- (ii) $(f_1, \dots, f_s)$ is an ideal for any polynomials $f_1, \dots, f_s \in S$. Then, we shall say that $(f_1, \dots, f_s)$ is the ideal generated by the polynomials $f_1, \dots, f_s$.
- (iii) The intersection $I \cap J$ of two ideals $I, J \subset S$ is an ideal.
- (iv) The product $I \cdot J$ (also written IJ) of two ideals $I, J \subset S$ is an ideal.
- (v) (A) is an ideal for any set of polynomials $A \subset S$.

PROOF. (i) We check the three conditions of [Definition 1.2.1](#). Notice that $0 \in I + J$ because by assumption $0 \in I$ and $0 \in J$, and, of course, $0 = 0 + 0$. Let $f, g \in I + J$. By definition, $f = f_1 + g_1$ and $g = f_2 + g_2$ with $f_1, f_2 \in I$ and $g_1, g_2 \in J$. Then we can write

$$f + g = (f_1 + f_2) + (g_1 + g_2).$$

As I and J are ideals, we have $f_1 + f_2 \in I$ and $g_1 + g_2 \in J$, and thus $f + g \in I + J$. Similarly, for any $h \in S$, we obtain that $hf = hf_1 + hg_1$ is in $I + J$ because I and J are ideals. This completes the proof that $I + J$ is an ideal.

(ii) It is easy to check that (f_i) is an ideal for each $1 \leq i \leq s$. Notice that $(f_1, \dots, f_s) = \sum_{i=1}^s (f_i)$ (see [Exercise 1.8](#)). Therefore part (i) implies that $(f_1, \dots, f_s)$ is an ideal.

(iii) Left as an exercise in [Exercise 1.1](#).

(iv) Left as an exercise in [Exercise 1.2](#).

(v) Left as an exercise in [Exercise 1.3](#). □

Given an ideal $I \subset S$, the *radical* of I is given by

$$\sqrt{I} := \{f \in S \mid f^k \in I \text{ for some } k \geq 1\}.$$

We say I is a *radical ideal* if $I = \sqrt{I}$. We shall see that radical ideals are the algebraic counterpart of varieties. We leave the proof of some properties of radicals as an exercise (see [Exercise 1.9](#)).

EXAMPLE 1.2.4. Let $S = \mathbb{k}[x, y, z]$.

- (i) If $I = (x, y)$ and $J = (z^2 + x)$, then $I + J = (x, y, z^2 + x) = (x, y, z^2)$.
- (ii) If $I = (xy)$ and $J = (x^2)$, then $I \cap J = (x^2y)$ and $IJ = (x^3y)$.
- (iii) If $I = (x^2, y^2)$, then $\sqrt{I} = (x, y)$.
- (iv) If $I = (x^2)$ and $J = (y^2)$, then $I \cap J = (x^2y^2) = IJ$.

Given two ideals $I, J \subset S$, we have the *ideal quotient*

$$I : J := \{f \in S \mid fg \in I \text{ for all } g \in J\}$$

and the *saturation ideal*

$$I : J^\infty := \bigcup_{k=1}^{\infty} I : J^k.$$

The fact that $I : J$ is an ideal follows straightforwardly by checking the conditions of [Definition 1.2.1](#). For a polynomial $f \in S$, we use $I : f$ as shorthand for $I : (f)$. Since $J^{k+1} \subset J^k$, we get $I : J^k \subset I : J^{k+1}$ (see [Exercise 1.10\(i\)](#)), and then [Exercise 1.6](#) implies that $I : J^\infty$ is an ideal.

The following classes of ideals will play an important role.

DEFINITION 1.2.5. Let $I \subset S$ be a proper ideal.

- (i) I is *prime* if, whenever $f, g \in S$ and $fg \in I$, then either $f \in I$ or $g \in I$.
- (ii) I is *maximal* if it is maximal with respect to inclusion among proper ideals (i.e., if $J \subset S$ is another ideal with $I \subset J$, then either $I = J$ or $J = S$).
- (iii) I is *primary* if whenever $f, g \in S$ and $fg \in I$, then either $f \in I$ or some power $g^m \in I$ for some $m > 0$.

LEMMA 1.2.6. *A prime ideal is radical.*

PROOF. Let $P \subset S$ be a prime ideal. Let $f \in \sqrt{P}$, that is, $f^k \in P$ for some $k > 0$. By the definition of prime ideal, since $f^k = f \cdot f^{k-1} \in P$, we have $f \in P$ or $f^{k-1} \in P$. Thus inductively we obtain $f \in P$. $\square$

LEMMA 1.2.7. *A maximal ideal is a prime ideal.*

PROOF. Let $I \subset S$ be a maximal ideal. Let $f, g \in S$ with $fg \in I$. Suppose that $f \notin I$. Then $I : f$ is a proper ideal because $f \notin I$ is equivalent to $1 \notin I : f$. Since $I : f \supset I$ (see Exercise 1.10(ii)), the maximality of I yields the equality $I : f = I$. As $fg \in I$, it follows that $g \in I : f = I$. Therefore, I is a prime ideal. $\square$

LEMMA 1.2.8. *If I is a primary ideal, then $\sqrt{I}$ is prime and is the smallest prime ideal containing I .*

PROOF. Let $P = \sqrt{I}$. Let $f, g \in S$ with $fg \in \sqrt{I}$, that is, there is some $k > 0$ such that $f^k g^k \in I$. Applying the definition of primary ideal to the elements f^k and g^k , it follows that either $f^k \in I$ or $g^{km} \in I$ for some $m > 0$. This means that either $f \in \sqrt{I}$ or $g \in \sqrt{I}$. So P is a prime ideal.

Let $P' \subset S$ be a prime ideal containing I . Then it follows that $P' = \sqrt{P'} \supset \sqrt{I} = P$. This means that P is the smallest prime containing I . $\square$

DEFINITION 1.2.9. If $I \subset S$ is primary and $P = \sqrt{I}$, then we say that I is *P-primary*.

1.3. Zariski topology

We start with the definition of varieties.

DEFINITION 1.3.1. Let $I \subset S = \mathbb{k}[x_1, \dots, x_n]$ be an ideal. The *affine variety* defined by I is given by

$$V(I) := \{(a_1, \dots, a_n) \in \mathbb{k}^n \mid f(a_1, \dots, a_n) = 0 \text{ for all } f \in I\} \subset \mathbb{A}_{\mathbb{k}}^n.$$

Notice that if $f_1, \dots, f_s \in S$ are polynomials and $I = (f_1, \dots, f_s)$ is the ideal generated by them, then clearly $V(I) = V(f_1, \dots, f_s)$. The Hilbert basis theorem will later show that any ideal in S

is generated by finitely many polynomials; consequently, every variety of the form $V(I)$ can be written as $V(f_1, \dots, f_s)$. It is clear from the definition that for any two ideals $I, J \subset S$, if $I \subset J$, then $V(I) \supset V(J)$ (i.e., $V(-)$ reverses inclusions).

DEFINITION 1.3.2. A *topology* on a set X is given by declaring some subsets of X to be *closed*, such that the following properties hold:

- (a) The empty set $\emptyset$ and the whole space X are closed.
- (b) Arbitrary intersections of closed sets are closed.
- (c) Finite unions of closed sets are closed.

Let $\{I_\lambda\}_{\lambda \in \Lambda}$ be a (possibly infinite) collection of ideals in S . Then we can consider the sum and the intersection

$$\sum_{\lambda \in \Lambda} I_\lambda := \left(\bigcup_{\lambda} I_\lambda \right) \subset S \quad \text{and} \quad \bigcap_{\lambda \in \Lambda} I_\lambda \subset S.$$

The next lemma describes how the operation $V(-)$ behaves with respect to sums and intersections of ideals.

LEMMA 1.3.3. (i) Let $\{I_\lambda\}_{\lambda \in \Lambda}$ be a (possibly infinite) collection of ideals in S . Then we have the equality

$$V\left(\sum_{\lambda \in \Lambda} I_\lambda\right) = \bigcap_{\lambda \in \Lambda} V(I_\lambda) \subset \mathbb{A}_{\mathbb{k}}^n.$$

(ii) Let $I_1, \dots, I_k$ be ideals in S . Then we have the equality

$$V\left(\prod_{j=1}^k I_j\right) = V\left(\bigcap_{j=1}^k I_j\right) = \bigcup_{j=1}^k V(I_j) \subset \mathbb{A}_{\mathbb{k}}^n.$$

PROOF. Let $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{k}^n$.

(i) We have that $\alpha \in \bigcap_{\lambda \in \Lambda} V(I_\lambda)$ if and only if $f_\lambda(\alpha) = 0$ for all $f_\lambda \in I_\lambda$ and $\lambda \in \Lambda$. Recall that a polynomial f in $\sum_{\lambda \in \Lambda} I_\lambda$ is by definition of the form

$$f = f_{\lambda_1} + \dots + f_{\lambda_k}$$

with $f_{\lambda_i} \in I_{\lambda_i}$ and $\lambda_1, \dots, \lambda_k \in \Lambda$. As a consequence, we obtain the equivalence

$$f(\alpha) = 0 \text{ for all } f \in \sum_{\lambda \in \Lambda} I_\lambda \quad \Longleftrightarrow \quad f_\lambda(\alpha) = 0 \text{ for all } f_\lambda \in I_\lambda \text{ and } \lambda \in \Lambda.$$

This shows the required equality.

(ii) By induction we can assume that $k = 2$ (see [Exercise 1.8](#)). Since $I_1 I_2 \subset I_1 \cap I_2 \subset I_j$, we obtain

$$V(I_1 I_2) \supset V(I_1 \cap I_2) \supset V(I_j) \quad \text{for } 1 \leq j \leq 2.$$

Taking the union yields the inclusions $V(I_1 I_2) \supset V(I_1 \cap I_2) \supset V(I_1) \cup V(I_2)$. Therefore, to conclude the proof it suffices to show the reverse inclusion $V(I_1) \cup V(I_2) \supset V(I_1 I_2)$.

Suppose that $\alpha \in V(I_1 I_2)$. Hence, by definition, $(f \cdot g)(\alpha) = f(\alpha)g(\alpha) = 0$ for all $f \in I_1$ and $g \in I_2$. If $\alpha \in V(I_1)$, then $\alpha \in V(I_1) \subset V(I_1) \cup V(I_2)$ and we are done. Thus we may assume $\alpha \notin V(I_1)$ and so there is a polynomial $f' \in I_1$ such that $f'(\alpha) \neq 0$. For all $g \in I_2$, since $f'(\alpha)g(\alpha) = 0$, it follows that $g(\alpha) = 0$. This means precisely that $\alpha \in V(I_2)$. Therefore, we proved the inclusion $V(I_1 I_2) \subset V(I_1) \cup V(I_2)$, and this concludes the proof. $\square$

DEFINITION-PROPOSITION 1.3.4. The vector space $\mathbb{k}^n$ has a topology where the closed subsets are affine varieties. This topology is called the *Zariski topology*. The space $\mathbb{k}^n$, when endowed with the Zariski topology, is called the *affine n-space* $\mathbb{A}_{\mathbb{k}}^n$.

PROOF. We need to show that algebraic varieties satisfy the three conditions of [Definition 1.3.2](#).

First we check property (a). The empty set $\emptyset = V(S)$ is the set of zeroes of the unit ideal. The whole space $\mathbb{A}_{\mathbb{k}}^n = V(0)$ is the set of zeroes of the zero ideal.

Properties (b) and (c) follow from [Lemma 1.3.3](#). $\square$

COROLLARY 1.3.5. Suppose that $\mathbb{k}$ is an infinite field. If $U_1, U_2 \subset \mathbb{A}_{\mathbb{k}}^n$ are two nonempty open subsets, then $U_1 \cap U_2 \neq \emptyset$.

PROOF. For $i = 1, 2$, write $\mathbb{A}_{\mathbb{k}}^n \setminus U_i = V(I_i)$ for some ideal $I_i \subset S$, and choose a point $\alpha_i \in U_i$. Since $\alpha_i \notin V(I_i)$, there is a polynomial $f_i \in I_i$ such that $f_i(\alpha_i) \neq 0$. Thus $f_i \neq 0$ and the principal open subset

$$D(f_i) := \{\alpha \in \mathbb{A}_{\mathbb{k}}^n \mid f_i(\alpha) \neq 0\}$$

is contained in U_i . The polynomial $f_1 f_2$ is nonzero because S is an integral domain. By [Proposition 1.1.7](#), it cannot yield the zero function on $\mathbb{k}^n$. As a consequence, there is a point $\alpha \in \mathbb{A}_{\mathbb{k}}^n$ such that $f_1(\alpha)f_2(\alpha) \neq 0$. Then $\alpha \in D(f_1) \cap D(f_2) \subset U_1 \cap U_2$, as desired. $\square$

The following simple example shows that in general the operation $V(-)$ does not commute with arbitrary intersections of ideals.

EXAMPLE 1.3.6. Let $S = \mathbb{k}[x]$ and consider the infinite collection of ideals $I_j = (x^j) \subset S$ for all $j \geq 1$. Notice that $\bigcap_{j \geq 1} I_j = 0$ is the zero ideal. On the other hand, we have $V(I_j) = \{0\} \subset \mathbb{A}_{\mathbb{k}}^1$ for all $j \geq 1$. Therefore we obtain

$$V\left(\bigcap_{j \geq 1} I_j\right) = \mathbb{A}_{\mathbb{k}}^1 \neq \{0\} = \bigcup_{j \geq 1} V(I_j).$$

This shows that the finiteness assumption in [Lemma 1.3.3\(ii\)](#) is essential.

We now introduce the “inverse” operation to $V(-)$.

DEFINITION 1.3.7. Let $X \subset \mathbb{A}_{\mathbb{k}}^n$ be an affine variety. Then the *ideal* of X is given by

$$I(X) := \{f \in S \mid f(a) = 0 \text{ for all } a \in X\} \subset S$$

(see [Exercise 1.11](#)).

EXAMPLE 1.3.8. In general, we trivially have $I(\emptyset) = S$. If the field $\mathbb{k}$ is infinite, then we have $I(\mathbb{A}_{\mathbb{k}}^n) = 0$. Also, see [Exercise 1.12](#).

Let $I \subset S$ be an ideal and $X \subset \mathbb{A}_{\mathbb{k}}^n$ be a variety. Then we can perform the following operations

$$I \subset S \quad \mapsto \quad V(I) \subset \mathbb{A}_{\mathbb{k}}^n \quad \mapsto \quad I(V(I)) \subset S$$

and

$$X \subset \mathbb{A}_{\mathbb{k}}^n \quad \mapsto \quad I(X) \subset S \quad \mapsto \quad V(I(X)) \subset \mathbb{A}_{\mathbb{k}}^n.$$

These operations satisfy the following:

LEMMA 1.3.9. *Let $I \subset S$ be an ideal and $X \subset \mathbb{A}_{\mathbb{k}}^n$ be a variety. Then:*

- (i) $I \subset I(V(I))$ and the inclusion can be strict (see [Example 1.3.10](#)).
- (ii) $X = V(I(X))$.

PROOF. (i) Let $f \in I$. We need to show that $f(a) = 0$ for all $a \in V(I) \subset \mathbb{A}_{\mathbb{k}}^n$. Since $a \in V(I)$ if and only if $g(a) = 0$ for all $g \in I$, it clearly follows that $f(a) = 0$. Hence $I \subset I(V(I))$.

(ii) First we show $X \subset V(I(X))$. Let $a \in X$. We need to show that $f(a) = 0$ for all $f \in I(X) \subset S$. Since $f \in I(X)$ if and only if $f(b) = 0$ for all $b \in X$, it clearly follows that $f(a) = 0$. Hence $X \subset V(I(X))$.

We now show the reverse inclusion $X \supset V(I(X))$. By definition we can write $X = V(J) \subset \mathbb{A}_{\mathbb{k}}^n$ for some ideal $J \subset S$. From part (i) above, we get $I(X) = I(V(J)) \supset J$. Then the reverse inclusion property of $V(-)$ gives

$$V(I(X)) \subset V(J) = X.$$

So the equality $X = V(I(X))$ follows. □

EXAMPLE 1.3.10. In $S = \mathbb{k}[x]$, we have the strict inclusion $(x^2) \subsetneq I(V(x^2)) = (x)$.

1.4. Polynomial rings in one variable

We now study in more detail the polynomial ring $\mathbb{k}[x]$ in one variable. The degree of a polynomial gives $\mathbb{k}[x]$ a division algorithm analogous to division with remainder in $\mathbb{Z}$. This makes the ideals and the factorization of polynomials in one variable particularly concrete.

DEFINITION 1.4.1. Let

$$f = c_d x^d + c_{d-1} x^{d-1} + \cdots + c_0 \in \mathbb{k}[x]$$

be a nonzero polynomial of degree d , so that $c_d \neq 0$. The coefficient c_d is called the *leading coefficient* of f and is denoted by $LC(f)$. We say that f is *monic* if $LC(f) = 1$.

We begin with some elementary consequences of the degree.

LEMMA 1.4.2. *Let $f, g \in \mathbb{k}[x]$ be nonzero polynomials. Then:*

- (i) $\deg(fg) = \deg(f) + \deg(g)$.
- (ii) *The units of $\mathbb{k}[x]$ are exactly the nonzero elements of $\mathbb{k}$.*
- (iii) *If g divides f , then $\deg(g) \leq \deg(f)$.*

PROOF. Write $f = c_d x^d + \dots$ and $g = e_m x^m + \dots$, where $c_d, e_m \neq 0$. The coefficient of x^{d+m} in fg is $c_d e_m$, which is nonzero because $\mathbb{k}$ is a field. This proves (i).

If $fg = 1$, then part (i) gives $\deg(f) + \deg(g) = 0$, and so both f and g are nonzero constants. Conversely, every nonzero element of $\mathbb{k}$ has an inverse in $\mathbb{k}$ and is thus a unit of $\mathbb{k}[x]$. This proves (ii). Finally, if $f = gh$ for some nonzero polynomial h , then part (i) yields $\deg(f) = \deg(g) + \deg(h) \geq \deg(g)$, proving (iii). $\square$

The next theorem is the basic computational tool in $\mathbb{k}[x]$.

THEOREM 1.4.3 (Division algorithm). *Let $f, g \in \mathbb{k}[x]$ with $g \neq 0$. Then there are unique polynomials $q, r \in \mathbb{k}[x]$ such that*

$$f = qg + r$$

and either $r = 0$ or $\deg(r) < \deg(g)$. We call q the quotient and r the remainder obtained when dividing f by g .

PROOF. We first show existence. If $f = 0$ or $\deg(f) < \deg(g)$, we can take $q = 0$ and $r = f$. Suppose that $\deg(f) \geq \deg(g)$. Write $\deg(f) = d$, $\deg(g) = m$, $LC(f) = c$ and $LC(g) = e$. The polynomial

$$f_1 := f - \frac{c}{e} x^{d-m} g$$

is either zero or has degree strictly smaller than d . Repeating this step eventually gives polynomials $q, r \in \mathbb{k}[x]$ such that $f = qg + r$ and either $r = 0$ or $\deg(r) < \deg(g)$. The process terminates because the degree decreases at each step and a strictly decreasing sequence of nonnegative integers must be finite.

For uniqueness, suppose that

$$f = qg + r = q'g + r'$$

with both r and r' either zero or of degree smaller than $\deg(g)$. Then $(q - q')g = r' - r$. If $q - q' \neq 0$, Lemma 1.4.2 gives

$$\deg((q - q')g) \geq \deg(g),$$

whereas $r' - r$ is either zero or has degree smaller than $\deg(g)$, which is impossible. Thus $q = q'$ and consequently $r = r'$. $\square$

EXAMPLE 1.4.4. In $\mathbb{Q}[x]$, divide $f = 2x^4 + 3x^3 - x + 5$ by $g = x^2 + x - 1$. Successively canceling the leading terms gives

$$2x^4 + 3x^3 - x + 5 = (2x^2 + x + 1)(x^2 + x - 1) + (6 - x).$$

Thus the quotient is $2x^2 + x + 1$ and the remainder is $6 - x$.

Division by a linear polynomial gives a useful test for roots.

COROLLARY 1.4.5 (Remainder theorem and factor theorem). *Let $f \in \mathbb{k}[x]$ and $a \in \mathbb{k}$. The remainder obtained when dividing f by $x - a$ is $f(a)$. As a consequence,*

$$f(a) = 0 \iff x - a \text{ divides } f.$$

PROOF. By Theorem 1.4.3, we can write $f = q(x - a) + r$, where r is a constant. Evaluating at a gives $f(a) = r$. Therefore $f(a) = 0$ if and only if the remainder is zero, which is equivalent to $x - a$ dividing f . $\square$

The factor theorem also gives another proof of the bound in Corollary 1.1.9. Indeed, if $a_1, \dots, a_s \in \mathbb{k}$ are distinct roots of a nonzero polynomial f , then

$$\prod_{i=1}^s (x - a_i) \text{ divides } f.$$

To see this, first factor out $x - a_1$. For $i > 1$, the equality $0 = f(a_i) = (a_i - a_1)g(a_i)$ shows that a_i remains a root of the quotient g . Continuing in this way gives the claimed divisibility. It follows from Lemma 1.4.2 that $s \leq \deg(f)$.

We next describe the analogue of the greatest common divisor of two integers.

DEFINITION 1.4.6. Let $f, g \in \mathbb{k}[x]$ be polynomials, not both zero. A polynomial $d \in \mathbb{k}[x]$ is a *greatest common divisor* of f and g if:

- (i) d divides both f and g .
- (ii) Every common divisor of f and g divides d .

Any two greatest common divisors differ by multiplication by a nonzero element of $\mathbb{k}$. We write $\gcd(f, g)$ for the unique monic greatest common divisor of f and g .

The division algorithm computes the greatest common divisor by repeatedly replacing a pair of polynomials with a divisor and a remainder.

ALGORITHM 1.4.7 (Euclidean algorithm). Let $f, g \in \mathbb{k}[x]$ be polynomials, not both zero. If one of them is zero, the monic scalar multiple of the other is their greatest common divisor.

Otherwise, after interchanging f and g if necessary, set $r_0 = f$ and $r_1 = g$ with $\deg(r_0) \geq \deg(r_1)$. Use the division algorithm repeatedly to write

$$r_{i-1} = q_i r_i + r_{i+1},$$

where either $r_{i+1} = 0$ or $\deg(r_{i+1}) < \deg(r_i)$. Stop when $r_{s+1} = 0$. The output is the monic scalar multiple of the last nonzero remainder r_s .

THEOREM 1.4.8 (Bézout's identity). *The Euclidean algorithm terminates and its output is $d = \gcd(f, g)$. Moreover, there are polynomials $a, b \in \mathbb{k}[x]$ such that*

$$d = af + bg.$$

As a consequence, $(f, g) = (d)$.

PROOF. If one of f and g is zero, all the assertions follow immediately by multiplying the other polynomial by the scalar that makes it monic. Thus assume that both f and g are nonzero. The nonzero remainders have strictly decreasing degrees, and so the algorithm must terminate. From the equality

$$r_{i+1} = r_{i-1} - q_i r_i,$$

it follows that the common divisors of r_{i-1} and r_i are exactly the common divisors of r_i and r_{i+1} . At the last step, the common divisors of r_s and $r_{s+1} = 0$ are precisely the divisors of r_s . Thus the monic scalar multiple d of r_s is $\gcd(f, g)$.

Each remainder r_i is a polynomial linear combination of f and g . This follows by induction from $r_0 = f$, $r_1 = g$ and the displayed recurrence. After multiplying the last nonzero remainder by the scalar that makes it monic, we obtain $d = af + bg$ for some $a, b \in \mathbb{k}[x]$. Thus $(d) \subset (f, g)$. Since d divides both f and g , the reverse inclusion $(f, g) \subset (d)$ also holds. $\square$

EXAMPLE 1.4.9. Let

$$f = x^4 - x^3 - x + 1 \quad \text{and} \quad g = x^3 - x$$

in $\mathbb{Q}[x]$. The successive divisions are

$$\begin{aligned} f &= (x-1)g + (x-1)^2, \\ g &= (x+2)(x-1)^2 + 2(x-1), \\ (x-1)^2 &= \frac{x-1}{2} 2(x-1). \end{aligned}$$

Therefore $\gcd(f, g) = x - 1$. Back-substitution gives the Bézout identity

$$x - 1 = -\frac{x+2}{2}f + \frac{x^2+x-1}{2}g.$$

In fact, every ideal in $\mathbb{k}[x]$ has one generator.

THEOREM 1.4.10. *Every ideal $I \subset \mathbb{k}[x]$ is principal. More precisely, every nonzero ideal has a unique monic generator.*

PROOF. The zero ideal is generated by 0. Suppose that $I \neq 0$ and choose a nonzero polynomial $d \in I$ of smallest degree. After multiplying d by the inverse of its leading coefficient, we may assume that d is monic. For any $f \in I$, the division algorithm gives $f = qd + r$, where either $r = 0$ or $\deg(r) < \deg(d)$. Since $r = f - qd \in I$, the minimality of $\deg(d)$ forces $r = 0$. Thus d divides every element of I , and hence $I = (d)$.

If $(d) = (e)$ with both d and e monic, then d divides e and e divides d . The degree formula implies that d and e differ by a nonzero constant, and the fact that both are monic gives $d = e$. $\square$

REMARK 1.4.11. The proof of [Theorem 1.4.10](#) does more than show that ideals are finitely generated: it shows that one generator always suffices. This is special to one variable. For example, the ideal $(x, y) \subset \mathbb{k}[x, y]$ cannot be generated by one polynomial. Indeed, a generator would divide both x and y ; the degree formula would force it either to be a unit or to make x and y scalar multiples of one another, and both alternatives are impossible.

We now turn to factorization.

DEFINITION 1.4.12. Two nonzero polynomials $f, g \in \mathbb{k}[x]$ are called *associates* if $f = cg$ for some $c \in \mathbb{k} \setminus \{0\}$. A nonconstant polynomial $p \in \mathbb{k}[x]$ is *irreducible over $\mathbb{k}$* if a factorization $p = fg$ in $\mathbb{k}[x]$ implies that either f or g is a nonzero constant.

Every polynomial of degree one is irreducible. The field of coefficients is essential in this definition. For example, $x^2 + 1$ is irreducible over $\mathbb{R}$ but factors as $(x - i)(x + i)$ over $\mathbb{C}$, whereas $x^2 - 2$ is irreducible over $\mathbb{Q}$ but factors over $\mathbb{R}$.

LEMMA 1.4.13 (Euclid's lemma). *Let $p \in \mathbb{k}[x]$ be irreducible. If p divides fg , then p divides f or p divides g .*

PROOF. Suppose that p does not divide f . Since p is irreducible, a monic greatest common divisor of p and f can only be 1 or the monic associate of p . Our assumption rules out the latter, and so $\gcd(p, f) = 1$. By Bézout's identity, there are $a, b \in \mathbb{k}[x]$ such that

$$1 = ap + bf.$$

Multiplying by g yields $g = agp + bfg$. Both terms on the right are divisible by p , and therefore p divides g . $\square$

THEOREM 1.4.14 (Unique factorization). *Every nonzero polynomial $f \in \mathbb{k}[x]$ can be written in the form*

$$f = c p_1^{e_1} \cdots p_s^{e_s},$$

where $c \in \mathbb{k} \setminus \{0\}$, the polynomials $p_1, \dots, p_s$ are distinct monic irreducible polynomials, and $e_1, \dots, e_s \geq 1$. This expression is unique up to the order of the factors.

PROOF. We prove existence by induction on $\deg(f)$. A nonzero constant already has the required form with no irreducible factors. If f is nonconstant and irreducible, divide it by its leading coefficient to obtain a monic irreducible factor. If f is reducible, write $f = gh$ with $0 < \deg(g), \deg(h) < \deg(f)$ and apply induction to g and h . Collecting associate factors and making them monic gives the displayed expression.

For uniqueness, suppose that two products of monic irreducibles give the same polynomial, after separating their leading constants. Any irreducible factor in the first product divides the second product, so repeated application of Lemma 1.4.13 shows that it divides one of the irreducible factors in the second product. The two monic irreducible polynomials must then be equal. Cancel this common factor and continue by induction on the degree. This proves that the factors and their exponents agree up to order. $\square$

Unique factorization gives a complete description of the prime and maximal ideals in one variable.

COROLLARY 1.4.15. *Let $p \in \mathbb{k}[x]$ be a nonconstant polynomial. Then the following conditions are equivalent:*

- (i) p is irreducible.
- (ii) (p) is a prime ideal.
- (iii) (p) is a maximal ideal.

As a consequence, every nonzero prime ideal of $\mathbb{k}[x]$ is maximal.

PROOF. If p is irreducible, Lemma 1.4.13 shows that (p) is prime. To see directly that (p) is maximal, let $(p) \subset I \subset \mathbb{k}[x]$. By Theorem 1.4.10, we have $I = (d)$ for some polynomial d . The inclusion $(p) \subset (d)$ means that d divides p . The irreducibility of p implies that either d is a unit, in which case $I = \mathbb{k}[x]$, or d is an associate of p , in which case $I = (p)$. Thus (p) is maximal.

Finally, if (p) is prime and $p = fg$, then $fg \in (p)$ implies that $f \in (p)$ or $g \in (p)$. If, say, $f \in (p)$, then p divides f . Together with $\deg(f) \leq \deg(p)$, the degree formula shows that f is an associate of p and that g is a unit. Thus the factorization is trivial, so p is irreducible. The last assertion follows from Theorem 1.4.10 and the equivalence just proved. $\square$

REMARK 1.4.16. The zero ideal is also prime because $\mathbb{k}[x]$ is an integral domain, but it is not maximal. For instance, $(0) \subsetneq (x) \subsetneq \mathbb{k}[x]$.

We next relate factorization to algebraic closure and to roots.

DEFINITION 1.4.17. A field $\mathbb{k}$ is *algebraically closed* if every nonconstant polynomial in $\mathbb{k}[x]$ has a root in $\mathbb{k}$.

PROPOSITION 1.4.18. *For a field $\mathbb{k}$, the following conditions are equivalent:*

- (i) *The field $\mathbb{k}$ is algebraically closed.*

(ii) Every irreducible polynomial in $\mathbb{k}[x]$ has degree one.

(iii) Every nonzero polynomial $f \in \mathbb{k}[x]$ of degree d can be written as

$$f = c \prod_{i=1}^d (x - \alpha_i)$$

for some $c \in \mathbb{k} \setminus \{0\}$ and $\alpha_1, \dots, \alpha_d \in \mathbb{k}$, where the elements α_i are not required to be distinct.

PROOF. Assume (i) and let $p \in \mathbb{k}[x]$ be irreducible. Since p has a root $\alpha \in \mathbb{k}$, the factor theorem shows that $x - \alpha$ divides p . Irreducibility then gives $\deg(p) = 1$, proving (ii). If (ii) holds, unique factorization expresses every nonzero polynomial as a product of linear factors, which gives (iii). Finally, (iii) clearly implies that every nonconstant polynomial has a root in $\mathbb{k}$, and so (i) holds. $\square$

For example, the fundamental theorem of algebra says precisely that $\mathbb{C}$ satisfies the conditions in [Proposition 1.4.18](#). It also follows from [Corollary 1.4.15](#) that, when $\mathbb{k}$ is algebraically closed, the maximal ideals of $\mathbb{k}[x]$ are exactly the ideals $(x - \alpha)$ with $\alpha \in \mathbb{k}$.

DEFINITION 1.4.19. Let $f \in \mathbb{k}[x]$ be a nonzero polynomial and let $\alpha \in \mathbb{k}$. The *multiplicity* of α as a root of f is the largest integer $m \geq 0$ such that $(x - \alpha)^m$ divides f . Thus α is a root of f if and only if its multiplicity is positive, and it is a *multiple root* if its multiplicity is at least two.

The largest integer in the definition exists because $(x - \alpha)^m$ cannot divide a nonzero polynomial of degree smaller than m . Equivalently, α has multiplicity m if and only if

$$f = (x - \alpha)^m h \quad \text{with} \quad h(\alpha) \neq 0.$$

Indeed, the condition $h(\alpha) \neq 0$ says by the factor theorem that $x - \alpha$ does not divide h .

PROPOSITION 1.4.20. Let $\alpha_1, \dots, \alpha_s \in \mathbb{k}$ be the distinct roots of a nonzero polynomial $f \in \mathbb{k}[x]$, and let m_i be the multiplicity of α_i . Then

$$\prod_{i=1}^s (x - \alpha_i)^{m_i} \text{ divides } f \quad \text{and} \quad \sum_{i=1}^s m_i \leq \deg(f).$$

If $\mathbb{k}$ is algebraically closed, then equality holds in the degree bound and

$$f = c \prod_{i=1}^s (x - \alpha_i)^{m_i}$$

for some $c \in \mathbb{k} \setminus \{0\}$.

PROOF. The polynomials $x - \alpha_i$ are distinct monic irreducible polynomials. Thus the divisibility and the degree bound follow from unique factorization and [Lemma 1.4.2](#). If $\mathbb{k}$ is algebraically closed, every irreducible factor of f is linear by [Proposition 1.4.18](#). Therefore the displayed product contains all the irreducible factors of f , and the final assertions follow. $\square$

The formal derivative detects multiple roots without first factoring the polynomial.

DEFINITION 1.4.21. Let $f = \sum_{i=0}^d c_i x^i \in \mathbb{k}[x]$. The *formal derivative* of f is the polynomial

$$f' = \sum_{i=1}^d i c_i x^{i-1}.$$

Here the integers appearing as coefficients are interpreted as elements of the field $\mathbb{k}$; in particular, the derivative of a constant polynomial is zero.

LEMMA 1.4.22. For all $f, g \in \mathbb{k}[x]$ and $c \in \mathbb{k}$, we have

- (i) $(f + g)' = f' + g'$ and $(cf)' = cf'$.
- (ii) $(fg)' = f'g + fg'$.

PROOF. Part (i) follows immediately from the definition. For part (ii), first check the identity on monomials of positive degree:

$$(x^i x^j)' = (i + j)x^{i+j-1} = ix^{i-1}x^j + x^i jx^{j-1}.$$

The identity is immediate when one of the monomials is constant. The general case follows by distributing products and using part (i). $\square$

PROPOSITION 1.4.23 (Derivative test for multiple roots). Let $f \in \mathbb{k}[x]$ be a nonzero polynomial and let $a \in \mathbb{k}$. Then a is a multiple root of f if and only if

$$f(a) = f'(a) = 0.$$

PROOF. Let m be the multiplicity of a and write $f = (x - a)^m h$ with $h(a) \neq 0$. If $m \geq 1$, the product rule gives

$$f' = m(x - a)^{m-1}h + (x - a)^m h'.$$

If $m \geq 2$, both f and f' vanish at a . If $m = 1$, then $f'(a) = h(a) \neq 0$. Finally, if $m = 0$, then $f(a) \neq 0$. These three cases prove the result. $\square$

DEFINITION 1.4.24. A nonzero polynomial $f \in \mathbb{k}[x]$ is *square-free over $\mathbb{k}$* if no square of a nonconstant polynomial in $\mathbb{k}[x]$ divides f . Equivalently, in the factorization from Theorem 1.4.14, every exponent e_i is equal to one.

THEOREM 1.4.25 (Square-free criterion). Let $f \in \mathbb{k}[x]$ be a nonzero polynomial. Then f has no multiple root in an algebraic closure of $\mathbb{k}$ if and only if

$$\gcd(f, f') = 1.$$

If $\mathbb{k}$ has characteristic zero, if $\mathbb{k}$ is finite, or if $\mathbb{k}$ is algebraically closed, these conditions are also equivalent to f being square-free over $\mathbb{k}$.

PROOF. In any field extension in which f splits into linear factors, [Proposition 1.4.23](#) shows that f and f' have a common root exactly when f has a multiple root. By Bézout's identity, $\gcd(f, f') = 1$ if and only if there exist $a, b \in \mathbb{k}[x]$ such that $af + bf' = 1$. Such an identity remains valid after extending the field and rules out a common root. Conversely, if $\gcd(f, f')$ is nonconstant, any root of this greatest common divisor in an algebraic closure is a common root of f and f' . This proves the first equivalence.

It remains to compare this condition with being square-free over $\mathbb{k}$. Write $f = c \prod_i p_i^{e_i}$ as in [Theorem 1.4.14](#). If every irreducible polynomial p_i has nonzero derivative, the product rule shows that p_i divides both f and f' exactly when $e_i \geq 2$. Therefore $\gcd(f, f') = 1$ exactly when every $e_i = 1$. Every nonconstant polynomial has nonzero derivative in characteristic zero. If $\mathbb{k}$ is algebraically closed, its irreducible polynomials are linear and also have nonzero derivative. Finally, suppose that $\mathbb{k}$ is finite of characteristic p . If an irreducible polynomial q had $q' = 0$, then q would have the form $\sum_j c_j x^{jp}$. The Frobenius map $c \mapsto c^p$ is a bijection on the finite set $\mathbb{k}$, so we could write

$$q = \left(\sum_j b_j x^j \right)^p$$

with $b_j^p = c_j$, contradicting the irreducibility of the nonconstant polynomial q . Thus the required condition on irreducible derivatives holds in all three stated cases. $\square$

The qualification in the last part of [Theorem 1.4.25](#) matters in positive characteristic. If $\mathbb{k}$ has characteristic $p > 0$, then $f' = 0$ precisely when all exponents occurring in f are divisible by p . Over a field in which not every element is a p -th power, such a nonconstant polynomial need not be a p -th power in $\mathbb{k}[x]$, even though it becomes a polynomial with multiple roots after passing to an algebraic closure.

In characteristic zero, the derivative also gives the polynomial with the same roots as f , but with all multiplicities removed.

COROLLARY 1.4.26. *Suppose that $\mathbb{k}$ has characteristic zero and let $f \in \mathbb{k}[x]$ be nonzero. Set $d = \gcd(f, f')$ and $f_{\text{sf}} = f/d$. Then f_{sf} is square-free and has the same roots as f in every field extension of $\mathbb{k}$. Moreover,*

$$\sqrt{(f)} = (f_{\text{sf}}).$$

PROOF. Write $f = c \prod_i p_i^{e_i}$ as a product of distinct monic irreducible polynomials. Since the characteristic is zero, the proof of [Theorem 1.4.25](#) shows that, up to a nonzero constant,

$$d = \prod_i p_i^{e_i-1} \quad \text{and hence} \quad f_{\text{sf}} = c' \prod_i p_i$$

for some $c' \in \mathbb{k} \setminus \{0\}$. The assertions about square-freeness and roots follow. Also, a polynomial h has a power in (f) if and only if every p_i divides h . This is equivalent to $h \in (\prod_i p_i) = (f_{\text{sf}})$, proving the equality of ideals. $\square$

We close the section by making the algebra–geometry correspondence completely explicit in one variable over an algebraically closed field.

THEOREM 1.4.27 (The algebra–geometry dictionary in one variable). *Suppose that $\mathbb{k}$ is algebraically closed and let $I \subset \mathbb{k}[x]$ be an ideal. Then*

$$I(V(I)) = \sqrt{I}.$$

More precisely, if I is a nonzero proper ideal, then there are distinct elements $\alpha_1, \dots, \alpha_s \in \mathbb{k}$ and integers $m_1, \dots, m_s \geq 1$ such that

$$I = \left(\prod_{i=1}^s (x - \alpha_i)^{m_i} \right), \quad V(I) = \{\alpha_1, \dots, \alpha_s\}, \quad I(V(I)) = \sqrt{I} = \left(\prod_{i=1}^s (x - \alpha_i) \right).$$

PROOF. If I is nonzero and proper, then by [Theorem 1.4.10](#) we can write $I = (f)$ with f monic and nonconstant, and [Proposition 1.4.20](#) gives

$$f = \prod_{i=1}^s (x - \alpha_i)^{m_i}$$

with the α_i distinct. It follows immediately that $V(I) = \{\alpha_1, \dots, \alpha_s\}$. A polynomial $h \in \mathbb{k}[x]$ vanishes at every α_i if and only if each $x - \alpha_i$ divides h . By unique factorization, this is equivalent to $\prod_i (x - \alpha_i)$ dividing h . Thus

$$I(V(I)) = \left(\prod_{i=1}^s (x - \alpha_i) \right).$$

Unique factorization also shows directly that this ideal is $\sqrt{(f)}$.

If $I = \mathbb{k}[x]$, then $V(I) = \emptyset$ and $I(V(I)) = \mathbb{k}[x] = \sqrt{I}$. If $I = (0)$, then $V(I) = \mathbb{A}_{\mathbb{k}}^1$ and $I(V(I)) = (0) = \sqrt{I}$ by [Proposition 1.1.7](#). Here we use that an algebraically closed field is infinite: if $\mathbb{k}$ were finite, the polynomial

$$1 + \prod_{\alpha \in \mathbb{k}} (x - \alpha)$$

would be nonconstant and would have no root in $\mathbb{k}$. □

1.5. Exercises

EXERCISE 1.1. *Show that the intersection $I \cap J$ of two ideals $I, J \subset S$ is an ideal. In fact, show that a (possibly infinite) intersection of ideals is an ideal.*

EXERCISE 1.2. *Show that the product $I \cdot J$ of two ideals $I, J \subset S$ is an ideal.*

EXERCISE 1.3. *Show that (A) is an ideal for any set of polynomials $A \subset S$.*

EXERCISE 1.4. *Let $I, J, K \subset S$ be ideals. Show that:*

- (i) $I \cdot (J + K) = IJ + IK$.

(ii) If $I \supset J$ or $I \supset K$, then $I \cap (J + K) = I \cap J + I \cap K$.

EXERCISE 1.5. Give an example where the union $I \cup J$ of two ideals $I, J \subset S$ is not an ideal.

EXERCISE 1.6. Let $\{I_j\}_{j \geq 1}$ be a sequence of ideals in S such that $I_j \subset I_{j+1}$ for all $j \geq 1$. Show that $\bigcup_{j \geq 1} I_j$ is an ideal.

EXERCISE 1.7. Let $I, J \subset S$ be ideals. Show that $I + J$ is the smallest ideal containing both I and J .

EXERCISE 1.8. Given three ideals $I, J, K \subset S$, show the following associativity equalities:

- (i) $(I + J) + K = I + (J + K)$.
- (ii) $(I \cap J) \cap K = I \cap (J \cap K)$.
- (iii) $(I \cdot J) \cdot K = I \cdot (J \cdot K)$.

Therefore, for a sequence $I_1, \dots, I_k \subset S$, we have well-defined ideals $\sum_{j=1}^k I_j \subset S$, $\bigcap_{j=1}^k I_j \subset S$ and $\prod_{j=1}^k I_j \subset S$ giving the sum, intersection and product, respectively.

EXERCISE 1.9. Let $I, J \subset S$ be ideals. Show that:

- (i) $\sqrt{I}$ is an ideal.
- (ii) $\sqrt{\sqrt{I}} = \sqrt{I}$.
- (iii) $\sqrt{I \cap J} = \sqrt{IJ} = \sqrt{I} \cap \sqrt{J}$.
- (iv) If $I \subset J$, then $\sqrt{I} \subset \sqrt{J}$.
- (v) $\sqrt{I} + \sqrt{J} \subset \sqrt{I + J}$ and $\sqrt{\sqrt{I} + \sqrt{J}} = \sqrt{I + J}$.
- (vi) $V(\sqrt{I}) = V(I) \subset \mathbb{A}_{\mathbb{k}}^n$.

EXERCISE 1.10. Let $I, J, K \subset S$ be ideals. Show that:

- (i) If $J \subset K$, then $I : J \supset I : K$.
- (ii) $I \subset (I : J)$.
- (iii) $(I : J) \cdot J \subset I$.
- (iv) $(I : J) : K = (I : JK) = (I : K) : J$.
- (v) If $I_1, \dots, I_s \subset S$ are ideals, then $\left(\bigcap_{j=1}^s I_j\right) : J = \bigcap_{j=1}^s (I_j : J)$.
- (vi) If $J_1, \dots, J_s \subset S$ are ideals, then $I : \left(\sum_{i=1}^s J_i\right) = \bigcap_{i=1}^s (I : J_i)$.

EXERCISE 1.11. Let $X \subset \mathbb{A}_{\mathbb{k}}^n$ be an affine variety. Show that $I(X) \subset S$ is an ideal.

EXERCISE 1.12. Let $p > 1$ be a prime number, $\mathbb{k} = \mathbb{Z}/p\mathbb{Z}$ be the field of p elements and $S = \mathbb{k}[x]$. Show that $I(\mathbb{A}_{\mathbb{k}}^1) = (x^p - x) \subset S$.

EXERCISE 1.13. Show that the following subsets are not affine varieties:

- (i) $X = \mathbb{Z}^n \subset \mathbb{A}_{\mathbb{R}}^n$.
- (ii) $X = \mathbb{A}_{\mathbb{R}}^n \setminus \{(0, \dots, 0)\} \subset \mathbb{A}_{\mathbb{R}}^n$.
- (iii) $X = \{(a, b) \in \mathbb{A}_{\mathbb{R}}^2 \mid b = \sin(a)\} \subset \mathbb{A}_{\mathbb{R}}^2$.

(iv) $X = \{(a, b) \in \mathbb{A}_{\mathbb{R}}^2 \mid b = e^a\} \subset \mathbb{A}_{\mathbb{R}}^2$.

EXERCISE 1.14. Let $X \subset \mathbb{A}_{\mathbb{k}}^n$ and $Y \subset \mathbb{A}_{\mathbb{k}}^m$ be affine varieties. Show that the Cartesian product $X \times Y \subset \mathbb{A}_{\mathbb{k}}^n \times \mathbb{A}_{\mathbb{k}}^m \cong \mathbb{A}_{\mathbb{k}}^{n+m}$ is also an affine variety.

EXERCISE 1.15. Identify $\mathbb{A}_{\mathbb{k}}^{mn}$ with the space of $m \times n$ matrices, and let $r \geq 0$. Show that the set of matrices with rank $\leq r$ is an affine variety in $\mathbb{A}_{\mathbb{k}}^{mn}$.

EXERCISE 1.16. Show that a finite set of points in $\mathbb{A}_{\mathbb{k}}^n$ is an affine variety.

EXERCISE 1.17. Show that the only affine varieties in $\mathbb{A}_{\mathbb{k}}^1$ are (a) the empty set, (b) the whole space $\mathbb{A}_{\mathbb{k}}^1$, and (c) a finite set of points.

EXERCISE 1.18. This exercise shows that affine varieties are not stable under projections. Consider the affine variety $C = \{(a, b) \in \mathbb{A}_{\mathbb{R}}^2 \mid a^2 + b^2 = 1\}$ (the circle of radius 1) and the natural projection $\pi: \mathbb{A}_{\mathbb{R}}^2 \rightarrow \mathbb{A}_{\mathbb{R}}^1$, $(a, b) \mapsto a$ that forgets the second coordinate. Show that the set $\pi(C) \subset \mathbb{A}_{\mathbb{R}}^1$ is not an affine variety.

CHAPTER 2

Algebra and combinatorics of monomial ideals

Throughout this chapter, we continue using the previous notation: $S = \mathbb{k}[x_1, \dots, x_n]$ is a polynomial ring in n variables over a field $\mathbb{k}$. We use the notation $\mathbf{x}^{\mathbf{a}} = x_1^{a_1} \cdots x_n^{a_n}$ for any $\mathbf{a} := (a_1, \dots, a_n) \in \mathbb{N}^n$.

2.1. Basic properties of monomial ideals

Denote by

$$\text{Mon}(S) := \{\mathbf{x}^{\mathbf{a}} = x_1^{a_1} \cdots x_n^{a_n} \mid \mathbf{a} = (a_1, \dots, a_n) \in \mathbb{N}^n\}$$

the set of monomials of S . For a polynomial $f = \sum_{\mathbf{a} \in \mathbb{N}^n} c_{\mathbf{a}} \mathbf{x}^{\mathbf{a}} \in S$, its *support* is the finite set

$$\text{supp}(f) := \{\mathbf{x}^{\mathbf{a}} \in \text{Mon}(S) \mid c_{\mathbf{a}} \neq 0\}.$$

Thus, $\text{supp}(0) = \emptyset$.

DEFINITION 2.1.1. An ideal $I \subset S$ is called a *monomial ideal* if it is generated by monomials.

The divisibility of monomials can be read directly from their exponent vectors. If $\mathbf{a} = (a_1, \dots, a_n)$ and $\mathbf{b} = (b_1, \dots, b_n)$ belong to $\mathbb{N}^n$, then

$$\mathbf{x}^{\mathbf{a}} \mid \mathbf{x}^{\mathbf{b}} \iff a_i \leq b_i \text{ for all } 1 \leq i \leq n.$$

Indeed, in this case $\mathbf{x}^{\mathbf{b}} = \mathbf{x}^{\mathbf{b}-\mathbf{a}} \mathbf{x}^{\mathbf{a}}$. As a consequence, a collection of monomials is the collection of all monomials in an ideal precisely when it is closed under taking monomial multiples.

The next result is the basic reason that membership in a monomial ideal can be decided term by term.

THEOREM 2.1.2. *Let $I \subset S$ be a monomial ideal. The set of monomials belonging to I is a $\mathbb{k}$ -basis of I .*

PROOF. Let $\mathcal{M} \subset \text{Mon}(S)$ be a set of monomial generators of I . Every element $f \in I$ can be written as a finite sum

$$f = h_1 u_1 + \cdots + h_r u_r,$$

where $u_1, \dots, u_r \in \mathcal{M}$ and $h_1, \dots, h_r \in S$. After expanding each h_i as a linear combination of monomials, we see that f is a $\mathbb{k}$ -linear combination of monomials divisible by one of the u_i . All these monomials belong to I . Thus the monomials in I span I . They are linearly independent because all monomials form a $\mathbb{k}$ -basis of S . □

COROLLARY 2.1.3. Let $I \subset S$ be an ideal. The following conditions are equivalent:

- (a) I is a monomial ideal.
- (b) For every $f \in S$, one has $f \in I$ if and only if $\text{supp}(f) \subset I$.

PROOF. Suppose first that I is a monomial ideal. If $f \in I$, then the uniqueness of the expression of f in the monomial basis, together with Theorem 2.1.2, gives $\text{supp}(f) \subset I$. The converse follows because an ideal is closed under finite sums and multiplication by elements of $\mathbb{k}$.

Now suppose that (b) holds and let $\mathcal{N}$ be the set of monomials belonging to I . Every $f \in I$ is a $\mathbb{k}$ -linear combination of elements of $\mathcal{N}$, and hence $I = (\mathcal{N})$ is a monomial ideal. $\square$

COROLLARY 2.1.4. Let I be a monomial ideal. The residue classes of the monomials not belonging to I form a $\mathbb{k}$ -basis of S/I . These monomials are called standard monomials of I .

PROOF. The monomials of S are the disjoint union of the monomials belonging to I and the standard monomials of I . Therefore, Theorem 2.1.2 gives a decomposition of $\mathbb{k}$ -vector spaces

$$S = I \oplus \bigoplus_{u \in \text{Mon}(S) \setminus I} \mathbb{k}u.$$

Passing to S/I proves the assertion. $\square$

EXAMPLE 2.1.5. Let $S = \mathbb{k}[x, y]$ and $I = (x^2, y^2)$. The standard monomials of I are $1, x, y, xy$. Therefore, Corollary 2.1.4 gives the decomposition of $\mathbb{k}$ -vector spaces

$$S/I = \mathbb{k}\bar{1} \oplus \mathbb{k}\bar{x} \oplus \mathbb{k}\bar{y} \oplus \mathbb{k}\bar{xy}.$$

The set of monomials belonging to a monomial ideal can be described explicitly from any monomial system of generators.

PROPOSITION 2.1.6. Let $\mathcal{M} \subset \text{Mon}(S)$ be a system of generators of the monomial ideal I . Then a monomial $v \in \text{Mon}(S)$ belongs to I if and only if $u \mid v$ for some $u \in \mathcal{M}$.

PROOF. If $u \mid v$ for some $u \in \mathcal{M}$, then $v \in I$. Conversely, the proof of Theorem 2.1.2 shows that I is spanned by the monomial multiples of the elements of $\mathcal{M}$. Since distinct monomials are linearly independent, a monomial $v \in I$ must itself be one of these multiples. $\square$

EXAMPLE 2.1.7. Let $I = (x^4, x^2y^3, y^6) \subset \mathbb{k}[x, y]$. Then $x^a y^b \in I$ if and only if

$$a \geq 4, \quad \text{or} \quad a \geq 2 \text{ and } b \geq 3, \quad \text{or} \quad b \geq 6.$$

For instance, x^5y and x^2y^8 belong to I , whereas x^3y^2 is standard. It follows from Corollary 2.1.3 that

$$f = 2x^5y - x^3y^2 + x^2y^8$$

does not belong to I , because one monomial in its support is standard.

There is also a useful grading which remembers every exponent, rather than only the total degree. For $\mathbf{a} \in \mathbb{N}^n$, set $S_{\mathbf{a}} = \mathbb{k}\mathbf{x}^{\mathbf{a}}$. Then

$$S = \bigoplus_{\mathbf{a} \in \mathbb{N}^n} S_{\mathbf{a}}$$

and $S_{\mathbf{a}}S_{\mathbf{b}} \subset S_{\mathbf{a}+\mathbf{b}}$. We call this the $\mathbb{N}^n$ -grading of S . An ideal $I \subset S$ is $\mathbb{N}^n$ -graded if

$$I = \bigoplus_{\mathbf{a} \in \mathbb{N}^n} (I \cap S_{\mathbf{a}}).$$

PROPOSITION 2.1.8. *An ideal of S is monomial if and only if it is $\mathbb{N}^n$ -graded.*

PROOF. Suppose that I is monomial and let $f = \sum_{\mathbf{a}} c_{\mathbf{a}}\mathbf{x}^{\mathbf{a}} \in I$. By [Corollary 2.1.3](#), each nonzero component $c_{\mathbf{a}}\mathbf{x}^{\mathbf{a}}$ belongs to I . Thus I is $\mathbb{N}^n$ -graded.

Conversely, suppose that I is $\mathbb{N}^n$ -graded. Every $f \in I$ is the sum of its multihomogeneous components, and every nonzero multihomogeneous component is a scalar multiple of a monomial belonging to I . Hence the monomials in I generate I . $\square$

In particular, a monomial ideal is homogeneous for the usual grading by total degree. The finer $\mathbb{N}^n$ -grading explains why many operations on monomial ideals reduce to divisibility computations; we shall make those computations explicit below.

2.2. Dickson's lemma

In this section, we present one of the central results we shall need.

Let $\mathcal{M}$ be a nonempty subset of $\text{Mon}(S)$. A monomial $\mathbf{x}^{\mathbf{a}} \in \mathcal{M}$ is said to be a minimal element of $\mathcal{M}$ with respect to divisibility if whenever $\mathbf{x}^{\mathbf{b}} \mid \mathbf{x}^{\mathbf{a}}$ with $\mathbf{x}^{\mathbf{b}} \in \mathcal{M}$, then $\mathbf{x}^{\mathbf{b}} = \mathbf{x}^{\mathbf{a}}$. Let $\mathcal{M}^{\min}$ denote the set of minimal elements of $\mathcal{M}$.

We shall repeatedly use the following elementary observation. Every $u \in \mathcal{M}$ is divisible by an element of $\mathcal{M}^{\min}$. Indeed, if u is not minimal, replace it by a proper divisor which belongs to $\mathcal{M}$. The total degree decreases at every replacement, so this process must stop.

The following result tells us that the set of monomials $\text{Mon}(S)$ is “almost well-ordered” with respect to divisibility.

THEOREM 2.2.1 (Dickson lemma). *Let $\mathcal{M}$ be a nonempty subset of $\text{Mon}(S)$. Then $\mathcal{M}^{\min}$ is a finite set.*

PROOF. We prove Dickson's lemma by induction on n , the number of variables of S . If $n = 1$, then $\mathcal{M}$ consists of certain powers of x_1 , and the set of minimal elements of $\mathcal{M}$ is the set $\{x_1^c\}$, where c is the smallest number such that $x_1^c \in \mathcal{M}$.

Now assume that $n \geq 2$. Let $\mathcal{N}$ be the set of monomials

$$\mathcal{N} := \{\mathbf{x}^{\mathbf{c}} = x_1^{c_1} \cdots x_{n-1}^{c_{n-1}} \in \mathbb{k}[x_1, \dots, x_{n-1}] \mid \mathbf{x}^{\mathbf{c}}x_n^d \in \mathcal{M} \text{ for some } d \geq 0\}.$$

By the induction hypothesis, the set $\mathcal{N}^{\min}$ of minimal elements of $\mathcal{N}$ is finite, say $\mathcal{N}^{\min} = \{\mathbf{x}^{\mathbf{c}_1}, \dots, \mathbf{x}^{\mathbf{c}_r}\} \subset \mathbb{k}[x_1, \dots, x_{n-1}]$ for some $\mathbf{c}_1, \dots, \mathbf{c}_r \in \mathbb{N}^{n-1}$. For each $\mathbf{x}^{\mathbf{c}_i}$, choose the smallest $\alpha_i \geq 0$ such that $\mathbf{x}^{\mathbf{c}_i} x_n^{\alpha_i} \in \mathcal{M}$. We claim that $\mathbf{x}^{\mathbf{c}_i} x_n^{\alpha_i} \in \mathcal{M}^{\min}$. If $\mathbf{x}^{\mathbf{d}} x_n^e \in \mathcal{M}$ divides $\mathbf{x}^{\mathbf{c}_i} x_n^{\alpha_i}$, then $\mathbf{x}^{\mathbf{d}} \in \mathcal{N}$ divides $\mathbf{x}^{\mathbf{c}_i}$. The minimality of $\mathbf{x}^{\mathbf{c}_i}$ in $\mathcal{N}$ gives $\mathbf{x}^{\mathbf{d}} = \mathbf{x}^{\mathbf{c}_i}$, and the choice of α_i then gives $e = \alpha_i$. Let $\alpha = \max\{\alpha_1, \dots, \alpha_r\}$, and for each b with $0 \leq b < \alpha$ let

$$\mathcal{N}_b := \{\mathbf{x}^{\mathbf{c}} \in \mathbb{k}[x_1, \dots, x_{n-1}] \mid \mathbf{x}^{\mathbf{c}} x_n^b \in \mathcal{M}\}.$$

Again by the induction hypothesis, $\mathcal{N}_b^{\min}$ is finite when $\mathcal{N}_b$ is nonempty; if $\mathcal{N}_b$ is empty, so is $\mathcal{N}_b^{\min}$. We use the notation

$$\mathcal{N}_b^{\min} x_n^b := \{\mathbf{x}^{\mathbf{c}} x_n^b \mid \mathbf{x}^{\mathbf{c}} \in \mathcal{N}_b^{\min}\}.$$

We claim that

$$\mathcal{M}^{\min} \subset \{\mathbf{x}^{\mathbf{c}_1} x_n^{\alpha_1}, \dots, \mathbf{x}^{\mathbf{c}_r} x_n^{\alpha_r}\} \cup \bigcup_{b=0}^{\alpha-1} \mathcal{N}_b^{\min} x_n^b.$$

Since the right-hand side of this inclusion is a finite set, the assertion of the theorem follows from this claim.

In order to prove the claim, let $u = \mathbf{x}^{\mathbf{c}} x_n^d$ be a monomial in $\mathcal{M}^{\min}$. If $d \geq \alpha$, then some monomial in $\{\mathbf{x}^{\mathbf{c}_1} x_n^{\alpha_1}, \dots, \mathbf{x}^{\mathbf{c}_r} x_n^{\alpha_r}\}$ divides u . If $0 \leq d < \alpha$, then u is divisible by a monomial in $\mathcal{N}_d^{\min} x_n^d$. In both cases the divisor belongs to $\mathcal{M}$, and hence the minimality of u implies that u is equal to that divisor. This completes the proof of the theorem. $\square$

The next corollary tells us that monomial ideals are finitely generated. This is a special case of Hilbert basis theorem for monomial ideals.

COROLLARY 2.2.2 (Hilbert basis theorem for monomial ideals). *Let I be a monomial ideal. Then each set of monomial generators of I contains a finite set which generates I .*

PROOF. Let $\mathcal{M}$ be a set of monomial generators of I . If $I = (0)$, the empty subset of $\mathcal{M}$ generates I . Thus assume that $I \neq (0)$. By [Theorem 2.2.1](#), the set of minimal elements of $\mathcal{M}$ is finite. Every element of $\mathcal{M}$ is divisible by a minimal element of $\mathcal{M}$, so this finite set generates I . $\square$

PROPOSITION 2.2.3. *Each monomial ideal has a unique minimal monomial set of generators. More precisely, let G denote the set of monomials in I which are minimal with respect to divisibility. Then G is the unique minimal set of monomial generators.*

PROOF. If $I = (0)$, then $G = \emptyset$, and the statement is clear. Assume $I \neq (0)$. Dickson's lemma shows that G is finite, and the elementary observation preceding [Theorem 2.2.1](#) shows that every monomial in I is divisible by an element of G . Thus [Proposition 2.1.6](#) gives $I = (G)$.

Let $\mathcal{N}$ be any monomial set of generators of I . For $u \in G$, [Proposition 2.1.6](#) gives an element $v \in \mathcal{N}$ which divides u . Since $v \in I$ and u is minimal among the monomials of I , we have $u = v$.

Therefore $G \subseteq \mathcal{N}$. It follows that G is contained in every monomial system of generators, and hence it is the unique minimal one. $\square$

The uniqueness in the previous proposition is special to monomial systems of generators.

EXAMPLE 2.2.4. Let $S = \mathbb{k}[x, y]$ and consider the nonmonomial ideal

$$I = ((x+1)y^2, (x+1)(y+1)).$$

If the first displayed generator belonged to the ideal generated by the second, then cancellation of the nonzero factor $x+1$ would imply that $y+1$ divides y^2 . Similarly, if the second belonged to the ideal generated by the first, then y^2 would divide $y+1$. Neither divisibility holds, and so the displayed system of generators is minimal with respect to inclusion. On the other hand, the equality

$$1 = y^2 - (y-1)(y+1)$$

shows that $(y^2, y+1) = S$. It follows that

$$I = (x+1)(y^2, y+1) = (x+1).$$

Thus the same ideal has an inclusion-minimal system of generators with two elements and another one with only one element.

As a consequence we make the following definition.

DEFINITION 2.2.5. The unique minimal set of monomial generators of the monomial ideal I is denoted by $G(I) \subset \text{Mon}(S)$.

COROLLARY 2.2.6. Let $I, J \subset S$ be monomial ideals. Then $I \subseteq J$ if and only if, for every $u \in G(I)$, there exists $v \in G(J)$ such that $v \mid u$.

PROOF. This follows immediately from Proposition 2.1.6. $\square$

As another consequence of Dickson's lemma we have a "Noetherian property" for monomial ideals.

PROPOSITION 2.2.7. Each ascending sequence of monomial ideals $I_1 \subset I_2 \subset \cdots \subset I_l \subset \cdots$ in S terminates, that is, there exists an integer k such that $I_l = I_k$ for all $l \geq k$.

PROOF. Let $\mathcal{M} = \bigcup_{j=1}^{\infty} G(I_j)$. If $\mathcal{M} = \emptyset$, then every I_j is zero and there is nothing to prove. According to Dickson's lemma (see Theorem 2.2.1), the set $\mathcal{M}^{\min}$ is finite. Hence there is an integer k such that $\mathcal{M}^{\min} \subset \bigcup_{j=1}^k G(I_j)$. Now let $l \geq k$ and let u be a monomial in I_l . Then there exists $v \in \mathcal{M}^{\min}$ which divides u . This implies that $u \in \bigcup_{j=1}^k I_j = I_k$, as desired. $\square$

2.3. Algebraic operations on monomial ideals

The usual operations on ideals have especially simple descriptions for monomial ideals. Given two monomials $u = x_1^{a_1} \cdots x_n^{a_n}$ and $v = x_1^{b_1} \cdots x_n^{b_n}$, we have the following explicit descriptions

$$\gcd(u, v) = x_1^{\min\{a_1, b_1\}} \cdots x_n^{\min\{a_n, b_n\}}$$

and

$$\text{lcm}(u, v) = x_1^{\max\{a_1, b_1\}} \cdots x_n^{\max\{a_n, b_n\}}.$$

PROPOSITION 2.3.1. *Let $I = (u_1, \dots, u_r)$ and $J = (v_1, \dots, v_s)$ be monomial ideals. Then*

$$I + J = (u_1, \dots, u_r, v_1, \dots, v_s)$$

and

$$IJ = (u_i v_j \mid 1 \leq i \leq r, 1 \leq j \leq s).$$

In particular, sums and products of monomial ideals are monomial ideals.

PROOF. The assertion about the sum follows immediately from the definition of the ideal generated by a set. By the definition of a product of ideals, IJ is generated by all products fg with $f \in I$ and $g \in J$. Writing $f = \sum_i a_i u_i$ and $g = \sum_j b_j v_j$, we see that every such product belongs to the ideal generated by the $u_i v_j$. The reverse inclusion is clear because $u_i v_j \in IJ$. $\square$

Iterating the product formula, one obtains

$$I^q = (u_{i_1} \cdots u_{i_q} \mid 1 \leq i_1, \dots, i_q \leq r)$$

for every $q \geq 1$. These displayed generating sets need not be minimal: after carrying out an operation, one obtains the minimal monomial generators by deleting every generator divisible by another generator in the list.

For arbitrary ideals, finding generators of an intersection can be difficult. For monomial ideals, this process is explicit.

LEMMA 2.3.2. *Let I and J be monomial ideals. Then $I \cap J$ is a monomial ideal, and*

$$\mathcal{G} = \{\text{lcm}(u, v) \mid u \in G(I) \text{ and } v \in G(J)\}$$

is a set of generators of $I \cap J$.

PROOF. Let $f \in I \cap J$. By [Corollary 2.1.3](#), since I and J are monomial ideals, it follows that $\text{supp}(f) \subset I \cap J$. Again applying [Corollary 2.1.3](#), we see that $I \cap J$ is a monomial ideal.

Let $w \in \text{Mon}(S)$ be a monomial in $I \cap J$. Due to [Proposition 2.1.6](#), there exist $u \in G(I)$ and $v \in G(J)$ such that $u \mid w$ and $v \mid w$. It follows that $\text{lcm}(u, v)$ divides w . Since $\text{lcm}(u, v) \in I \cap J$ for all $u \in G(I)$ and $v \in G(J)$, we conclude that $\mathcal{G}$ is indeed a set of generators of $I \cap J$. $\square$

Similarly, computing ideal quotients is difficult in general. However, for monomial ideals, the task is quite explicit.

LEMMA 2.3.3. *Let I and J be monomial ideals. Then $I : J$ is a monomial ideal, and*

$$I : J = \bigcap_{v \in G(J)} I : v.$$

Moreover, for any monomial $v \in \text{Mon}(S)$, the set $\{u / \gcd(u, v) \mid u \in G(I)\}$ generates $I : v$. Here an intersection indexed by the empty set is understood to be S .

PROOF. Let $f \in I : J$. Then $fv \in I$ for all $v \in G(J)$. In view of Corollary 2.1.3 we have $\text{supp}(f)v = \text{supp}(fv) \subset I$. This implies that $\text{supp}(f) \subset I : J$. Thus Corollary 2.1.3 yields that $I : J$ is a monomial ideal.

The given presentation follows from Exercise 1.10. It is clear that $\{u / \gcd(u, v) \mid u \in G(I)\} \subset I : v$. So now let $w \in \text{Mon}(S)$ be a monomial in $I : v$. Then there exists $u \in G(I)$ such that u divides wv . This implies that $u / \gcd(u, v)$ divides w , as desired. $\square$

We next make saturation and radicals equally explicit. For a monomial $u = x_1^{a_1} \cdots x_n^{a_n}$, set

$$\sqrt{u} := \prod_{a_i > 0} x_i.$$

Thus $\sqrt{u}$ is the squarefree monomial with the same variables as u . More generally, if $A \subseteq [n]$, set

$$u^{(A)} := \prod_{i \notin A} x_i^{a_i};$$

this is the monomial obtained from u by removing all factors involving the variables x_i with $i \in A$.

PROPOSITION 2.3.4. *Let $I \subset S$ be a monomial ideal and let $v \in \text{Mon}(S)$. Set $A = \{i \in [n] \mid x_i \mid v\}$. Then*

$$I : v^\infty = (u^{(A)} \mid u \in G(I)).$$

If $J \subset S$ is another monomial ideal, then

$$I : J^\infty = \bigcap_{v \in G(J)} I : v^\infty.$$

In particular, both $I : v^\infty$ and $I : J^\infty$ are monomial ideals.

PROOF. By Lemma 2.3.3, the ideal $I : v^q$ is generated by

$$\left\{ \frac{u}{\gcd(u, v^q)} \mid u \in G(I) \right\}.$$

For all sufficiently large q , the monomial $\gcd(u, v^q)$ contains the full power of every variable of u whose index belongs to A . For such q , one has $u/\gcd(u, v^q) = u^{(A)}$ simultaneously for all $u \in G(I)$. Since $I : v \subseteq I : v^2 \subseteq \cdots$, the first formula follows.

The inclusion $I : J^\infty \subseteq \bigcap_{v \in G(J)} I : v^\infty$ is clear. For the reverse inclusion, let f belong to the intersection. For each $v \in G(J)$, choose $q_v \geq 1$ such that $fv^{q_v} \in I$. The set $G(J)$ is finite, so we may set

$$q = 1 + \sum_{v \in G(J)} (q_v - 1).$$

Every monomial generator of J^q is a product of q elements of $G(J)$. In such a product, some $v \in G(J)$ occurs at least q_v times. It follows that multiplying this product by f gives an element of I . Therefore $fJ^q \subseteq I$, and so $f \in I : J^\infty$. $\square$

PROPOSITION 2.3.5. *Let $I \subset S$ be a monomial ideal. Then*

$$\sqrt{I} = (\sqrt{u} \mid u \in G(I)).$$

In particular, the radical of a monomial ideal is a monomial ideal, and I is radical if and only if every monomial in $G(I)$ is squarefree.

PROOF. We first observe that every ideal generated by squarefree monomials is radical. Let $R = (m_1, \dots, m_r)$, where every m_j is squarefree. The cases $R = (0)$ and $R = S$ are clear, so assume that $r \geq 1$ and that every m_j is nonconstant. For each choice of a variable x_{i_j} dividing m_j , let

$$P = (x_{i_1}, \dots, x_{i_r}).$$

The ideal P is prime, because its quotient is a polynomial ring over $\mathbb{k}$, and $R \subseteq P$. We claim that R is the intersection of all the primes obtained in this way. Indeed, if a monomial w does not belong to R , then $m_j \nmid w$ for every j . Since m_j is squarefree, for every j we can choose a variable x_{i_j} which divides m_j but not w . The corresponding prime P contains R but not w . Thus R and the intersection of these primes contain exactly the same monomials; [Theorem 2.1.2](#) gives equality. An intersection of prime ideals is radical, so the observation follows.

Now set $R = (\sqrt{u} \mid u \in G(I))$. Every generator u of I is a multiple of $\sqrt{u}$, so $I \subseteq R$. The preceding observation shows that R is radical, and hence $\sqrt{I} \subseteq R$. Conversely, for each $u \in G(I)$, a sufficiently large power of $\sqrt{u}$ is divisible by u . Thus $\sqrt{u} \in \sqrt{I}$, which gives $R \subseteq \sqrt{I}$ and proves the formula.

If every element of $G(I)$ is squarefree, the formula gives $\sqrt{I} = I$. Conversely, if I is radical and $u \in G(I)$, then $\sqrt{u} \in \sqrt{I} = I$ and $\sqrt{u} \mid u$. The minimality of u gives $u = \sqrt{u}$. $\square$

EXAMPLE 2.3.6. Let

$$I = (x^4, x^2y^2, y^5), \quad J = (x^3y, y^4) \subset \mathbb{k}[x, y].$$

After deleting redundant generators, the formulas above give

$$\begin{aligned} G(I + J) &= \{x^4, x^3y, x^2y^2, y^4\}, \\ G(IJ) &= \{x^7y, x^5y^3, x^4y^4, x^2y^6, y^9\}, \\ G(I \cap J) &= \{x^4y, x^3y^2, x^2y^4, y^5\}. \end{aligned}$$

For $v = xy^2$, the colon formula yields

$$I : v = (x^3, x, y^3) = (x, y^3),$$

and the radical formula yields $\sqrt{I} = (x, y)$.

EXAMPLE 2.3.7. Let $L = (x^2y, xz^2) \subset \mathbb{k}[x, y, z]$. Removing all powers of x from the generators gives

$$L : x^\infty = (y, z^2).$$

On the other hand, removing all powers of y gives

$$L : y^\infty = (x^2, xz^2),$$

and $\sqrt{L} = (xy, xz) = x(y, z)$. These computations illustrate that saturation removes the factors supported on the saturating monomial, whereas taking a radical keeps each variable but lowers every positive exponent to one.

2.4. Primary decomposition of monomial ideals

In this section, we discuss the notion of primary decomposition for monomial ideals. A decomposition of an ideal $I \subset S$ as an intersection $I = \bigcap_{i=1}^m Q_i$ of ideals is called *irredundant* if none of the ideals Q_i can be omitted in this presentation. We call a monomial of the form x_i^a , with $a \geq 1$, a *pure power*. The following important theorem does the job for monomial ideals.

THEOREM 2.4.1. *Let $I \subset S = \mathbb{k}[x_1, \dots, x_n]$ be a proper monomial ideal. Then $I = \bigcap_{i=1}^m Q_i$, where each Q_i is generated by pure powers of the variables. In other words, each Q_i is of the form $(x_{i_1}^{a_1}, \dots, x_{i_k}^{a_k}) \subset S$. Moreover, an irredundant decomposition of this form is unique.*

PROOF. If $I = (0)$, take the one-term decomposition $I = (0)$, where (0) is generated by the empty set of pure powers. Thus assume $I \neq (0)$. Let $G(I) = \{u_1, \dots, u_r\}$, and suppose some u_i is not a pure power, say u_1 . Choose a variable x_j dividing u_1 , and write $u_1 = vw$, where v is the full power of x_j occurring in u_1 . Then v and w are coprime monomials with $v \neq 1 \neq w$. We claim that $I = I_1 \cap I_2$ where $I_1 = (v, u_2, \dots, u_r)$ and $I_2 = (w, u_2, \dots, u_r)$. Indeed, since v and w are coprime, Lemma 2.3.2 yields

$$I_1 \cap I_2 = (\text{lcm}(v, w), u_2, \dots, u_r) = (u_1, u_2, \dots, u_r) = I.$$

Here all the other least common multiples supplied by Lemma 2.3.2 are multiples of some u_i with $i \geq 2$.

If either $G(I_1)$ or $G(I_2)$ contains an element which is not a pure power, we proceed as before. This process terminates. Indeed, for a nonzero proper monomial ideal L , set

$$c(L) = \sum_{u \in G(L)} (|\{i \mid x_i \mid u\}| - 1).$$

The integer $c(L)$ is nonnegative, and it is zero precisely when L is generated by pure powers. In the above splitting, $c(I_1) < c(I)$ and $c(I_2) < c(I)$: the generator u_1 is replaced respectively by the pure power v and by the monomial w , whose support has one fewer variable, and passing to a minimal generating set can only delete terms from the displayed sum. Induction on $c(I)$ therefore gives, after finitely many steps, a presentation of I as an intersection of monomial ideals generated by pure powers. By omitting those ideals which contain the intersection of the others we end up with an irredundant intersection.

So it remains to show uniqueness. Let $Q_1 \cap \cdots \cap Q_r = I = Q'_1 \cap \cdots \cap Q'_s$ be two irredundant intersections of ideals generated by pure powers. We will show that for each $1 \leq i \leq r$ there exists $1 \leq j \leq s$ such that $Q'_j \subseteq Q_i$. By symmetry we then also have that for each $1 \leq k \leq s$ there exists $1 \leq l \leq r$ such that $Q_l \subseteq Q'_k$. This will then imply that $r = s$ and $\{Q_1, \dots, Q_r\} = \{Q'_1, \dots, Q'_s\}$.

Let $1 \leq i \leq r$. We may assume that $Q_i = (x_1^{a_1}, \dots, x_k^{a_k})$. Suppose that $Q'_j \not\subseteq Q_i$ for all $1 \leq j \leq s$. Then for each j there exists $x_{\ell_j}^{b_j} \in Q'_j \setminus Q_i$. It follows that either $\ell_j > k$ or $b_j < a_{\ell_j}$. Let

$$u = \text{lcm}\{x_{\ell_1}^{b_1}, \dots, x_{\ell_s}^{b_s}\}.$$

We have $u \in \bigcap_{j=1}^s Q'_j = I \subseteq Q_i$. If $k = 0$, then $Q_i = (0)$ and this is already a contradiction. Therefore there exists $1 \leq c \leq k$ such that $x_c^{a_c}$ divides u . But this is impossible: every pure power of x_c which occurs in the least common multiple has exponent strictly smaller than a_c . We have proved that for every i there is a j with $Q'_j \subseteq Q_i$.

By symmetry, for this j there is an ℓ with $Q_\ell \subseteq Q'_j$. Thus $Q_\ell \subseteq Q_i$. The irredundancy of $Q_1 \cap \cdots \cap Q_r$ forces $\ell = i$; otherwise the component Q_i could be omitted. As a consequence,

$$Q_i = Q'_j.$$

It follows that every component in the first decomposition occurs in the second, and symmetry gives the reverse inclusion. Therefore $r = s$ and the two collections of components are equal. $\square$

The proof gives an effective procedure: split one mixed monomial generator at a time, compute each intersection by least common multiples, and remove redundant components at the end.

EXAMPLE 2.4.2. Let $I = (x^2y, xy^2) \subset \mathbb{k}[x, y]$. First split the common mixed behavior as follows:

$$I = (x^2, y^2) \cap (xy).$$

The remaining mixed principal ideal satisfies $(xy) = (x) \cap (y)$. Hence

$$I = (x) \cap (y) \cap (x^2, y^2)$$

is an irredundant decomposition into ideals generated by pure powers. Indeed, y^2 belongs to $(y) \cap (x^2, y^2)$ but not to (x) ; similarly, x^2 detects the component (y) and xy detects the component (x^2, y^2) .

For Artinian monomial ideals, that is, monomial ideals I with $\dim_{\mathbb{k}}(S/I) < \infty$, there is a direct description of the decomposition in terms of standard monomials. A standard monomial m of I is called *maximal* if every proper monomial multiple of m belongs to I .

PROPOSITION 2.4.3. *Let $I \subset S$ be a proper Artinian monomial ideal, and let $\mathcal{C}(I)$ be the set of its maximal standard monomials. If $m = x_1^{a_1} \cdots x_n^{a_n} \in \mathcal{C}(I)$, set*

$$Q_m = (x_1^{a_1+1}, \dots, x_n^{a_n+1}).$$

Then

$$I = \bigcap_{m \in \mathcal{C}(I)} Q_m,$$

and this decomposition is irredundant.

PROOF. Since I is Artinian, it has only finitely many standard monomials by [Corollary 2.1.4](#). Thus every standard monomial divides a maximal standard monomial. For $u = x_1^{b_1} \cdots x_n^{b_n}$, one has

$$u \notin Q_m \iff b_i \leq a_i \text{ for all } i \iff u \mid m.$$

It follows that $u \notin \bigcap_{m \in \mathcal{C}(I)} Q_m$ if and only if u divides some maximal standard monomial of I . This is equivalent to u being standard: every divisor of a standard monomial is standard, and every standard monomial divides a maximal one. Therefore $I = \bigcap_m Q_m$.

Distinct maximal standard monomials are incomparable under divisibility. Fix $m \in \mathcal{C}(I)$. We have $m \notin Q_m$, while $m \in Q_{m'}$ for every $m' \in \mathcal{C}(I)$ with $m' \neq m$; otherwise $m \mid m'$, contradicting maximality. Thus m belongs to the intersection of all components other than Q_m but not to Q_m . Consequently no component can be omitted. $\square$

EXAMPLE 2.4.4. Consider the Artinian ideal

$$I = (x^3, x^2y^2, y^4) \subset \mathbb{k}[x, y].$$

Its maximal standard monomials are x^2y and xy^3 . The preceding proposition gives

$$I = (x^3, y^2) \cap (x^2, y^4).$$

This equality can also be checked immediately by taking the least common multiples of the generators of the two components and deleting the redundant monomial x^3y^4 .

A proper monomial ideal is called *irreducible* if it cannot be written as a proper intersection of two other monomial ideals. Equivalently, I is irreducible if whenever $I = I_1 \cap I_2$ then either $I = I_1$ or $I = I_2$. It is called *reducible* if it is not irreducible.

COROLLARY 2.4.5. *A proper monomial ideal is irreducible if and only if it is generated by pure powers of the variables.*

PROOF. Let $Q = (x_{i_1}^{a_1}, \dots, x_{i_k}^{a_k})$. Suppose $Q = I \cap J$, where I and J are monomial ideals properly containing Q . Choose monomials $u \in I \setminus Q$ and $v \in J \setminus Q$. For every $1 \leq j \leq k$, the exponent of x_{i_j} in both u and v is strictly smaller than a_j . The same is therefore true of $\text{lcm}(u, v)$, and so $\text{lcm}(u, v) \notin Q$. On the other hand, $\text{lcm}(u, v) \in I \cap J = Q$, a contradiction.

Conversely, suppose that Q is not generated by pure powers. Then $G(Q)$ contains a mixed monomial $u = vw$ with $\gcd(v, w) = 1$ and $v \neq 1 \neq w$. As in the proof of [Theorem 2.4.1](#), replacing u by v and by w gives monomial ideals I and J such that $Q = I \cap J$. Neither v nor w belongs to Q : otherwise a minimal generator of Q would divide a proper divisor of u , contradicting the minimality of u . Hence both I and J properly contain Q , and Q is reducible. $\square$

LEMMA 2.4.6. *Let $I \subset S$ be an irreducible monomial ideal. Then I is primary.*

PROOF. By [Corollary 2.4.5](#), we may write

$$I = (x_{i_1}^{a_1}, \dots, x_{i_k}^{a_k}).$$

Set $A = S/I$ and $B = \mathbb{k}[x_j \mid j \notin \{i_1, \dots, i_k\}]$. The residue classes of the monomials

$$x_{i_1}^{b_1} \cdots x_{i_k}^{b_k}, \quad 0 \leq b_j < a_j,$$

form a basis of the free B -module A . Let $n = (x_{i_1}, \dots, x_{i_k})A$. If

$$N = 1 + \sum_{j=1}^k (a_j - 1),$$

then $n^N = 0$.

We claim that every element of A outside n is a nonzerodivisor. Let $\bar{g} \notin n$. We can write $\bar{g} = h + r$, where $0 \neq h \in B$ and $r \in n$. Since $r^N = 0$, we have

$$(h + r)(h^{N-1} - h^{N-2}r + \cdots + (-r)^{N-1}) = h^N.$$

Multiplication by h^N is injective on A , because A is a free module over the domain B . It follows from the displayed equality that multiplication by $\bar{g}$ is also injective, proving the claim.

Now let $f, g \in S$ with $fg \in I$ and $f \notin I$. In A we have $\bar{f} \neq 0$ and $\bar{f}\bar{g} = 0$. The claim implies that $\bar{g} \in n$, and consequently $\bar{g}^N = 0$. Thus $g^N \in I$, which proves that I is primary. $\square$

REMARK 2.4.7. Corollary 2.4.5 and Lemma 2.4.6 yield that an ideal $Q = (x_{i_1}^{a_1}, \dots, x_{i_k}^{a_k}) \subset S$ generated by pure powers of the variables is primary with respect to the prime ideal $\sqrt{Q} = (x_{i_1}, \dots, x_{i_k}) \subset S$.

REMARK 2.4.8. Theorem 2.4.1 in combination with Corollary 2.4.5 now says that each proper monomial ideal has a unique decomposition as an irredundant intersection of irreducible monomial ideals. Moreover, Lemma 2.4.6 tells us that such a decomposition is primary.

2.5. Hilbert functions of monomial ideals

In this section, we study the Hilbert functions of monomial ideals. Our main goal is to provide a self-contained and combinatorial proof of the existence of Hilbert polynomials and Hilbert series. Our proof will heavily depend on the decomposition of monomial ideals into ideals generated by pure powers of the variables (see Theorem 2.4.1) and on the fact that we have distributive law of sums over intersections in the case of monomial ideals (see Exercise 2.1). Since the polynomial ring in n variables should have dimension n , the following definition gives a purely algebraic notion of dimension for a monomial quotient.

DEFINITION-LEMMA 2.5.1. Let $I \subset S$ be a proper monomial ideal. From Theorem 2.4.1, let $I = \bigcap_{i=1}^m Q_i$ be an irredundant decomposition where each Q_i is generated by pure powers. If Q_i is generated by positive powers of exactly k_i distinct variables, set

$$\dim(S/Q_i) := n - k_i.$$

We define the *dimension* of S/I by

$$\dim(S/I) := \max \{ \dim(S/Q_i) \mid 1 \leq i \leq m \}.$$

PROOF. The irredundant decomposition is unique by the preceding remark, so the integer is well defined. If

$$Q_i = (x_{i_1}^{a_1}, \dots, x_{i_{k_i}}^{a_{k_i}}),$$

then S/Q_i is a finite free module over the polynomial ring in the remaining $n - k_i$ variables. This also explains the formula $\dim(S/Q_i) = n - k_i$. $\square$

DEFINITION 2.5.2. An S -module is an Abelian group M on which S acts linearly by a mapping $S \times M \rightarrow M$ that satisfies the axioms

- (i) $a(x + y) = ax + ay$.
- (ii) $(a + b)x = ax + bx$.
- (iii) $(ab)x = a(bx)$.
- (iv) $1x = x$.

for all $a, b \in S$ and $x, y \in M$.

We say that M is a *graded* S -module if it has a direct sum decomposition $M = \bigoplus_{i \in \mathbb{Z}} M_i$ as $\mathbb{k}$ -vector spaces and $S_i M_j \subseteq M_{i+j}$ for all $i, j \in \mathbb{Z}$, where $S_i = 0$ for $i < 0$. For a graded S -module M and $p \geq 0$, we consider the graded module $M(-p)$ with graded parts $M(-p)_i = M_{i-p}$. Of particular interest for us is the free graded module $S(-p)$ defined by $S(-p)_i = S_{i-p}$, i.e., for $p \geq 0$ the module $S(-p)$ is shifted p degrees with $S(-p)_p = S_0 = \mathbb{k}$. Given a monomial ideal $I \subset S$, from [Theorem 2.1.2](#) and [Corollary 2.1.4](#), we obtain that

$$I = \bigoplus_{j \geq 0} I_j \quad \text{and} \quad S/I = \bigoplus_{j \geq 0} [S/I]_j = \bigoplus_{j \geq 0} S_j/I_j.$$

These are graded S -modules whose graded parts are finite dimensional $\mathbb{k}$ -vector spaces. We will see that this also holds when I is a homogeneous ideal.

DEFINITION 2.5.3. Let M be a graded S -module with finite dimensional parts such that $M_k = 0$ for $k \ll 0$. The *Hilbert series* of M is the following Laurent series

$$\text{Hilb}_M(t) := \sum_{k \in \mathbb{Z}} \dim_{\mathbb{k}}(M_k) t^k.$$

Our primary case of interest is the Hilbert series of a quotient ring S/I by a homogeneous ideal I ; in this case, we have a power series $\text{Hilb}_{S/I}(t) = \sum_{k \geq 0} \dim_{\mathbb{k}}([S/I]_k) t^k \in \mathbb{N}[[t]]$ because $[S/I]_k = 0$ for $k < 0$.

Our goal is to prove the following important theorem.

THEOREM 2.5.4 (Hilbert). *Let $I \subset S = \mathbb{k}[x_1, \dots, x_n]$ be a proper monomial ideal, set $R = S/I$, and let $d = \dim(R)$. Then the following statements hold:*

(i) *We have the equality*

$$\text{Hilb}_{S/I}(t) = \frac{Q(t)}{(1-t)^d}$$

where $Q(t) \in \mathbb{Z}[t]$ is a polynomial with integer coefficients and $Q(1) \in \mathbb{Z}_+$ is a positive integer.

(ii) *If $d \geq 1$, there is a unique polynomial*

$$P_R(z) = e_0 \frac{z^{d-1}}{(d-1)!} + (\text{lower degree terms}) \in \mathbb{Q}[z]$$

(called the Hilbert polynomial of R) of degree $d-1$ such that $e_0 = Q(1)$ and

$$P_R(k) = \dim_{\mathbb{k}}(R_k)$$

for all $k \gg 0$. If $d = 0$, then S/I is finite dimensional and its Hilbert function is eventually zero; in this case we define the Hilbert polynomial of R to be the zero polynomial, and $Q(1) = \dim_{\mathbb{k}}(S/I)$.

DEFINITION 2.5.5. Let $I \subset S$ be a proper monomial ideal and set $R = S/I$. The *affine degree* of R is

$$\deg_{\text{aff}}(R) := Q(1),$$

where $Q(t)$ is the numerator in Theorem 2.5.4(i). When $d \geq 1$, this is the normalized leading coefficient of the Hilbert polynomial $P_R(z)$; when $d = 0$, it is $\dim_{\mathbb{k}}(R)$. Different monomial ideals can have the same radical while their quotient rings have different affine degrees.

To prove the above far-reaching theorem we shall need several technical results. We start with an important example.

EXAMPLE 2.5.6. In how many ways can you share k oranges with n people? This is a simple combinatorial problem whose solution is

$$\binom{k+n-1}{n-1} = \frac{(n+k-1)!}{(n-1)!k!}.$$

Equivalently, we can ask how many tuples $(a_1, \dots, a_n) \in \mathbb{N}^n$ are there such that

$$a_1 + \dots + a_n = k?$$

Since the monomials of degree k give a $\mathbb{k}$ -basis of S_k , we get

$$\dim_{\mathbb{k}}(S_k) = \binom{k+n-1}{n-1}.$$

On the other hand, the binomial theorem tells us that

$$\frac{1}{(1-t)^n} = \sum_{k=0}^{\infty} \binom{-n}{k} (-t)^k = \sum_{k=0}^{\infty} \binom{n+k-1}{k} t^k = \sum_{k=0}^{\infty} \binom{n+k-1}{n-1} t^k.$$

Therefore, we obtain the appealing fact

$$\text{Hilb}_S(t) = \frac{1}{(1-t)^n} \in \mathbb{N}[[t]].$$

Shifting makes a simple change to the Hilbert series.

LEMMA 2.5.7. Let M be a graded S -module with finite graded parts such that $M_k = 0$ for $k \ll 0$, and let $p \geq 0$. Then

$$\text{Hilb}_{M(-p)}(t) = t^p \text{Hilb}_M(t).$$

In particular, $\text{Hilb}_{S(-p)}(t) = \frac{t^p}{(1-t)^n}$.

PROOF. Making the following simple algebraic manipulation

$$\text{Hilb}_{M(-p)}(t) = \sum_{k \in \mathbb{Z}} \dim_{\mathbb{k}}(M_{k-p}) t^k = \sum_{k \in \mathbb{Z}} \dim_{\mathbb{k}}(M_k) t^{k+p} = t^p \text{Hilb}_M(t)$$

we get the equality. □

We say that an S -linear map $\varphi : M \rightarrow N$ between graded S -modules is *graded* if $\varphi(M_k) \subset N_k$ for all $k \in \mathbb{Z}$. Let L, M, N be S -modules. We say that we have a short exact sequence

$$0 \rightarrow L \xrightarrow{\varphi} M \xrightarrow{\psi} N \rightarrow 0;$$

if φ and ψ are S -linear maps, φ is injective, ψ is surjective and $\text{Ker}(\psi) = \text{Im}(\varphi)$. We say the short exact sequence is *graded* if the modules L, M, N and the maps φ, ψ are all graded.

The next lemma says that Hilbert series are additive.

LEMMA 2.5.8. *Let $0 \rightarrow L \rightarrow M \rightarrow N \rightarrow 0$ be a graded short exact sequence of graded S -modules with finite graded parts which vanish in all sufficiently negative degrees. Then*

$$\text{Hilb}_M(t) = \text{Hilb}_L(t) + \text{Hilb}_N(t).$$

PROOF. From the assumptions we get $M_k \cong L_k \oplus N_k$. So the required equality

$$\text{Hilb}_M(t) = \sum_{j \in \mathbb{Z}} \dim_{\mathbb{k}}(M_j) t^j = \sum_{j \in \mathbb{Z}} (\dim_{\mathbb{k}}(L_j) + \dim_{\mathbb{k}}(N_j)) t^j = \text{Hilb}_L(t) + \text{Hilb}_N(t)$$

follows. □

We now discuss an “inclusion-exclusion” type of argument that will allow us to express the Hilbert series of an intersection of monomial ideals. We start with two ideals.

LEMMA 2.5.9. *Let $I, J \subset S$ be two ideals. Then we have a natural short exact sequence*

$$0 \rightarrow S/(I \cap J) \xrightarrow{\varphi} S/I \oplus S/J \xrightarrow{\pi} S/(I + J) \rightarrow 0,$$

where $\varphi(f + I \cap J) = (f + I, f + J)$ and $\pi(f + I, g + J) = (f - g + I + J)$. Moreover, if I and J are monomial ideals (or homogeneous, as we will see later), then this short exact sequence is graded.

PROOF. For any $f + I + J \in S/(I + J)$, we have $\pi(f + I, 0 + J) = f + I + J$; thus the map π is surjective. For any $f + I \cap J \in S/(I \cap J)$, if $\varphi(f + I \cap J) = (f + I, f + J) = (0 + I, 0 + J)$, then $f \in I \cap J$; thus the map φ is injective.

The inclusion $\text{Im}(\varphi) \subset \text{Ker}(\pi)$ is clear by construction. On the other hand, let $(f + I, g + J) \in \text{Ker}(\pi)$. This means that $f - g \in I + J$, and so we can write $f - g = -a + b$ with $a \in I$ and $b \in J$. Since $f + a = g + b$, $a \in I$ and $b \in J$, it now follows that

$$\varphi(f + a + I \cap J) = (f + a + I, g + b + J) = (f + I, g + J);$$

hence $(f + I, g + J) \in \text{Im}(\varphi)$. Therefore, $\text{Im}(\varphi) = \text{Ker}(\pi)$ and it follows that we have a short exact sequence.

If I and J are monomial ideals, then the modules $S/(I \cap J)$, $S/I \oplus S/J$, $S/(I + J)$ are graded (see Theorem 2.1.2 and Corollary 2.1.4) and the maps φ and π are clearly graded. □

NOTATION 2.5.10. Given a sequence of monomial ideals $I_1, \dots, I_k \subset S$ and a nonempty subset $\mathfrak{J} \subseteq [k]$, we define the ideal $I_{\mathfrak{J}} := \sum_{i \in \mathfrak{J}} I_i$. We further specify

$$\text{IE}(\{I_1, \dots, I_k\})(t) := \sum_{\emptyset \neq \mathfrak{J} \subseteq [k]} (-1)^{|\mathfrak{J}|-1} \text{Hilb}_{S/I_{\mathfrak{J}}}(t) \in \mathbb{Z}[[t]].$$

The next result is only valid for monomial ideals and it uses the distributive law from [Exercise 2.1](#).

PROPOSITION 2.5.11. *Let $I_1, \dots, I_k \subset S$ be monomial ideals and consider the intersection $I = I_1 \cap \dots \cap I_k \subset S$. Then we have the equality*

$$\text{Hilb}_{S/I}(t) = \text{IE}(\{I_1, \dots, I_k\})(t).$$

PROOF. We proceed by induction on k . The case $k = 1$ is clear. Set $A = I_1 \cap \dots \cap I_{k-1}$. From [Lemma 2.5.9](#), we have the short exact sequence

$$0 \rightarrow S/I \rightarrow S/A \oplus S/I_k \rightarrow S/(A + I_k) \rightarrow 0.$$

The additivity of Hilbert series (see [Lemma 2.5.8](#)) gives the equality

$$\text{Hilb}_{S/I}(t) = \text{Hilb}_{S/A}(t) + \text{Hilb}_{S/I_k}(t) - \text{Hilb}_{S/(A+I_k)}(t).$$

Due to [Exercise 2.1](#), we have the distributive equality

$$(I_1 \cap \dots \cap I_{k-1}) + I_k = (I_1 + I_k) \cap \dots \cap (I_{k-1} + I_k).$$

Applying the inductive hypothesis to the two families $I_1, \dots, I_{k-1}$ and $I_1 + I_k, \dots, I_{k-1} + I_k$, we obtain

$$\begin{aligned} \text{Hilb}_{S/I}(t) &= \sum_{\emptyset \neq \mathfrak{J} \subseteq [k-1]} (-1)^{|\mathfrak{J}|-1} \text{Hilb}_{S/I_{\mathfrak{J}}}(t) + \text{Hilb}_{S/I_k}(t) \\ &\quad - \sum_{\emptyset \neq \mathfrak{J} \subseteq [k-1]} (-1)^{|\mathfrak{J}|-1} \text{Hilb}_{S/(I_{\mathfrak{J}}+I_k)}(t) \\ &= \sum_{\emptyset \neq \mathfrak{J} \subseteq [k-1]} (-1)^{|\mathfrak{J}|-1} \text{Hilb}_{S/I_{\mathfrak{J}}}(t) + \text{Hilb}_{S/I_k}(t) \\ &\quad + \sum_{\emptyset \neq \mathfrak{J} \subseteq [k-1]} (-1)^{|\mathfrak{J}|} \text{Hilb}_{S/I_{\mathfrak{J} \cup \{k\}}}(t) \\ &= \text{IE}(\{I_1, \dots, I_k\})(t). \end{aligned}$$

whence the result follows. $\square$

Another necessary ingredient is that of regular sequences. Given an S -module M , we say that $f \in S$ is regular (or a nonzerodivisor) on M if whenever $w \in M$ and $fw = 0 \in M$ then $w = 0 \in M$.

DEFINITION 2.5.12. A sequence of polynomials $f_1, \dots, f_m$ in S is *regular* if the following two conditions hold:

- (i) $(f_1, \dots, f_m) \neq S$.
- (ii) $f_1 \neq 0$ and f_i is regular on $S/(f_1, \dots, f_{i-1})$ for all $2 \leq i \leq m$.

By convention, the empty sequence is regular.

The lemma below gives another reason why ideals generated by pure powers of the variables are quite special.

PROPOSITION 2.5.13. Let $Q = (x_{i_1}^{a_1}, \dots, x_{i_k}^{a_k}) \subset S$ be an ideal generated by pure powers of the variables (with $i_1 < \dots < i_k$ and $a_j > 0$ for all $1 \leq j \leq k$). Then the following statements hold:

- (i) $x_{i_1}^{a_1}, \dots, x_{i_k}^{a_k}$ form a regular sequence.
- (ii) We have the equality

$$\text{Hilb}_{S/Q}(t) = \frac{\prod_{j=1}^k (1 + t + t^2 + \dots + t^{a_j-1})}{(1-t)^d}$$

where $d = n - k = \dim(S/Q)$.

PROOF. (i) By Lemma 2.3.3, we have

$$(x_{i_1}^{a_1}, \dots, x_{i_{j-1}}^{a_{j-1}}) : x_{i_j}^{a_j} = (x_{i_1}^{a_1}, \dots, x_{i_{j-1}}^{a_{j-1}}).$$

Hence Exercise 2.6 implies that $x_{i_1}^{a_1}, \dots, x_{i_k}^{a_k}$ form a regular sequence.

(ii) Let $1 \leq j \leq k$. Let $J = (x_{i_1}^{a_1}, \dots, x_{i_j}^{a_j}) \subset S$ and $J' = (x_{i_1}^{a_1}, \dots, x_{i_{j-1}}^{a_{j-1}}) \subset S$ (when $j = 1$, we have by convention $J' = 0$). We claim that we have a graded short exact sequence

$$0 \rightarrow (S/J')(-a_j) \xrightarrow{\varphi} S/J' \xrightarrow{\pi} S/J \rightarrow 0,$$

where $\varphi(f + J') = x_{i_j}^{a_j} f + J'$ and $\pi(f + J') = f + J$.

For any $f + J \in S/J$, we have $\pi(f + J') = f + J$; thus the map π is surjective. Notice that $\text{Im}(\varphi) = \text{Ker}(\pi)$ because for any $f + J' \in S/J'$ we have $\pi(f + J') = f + J = 0 + J$ if and only if $f + J' = x_{i_j}^{a_j} g + J'$ for some $g \in S$. We know that φ is injective because $x_{i_j}^{a_j}$ is regular on S/J' . Thus we indeed have a short exact sequence.

It remains to show that the short exact sequence is graded. All the considered modules are graded since we are dealing with monomial ideals. The map π is clearly graded. For any $\ell \in \mathbb{Z}$ and $f + J' \in [(S/J')(-a_j)]_\ell = [S/J']_{\ell-a_j}$ (choosing $f = 0$ or a homogeneous representative $f \in S$ with $\deg(f) = \ell - a_j$), we obtain

$$\varphi(f + J') = x_{i_j}^{a_j} f + J' \in [S/J']_\ell;$$

so φ is also graded. This concludes the proof that we have a graded short exact sequence.

By combining [Lemma 2.5.8](#) and [Lemma 2.5.7](#), we get

$$\text{Hilb}_{S/J}(t) = \text{Hilb}_{S/J'}(t) - t^{a_j} \text{Hilb}_{S/J'}(t) = (1 - t^{a_j}) \text{Hilb}_{S/J'}(t).$$

Finally, proceeding inductively and utilizing the initial computation of [Example 2.5.6](#), we obtain

$$\begin{aligned} \text{Hilb}_{S/Q}(t) &= \frac{\prod_{j=1}^k (1 - t^{a_j})}{(1 - t)^n} \\ &= \frac{\prod_{j=1}^k (1 - t)(1 + t + t^2 + \cdots + t^{a_j-1})}{(1 - t)^n} \\ &= \frac{\prod_{j=1}^k (1 + t + t^2 + \cdots + t^{a_j-1})}{(1 - t)^d}, \end{aligned}$$

as required. □

REMARK 2.5.14 (A very simple version of Bézout's theorem). Let $Q = (x_{i_1}^{a_1}, \dots, x_{i_k}^{a_k}) \subset S$. Then [Proposition 2.5.13\(ii\)](#) yields the formula

$$\deg_{\text{aff}}(S/Q) = a_1 \cdots a_k \neq 0.$$

Our last necessary ingredient is a structural result regarding power series. We first discuss a well-known characterization of *numerical polynomials*. Let $F : \mathbb{Z} \rightarrow \mathbb{Z}$ be a function. We define the *difference operator* by setting $(\Delta F)(n) := F(n+1) - F(n)$ for all $n \in \mathbb{Z}$. For all $i \geq 1$, we set $\Delta^i F = \Delta(\Delta^{i-1} F)$. Notice that $\Delta^i F$ is also a function from the integers to the integers.

LEMMA 2.5.15. *Let $d \geq 1$ and let $P(z) \in \mathbb{Q}[z]$ be a polynomial of degree $d-1$. Then the following conditions are equivalent:*

- (a) $P(k) \in \mathbb{Z}$ for all $k \in \mathbb{Z}$ (this means that $P(z)$ is a numerical polynomial).
- (b) There exist integers $a_0, \dots, a_{d-1} \in \mathbb{Z}$ such that

$$P(z) = \sum_{i=0}^{d-1} a_i \binom{z+i}{i}.$$

PROOF. The implication (b) $\Rightarrow$ (a) is clear.

Thus we concentrate on the implication (a) $\Rightarrow$ (b). Notice that the polynomials $\binom{z+i}{i}$ form a $\mathbb{Q}$ -basis of $\mathbb{Q}[z]$ (this can be proved for instance by utilizing the division algorithm on $\mathbb{Q}[z]$). Therefore we can write $P(z) = \sum_{i=0}^{d-1} a_i \binom{z+i}{i}$. We use the identity $\binom{z+i+1}{i} - \binom{z+i}{i} = \binom{z+i}{i-1}$. Thus we have

$$\Delta P(z) = \sum_{i=1}^{d-1} a_i \binom{z+i}{i-1},$$

and so applying the difference operator i -times yields

$$\Delta^i P(z) = \sum_{j=i}^{d-1} a_j \binom{z+j}{j-i}.$$

This implies that $a_i = \Delta^i P(-i-1) \in \mathbb{Z}$, as required. $\square$

LEMMA 2.5.16. *Let $d \geq 1$ and let $H(t) = \frac{Q(t)}{(1-t)^d} = \sum_{k=0}^{\infty} a_k t^k \in \mathbb{N}[[t]]$ be a power series where $Q(t) \in \mathbb{Z}[t]$ is a polynomial with integer coefficients and $Q(1) \neq 0$. Then the following statements hold:*

- (i) $Q(1) \in \mathbb{Z}_+$ is a positive integer.
- (ii) *There is a polynomial $P(z) = e_0 \frac{z^{d-1}}{(d-1)!} + (\text{lower degree terms}) \in \mathbb{Q}[z]$ of degree $d-1$ such that $e_0 = Q(1)$ and $P(k) = a_k$ for all $k \gg 0$.*

PROOF. By assumption, we have $Q(t) = \sum_{j=0}^m c_j t^j$ where each $c_j \in \mathbb{Z}$ is an integer. We can write $Q(t) = \sum_{j=0}^{\ell} e_j (1-t)^j$ with $e_j \in \mathbb{Z}$; when necessary, we set $e_j = 0$ for $j > \ell$. It is then clear that $e_0 = Q(1) = \sum_{j=0}^m c_j$ is a nonzero integer. We can make the expansion

$$H(t) = \frac{\sum_{j=0}^{\ell} e_j (1-t)^j}{(1-t)^d} = \sum_{j=0}^{d-1} \frac{e_j}{(1-t)^{d-j}} + \sum_{j=d}^{\ell} e_j (1-t)^{j-d}.$$

Let $H'(t) = \sum_{j=0}^{d-1} \frac{e_j}{(1-t)^{d-j}} = \sum_{k \geq 0} b_k t^k$. Notice that $b_k = a_k$ for $k \gg 0$ because $\sum_{j=d}^{\ell} e_j (1-t)^{j-d}$ is a polynomial and so it has finitely many terms. Due to [Example 2.5.6](#), we can expand

$$\begin{aligned} H'(t) &= \sum_{j=0}^{d-1} \frac{e_j}{(1-t)^{d-j}} \\ &= \sum_{j=0}^{d-1} e_j \sum_{k=0}^{\infty} \binom{k+d-j-1}{d-j-1} t^k \\ &= \sum_{k=0}^{\infty} \left(\sum_{j=0}^{d-1} e_j \binom{k+d-j-1}{d-j-1} \right) t^k. \end{aligned}$$

Consider the polynomial $P(z) = \sum_{j=0}^{d-1} e_j \binom{z+d-j-1}{d-j-1} \in \mathbb{Q}[z]$. By construction, $P(k) = b_k = a_k$ for $k \gg 0$ and

$$P(z) = e_0 \frac{z^{d-1}}{(d-1)!} + (\text{lower degree terms}) \in \mathbb{Q}[z].$$

Finally, notice that

$$e_0 = (d-1)! \lim_{k \rightarrow \infty} \frac{P(k)}{k^{d-1}} = (d-1)! \lim_{k \rightarrow \infty} \frac{a_k}{k^{d-1}} \geq 0.$$

Since e_0 is nonzero, it follows that $e_0 > 0$, and the result follows. $\square$

We now have all the ingredients to prove [Theorem 2.5.4](#).

PROOF OF THEOREM 2.5.4. Let $I \subset S = \mathbb{k}[x_1, \dots, x_n]$ be a proper monomial ideal, set $R = S/I$, and let $d = \dim(R)$. By Theorem 2.4.1, let $I = \bigcap_{j=1}^k I_j$ be the irredundant decomposition into ideals generated by pure powers of the variables. Following Notation 2.5.10, for each nonempty subset $\mathfrak{J} \subseteq [k]$, set $I_{\mathfrak{J}} = \sum_{j \in \mathfrak{J}} I_j \subset S$ and $d_{\mathfrak{J}} = \dim(S/I_{\mathfrak{J}})$. Notice that each $I_{\mathfrak{J}}$ is an ideal generated by pure powers. We also have $d_{\mathfrak{J}} \leq d$. By utilizing Proposition 2.5.11 and Proposition 2.5.13, we obtain

$$\begin{aligned} \text{Hilb}_{S/I}(t) &= \sum_{\emptyset \neq \mathfrak{J} \subseteq [k]} (-1)^{|\mathfrak{J}|-1} \text{Hilb}_{S/I_{\mathfrak{J}}}(t) \\ &= \sum_{\emptyset \neq \mathfrak{J} \subseteq [k]} (-1)^{|\mathfrak{J}|-1} \frac{Q_{\mathfrak{J}}(t)}{(1-t)^{d_{\mathfrak{J}}}} \\ &= \frac{\sum_{\emptyset \neq \mathfrak{J} \subseteq [k]} (-1)^{|\mathfrak{J}|-1} (1-t)^{d-d_{\mathfrak{J}}} Q_{\mathfrak{J}}(t)}{(1-t)^d} \\ &= \frac{Q(t)}{(1-t)^d}, \end{aligned}$$

where each $Q_{\mathfrak{J}}(t) \in \mathbb{Z}[t]$ is a polynomial with integer coefficients. Thus $Q(t) \in \mathbb{Z}[t]$ is also a polynomial with integer coefficients. If $d = 0$, every component I_j contains a positive power of every variable. Taking the largest of the finitely many exponents for each variable shows that each variable has a power in $I = \bigcap_j I_j$. Thus S/I is finite dimensional by Corollary 2.1.4. In this case $Q(t) = \text{Hilb}_{S/I}(t)$ is a polynomial and

$$Q(1) = \sum_{j \geq 0} \dim_{\mathbb{k}}([S/I]_j) = \dim_{\mathbb{k}}(S/I) > 0.$$

The Hilbert function is eventually zero, so its Hilbert polynomial is the zero polynomial. Thus assume from now on that $d \geq 1$.

We claim that $Q(1) \neq 0$. By contradiction, suppose $Q(1) = 0$. Then we can write

$$Q(t) = (1-t)^c Q'(t),$$

where $c \geq 1$, $Q'(t) \in \mathbb{Z}[t]$, and $Q'(1) \neq 0$. Set $e = d - c$, so that $\text{Hilb}_{S/I}(t) = Q'(t)/(1-t)^e$. After relabeling the components, we may assume that $\dim(S/I_1) = d$. Since $I \subseteq I_1$, we have

$$\dim_{\mathbb{k}}([S/I_1]_j) \leq \dim_{\mathbb{k}}([S/I]_j)$$

for all $j \geq 0$.

If $e \leq 0$, then $\text{Hilb}_{S/I}(t)$ is a polynomial, and hence its coefficients are eventually zero. This contradicts the displayed inequality, because Proposition 2.5.13 shows that $\dim_{\mathbb{k}}([S/I_1]_j) > 0$ for all sufficiently large j . If $e \geq 1$, Lemma 2.5.16 gives a polynomial $P_R(z)$ of degree $e - 1$ such that

$$P_R(j) = \dim_{\mathbb{k}}([S/I]_j)$$

for all $j \gg 0$. The same lemma applied to S/I_1 gives a polynomial $P_{S/I_1}(z)$ of degree $d - 1$ with positive leading coefficient such that

$$P_{S/I_1}(j) = \dim_{\mathbb{k}}([S/I_1]_j)$$

for all $j \gg 0$. Since $e - 1 < d - 1$, the inequality $P_{S/I_1}(j) \leq P_R(j)$ cannot hold for all sufficiently large j . This is again a contradiction, and proves the claim.

Now [Lemma 2.5.16](#) gives $Q(1) > 0$ and a polynomial $P_R(z)$ of degree $d - 1$ with the asserted leading term and eventual values. Such a polynomial is unique, because two polynomials which agree at all sufficiently large integers are equal. $\square$

2.6. MacLagan's theorem (an extension of Dickson's lemma)

Here we discuss an interesting extension of Dickson's lemma given by MacLagan [\[6\]](#). Given any collection of ideals $\mathcal{A} = \{I_\lambda\}_{\lambda \in \Lambda}$ in S , we denote by $\mathcal{A}^{\max}$ the collection of ideals in $\mathcal{A}$ that are maximal with respect to inclusion. We prove the following finiteness theorem.

THEOREM 2.6.1 (MacLagan [\[6\]](#)). *Let $\mathcal{A} = \{I_\lambda\}_{\lambda \in \Lambda}$ be a collection of monomial ideals in S . Then $\mathcal{A}^{\max}$ is a finite set.*

Notice that if each $I_\lambda \subset S$ is a principal monomial ideal (i.e., generated by one monomial), then the above theorem is precisely Dickson's lemma (see [Theorem 2.2.1](#)). This finiteness result is false for arbitrary ideals, as shown by the next easy example.

EXAMPLE 2.6.2. Let $S = \mathbb{k}[x]$ with $\mathbb{k}$ an infinite field. Consider the infinite collection of principal ideals $\mathcal{A} = \{I_\alpha\}_{\alpha \in \mathbb{k}}$ with $I_\alpha = (x - \alpha) \subset S$. Then $\mathcal{A}^{\max} = \mathcal{A}$.

An equivalent formulation of [Theorem 2.6.1](#) is given in the next result.

LEMMA 2.6.3. *[Theorem 2.6.1](#) holds if and only if every infinite collection $\mathcal{A}$ of monomial ideals in S contains an infinite chain $I_1 \supsetneq I_2 \supsetneq \cdots$.*

PROOF. Assume that [Theorem 2.6.1](#) holds. Let $\mathcal{A}$ be an infinite collection of monomial ideals. By the ascending chain condition in [Proposition 2.2.7](#), every element of $\mathcal{A}$ is contained in a maximal element of $\mathcal{A}$: otherwise one could construct an infinite strictly ascending chain inside $\mathcal{A}$. Since $\mathcal{A}^{\max}$ is finite and $\mathcal{A}$ is infinite, there is an ideal $I_1 \in \mathcal{A}^{\max}$ such that

$$\mathcal{A}_1 = \{I \in \mathcal{A} \mid I_1 \supsetneq I\}$$

is infinite. Applying the same argument to $\mathcal{A}_1$, we can choose $I_2 \in \mathcal{A}_1^{\max}$ such that

$$\mathcal{A}_2 = \{I \in \mathcal{A}_1 \mid I_2 \supsetneq I\}$$

is infinite. Proceeding inductively gives an infinite chain $I_1 \supsetneq I_2 \supsetneq \cdots$ of ideals in $\mathcal{A}$.

Conversely, suppose that every infinite collection of monomial ideals contains an infinite strictly descending chain. Let $\mathcal{A}$ be a collection of monomial ideals. If $\mathcal{A}^{\max}$ were infinite,

applying the condition to $\mathcal{A}^{\max}$ would produce an infinite strictly descending chain in $\mathcal{A}^{\max}$. This is impossible because distinct maximal elements are incomparable. Thus the proof is complete. $\square$

We say that a monomial ideal $I \subset S$ is *Artinian* if $\dim_{\mathbb{k}}(S/I) < \infty$ (see [Exercise 2.7](#)). By [Corollary 2.1.4](#), a monomial ideal $I \subset S$ is Artinian if and only if it has finitely many standard monomials.

LEMMA 2.6.4. *Let $\mathcal{A}$ be a collection of Artinian monomial ideals in S . Then $\mathcal{A}^{\max}$ is a finite set.*

PROOF. Suppose, for contradiction, that $\mathcal{B} = \mathcal{A}^{\max}$ is infinite. Choose $I_1 \in \mathcal{B}$. For each $I \in \mathcal{B} \setminus \{I_1\}$, since I and I_1 are incomparable, it follows that I contains some of the finite number of standard monomials of I_1 . As a consequence, there are infinitely many ideals in $\mathcal{B}$ whose intersection with the finite set of standard monomials of I_1 is the same nonempty set. We call this collection $\mathcal{B}_1$. Let $J_1 \subset S$ be the intersection of the ideals in $\mathcal{B}_1$. The shared standard monomials show that J_1 is nonzero, and [Corollary 2.1.3](#) shows that this intersection is a monomial ideal.

We will now build a strictly ascending chain of monomial ideals, which will be a contradiction by [Proposition 2.2.7](#). Suppose $\mathcal{B}_k$ and J_k have been chosen. Choose an ideal $I_{k+1} \in \mathcal{B}_k$. We can again find an infinite collection of ideals in $\mathcal{B}_k \setminus \{I_{k+1}\}$ which have the same nonempty intersection with the standard monomials of I_{k+1} . Let $\mathcal{B}_{k+1}$ be this collection, and let J_{k+1} be the intersection of the ideals in $\mathcal{B}_{k+1}$. Since $J_k = \bigcap_{I \in \mathcal{B}_k} I$, $J_{k+1} = \bigcap_{I \in \mathcal{B}_{k+1}} I$ and $\mathcal{B}_k \supset \mathcal{B}_{k+1}$, we clearly have the inclusion $J_{k+1} \supseteq J_k$. The shared set contains a monomial which belongs to J_{k+1} but not to I_{k+1} . Since $J_k \subseteq I_{k+1}$, this monomial does not belong to J_k . Therefore $J_{k+1} \supsetneq J_k$. Therefore, with this procedure we get an infinite strictly ascending chain of monomial ideals in S , which is impossible by [Proposition 2.2.7](#). Thus the proof is complete. $\square$

Let $I \subset S$ be a proper monomial ideal. By [Theorem 2.4.1](#), we have a unique irredundant decomposition

$$I = \bigcap_{i=1}^m Q_i$$

where each $Q_i \subset S$ is an ideal generated by pure powers. From [Remark 2.4.7](#), we know that each Q_i is primary with respect to an ideal generated by variables. For each subset $\mathfrak{J} = \{j_1, \dots, j_k\} \subseteq [n] = \{1, \dots, n\}$, let $\sigma_{\mathfrak{J}}(I)$ be the intersection of the Q_i 's that are primary with respect to $P_{\mathfrak{J}} = (x_{j_1}, \dots, x_{j_k}) \subset S$. By convention, we have $\sigma_{\mathfrak{J}}(I) = S$ if no Q_i is $P_{\mathfrak{J}}$ -primary. When $\sigma_{\mathfrak{J}}(I)$ is a proper ideal, it is $P_{\mathfrak{J}}$ -primary by [Exercise 2.8](#). After omitting the factors equal to S , we obtain the primary decomposition

$$I = \bigcap_{\substack{\mathfrak{J} \subseteq [n] \\ \sigma_{\mathfrak{J}}(I) \neq S}} \sigma_{\mathfrak{J}}(I).$$

Equivalently, one may intersect over all $\mathfrak{J} \subseteq [n]$, since the unit factors do not change the intersection. We call the decomposition by the proper factors the *standard primary decomposition* of I . Notice that the monomial generators of $\sigma_{\mathfrak{J}}(I)$ only involve the variables in the set $\{x_j \mid j \in \mathfrak{J}\}$ and that when we regard $\sigma_{\mathfrak{J}}(I)$ as an ideal in the polynomial subring $\mathbb{k}[x_j \mid j \in \mathfrak{J}]$ it becomes Artinian. For $I = S$, we set $\sigma_{\mathfrak{J}}(S) = S$ for every $\mathfrak{J} \subseteq [n]$; then the same intersection formula and the Artinian assertion remain valid.

PROOF OF THEOREM 2.6.1. By Lemma 2.6.3, we may assume that $\mathcal{A}$ is an infinite collection of monomial ideals in S , and we need to show the existence of an infinite chain $I_1 \supsetneq I_2 \supsetneq \cdots$ of ideals in $\mathcal{A}$.

For each $\mathfrak{J} \subseteq [n]$, we consider the following collection

$$\sigma_{\mathfrak{J}}(\mathcal{A}) := \{\sigma_{\mathfrak{J}}(I) \mid I \in \mathcal{A}\}$$

of Artinian monomial ideals in $\mathbb{k}[x_j \mid j \in \mathfrak{J}]$.

Let $\mathfrak{J} \subseteq [n]$ be any subset. Then we have the following two cases:

- (i) If $\sigma_{\mathfrak{J}}(\mathcal{A})$ is a finite set, then there is an infinite collection of ideals $\mathcal{A}' \subset \mathcal{A}$ such that $\sigma_{\mathfrak{J}}(I)$ is the same for all $I \in \mathcal{A}'$.
- (ii) If $\sigma_{\mathfrak{J}}(\mathcal{A})$ is an infinite set, then Lemma 2.6.4, the ascending chain condition, and the argument in the first half of Lemma 2.6.3 give an infinite family $\{I_k\}_{k \geq 1} \subset \mathcal{A}$ such that $\sigma_{\mathfrak{J}}(I_1) \supsetneq \sigma_{\mathfrak{J}}(I_2) \supsetneq \cdots$.

In either case, we obtain an infinite family $\{I_k\}_{k \geq 1} \subset \mathcal{A}$ such that $\sigma_{\mathfrak{J}}(I_1) \supsetneq \sigma_{\mathfrak{J}}(I_2) \supsetneq \cdots$ for the fixed subset $\mathfrak{J}$, although the inclusions need not be proper.

By running the above procedure over each $\mathfrak{J} \subseteq [n]$ and restricting at each step, we obtain an infinite family $\{I_k\}_{k \geq 1} \subset \mathcal{A}$ such that $\sigma_{\mathfrak{J}}(I_1) \supsetneq \sigma_{\mathfrak{J}}(I_2) \supsetneq \cdots$ for all $\mathfrak{J} \subseteq [n]$. Since $I_k = \bigcap_{\mathfrak{J} \subseteq [n]} \sigma_{\mathfrak{J}}(I_k)$, we have an infinite sequence $I_1 \supsetneq I_2 \supsetneq \cdots$ in $\mathcal{A}$. The ideals were chosen to be distinct, so every inclusion is proper and $I_1 \supsetneq I_2 \supsetneq \cdots$ is the required chain in $\mathcal{A}$. So the proof is complete. $\square$

An *antichain* is a collection of pairwise incomparable elements. The next immediate consequence is a useful form of the antichain property. For a collection $\mathcal{A}$, we denote by $\mathcal{A}^{\min}$ its collection of minimal elements.

COROLLARY 2.6.5. *Every antichain of monomial ideals in S is finite. As a consequence, for any collection $\mathcal{A}$ of monomial ideals, both $\mathcal{A}^{\min}$ and $\mathcal{A}^{\max}$ are finite.*

PROOF. The elements of an antichain are all maximal in that antichain, so its finiteness follows from Theorem 2.6.1. The maximal elements of $\mathcal{A}$ form an antichain, as do its minimal elements. $\square$

The theorem of MacLagan has some surprising consequences.

COROLLARY 2.6.6. *There are only finitely many monomial ideals in S with a given Hilbert series.*

PROOF. Let $H(t) = \sum_{k=0}^{\infty} a_k t^k \in \mathbb{N}[[t]]$ be a power series. Consider the collection

$$\mathcal{A} := \{I \subset S \text{ monomial ideal} \mid \text{Hilb}_{S/I}(t) = H(t)\}$$

of monomial ideals with Hilbert series equal to $H(t)$. Notice that, for any two monomial ideals $I \subseteq J$ in $\mathcal{A}$, we must have $I = J$ (indeed, for all $k \geq 0$, we have $I_k \subseteq J_k$ and $\dim_{\mathbb{k}}(I_k) = \binom{k+n-1}{n-1} - a_k = \dim_{\mathbb{k}}(J_k)$). Thus every ideal of $\mathcal{A}$ is maximal. Finally, by [Theorem 2.6.1](#), we obtain that $\mathcal{A} = \mathcal{A}^{\max}$ is a finite set. $\square$

REMARK 2.6.7. The finiteness result of [Corollary 2.6.6](#) is quite potent. It tells us that many challenging problems can be reduced to studying finitely many monomial ideals.

2.7. Exercises

EXERCISE 2.1. Let I, J and K be monomial ideals in S . Show that $I + (J \cap K) = (I + J) \cap (I + K)$.

EXERCISE 2.2. Let I, J and K be monomial ideals in S . Show that $I \cap (J + K) = (I \cap J) + (I \cap K)$.

EXERCISE 2.3. Show that a monomial ideal $I \subset S$ is a prime ideal if and only if I is generated by a subset of the variables.

EXERCISE 2.4. Let $I, J \subset S$ be monomial ideals. Show that $I : J^{\infty}$ is also a monomial ideal.

EXERCISE 2.5. Let $I \subset S$ be a monomial ideal. Show that $\sqrt{I}$ is also a monomial ideal.

EXERCISE 2.6. Let $f_1, \dots, f_m \in S$ be polynomials such that $(f_1, \dots, f_m) \neq S$. Show that $f_1, \dots, f_m$ form a regular sequence if and only if $f_1 \neq 0$ and $(f_1, \dots, f_{i-1}) : f_i = (f_1, \dots, f_{i-1})$ for $2 \leq i \leq m$.

EXERCISE 2.7. Let $I \subset S$ be a proper monomial ideal. Show that the following conditions are equivalent:

- (a) I is Artinian.
- (b) I is $\mathfrak{m}$ -primary where $\mathfrak{m} = (x_1, \dots, x_n)$ is the ideal generated by all variables.
- (c) For each $1 \leq i \leq n$, there is some $\alpha_i \geq 1$ such that $x_i^{\alpha_i} \in I$.

EXERCISE 2.8. Let $P \subset S$ be a prime ideal and I, J be two P -primary ideals. Show that $I \cap J$ is a P -primary ideal.

EXERCISE 2.9. Let $I \subset S = \mathbb{k}[x_1, \dots, x_n]$ and $J \subset T = \mathbb{k}[y_1, \dots, y_m]$ be proper monomial ideals, and set

$$U = \mathbb{k}[x_1, \dots, x_n, y_1, \dots, y_m], \quad K = IU + JU.$$

Show that

$$\dim(U/K) = \dim(S/I) + \dim(T/J) \quad \text{and} \quad \deg_{\text{aff}}(U/K) = \deg_{\text{aff}}(S/I) \deg_{\text{aff}}(T/J).$$

CHAPTER 3

Gröbner bases

Throughout this chapter, $S = \mathbb{k}[x_1, \dots, x_n]$ is a polynomial ring over a field $\mathbb{k}$. The main difficulty in working with an ideal $I \subset S$ is that an element of I can have many different presentations in terms of a chosen set of generators. Gröbner bases remedy this difficulty by replacing I with a monomial ideal which records the initial terms of all the elements of I . Once a Gröbner basis has been found, ideal membership, canonical representatives in S/I , and many other questions become explicit computations with monomials.

The purpose of this chapter is to develop this theory from the beginning. We shall prove the existence of finite Gröbner bases, Hilbert basis theorem, the multivariate division algorithm, Buchberger's criterion and algorithm, and the existence and uniqueness of reduced Gröbner bases. No material from an external reference is required for the results proved below.

3.1. Monomial orders

Let X be a set. A *partial order* on X is a binary relation $\leq$ over X which is *reflexive*, *antisymmetric* and *transitive*. That is, for all $a, b, c \in X$ we have

- $a \leq a$ (reflexivity);
- if $a \leq b$ and $b \leq a$, then $a = b$ (antisymmetry);
- if $a \leq b$ and $b \leq c$, then $a \leq c$ (transitivity).

It is common to write $a < b$ if $a \leq b$ and $a \neq b$. We also write $a \geq b$ ($a > b$), if $b \leq a$ ($b < a$). A typical example of a partially ordered set is the set of all subsets of a given set ordered by inclusion.

A partial order $\leq$ on X is called a *total order* if, for any two elements $a, b \in X$, one has $a \leq b$ or $b \leq a$.

We now define a total order on $\text{Mon}(S)$, the set of all monomials in S , which respects the multiplicative structure on this set.

DEFINITION 3.1.1. A *monomial order* on S is a total order $\leq$ on $\text{Mon}(S)$ with the properties:

- (i) $1 \leq u$ for all $u \in \text{Mon}(S)$.
- (ii) If $u, v \in \text{Mon}(S)$ and $u \leq v$, then $uw \leq vw$ for all $w \in \text{Mon}(S)$.

A monomial order satisfies the following two conditions.

LEMMA 3.1.2. Let $<$ be a monomial order on S . Then the following holds:

- (i) If $u, v \in \text{Mon}(S)$ with $u \mid v$, then $u \leq v$.
- (ii) (Artinian order) If $u_1, u_2, \dots$ is a sequence of monomials with $u_1 \geq u_2 \geq \dots$, then there exists an integer m such that $u_i = u_m$ for all $i \geq m$.

PROOF. (i) If $u \mid v$, then there exists a monomial w such that $v = uw$. Since $1 \leq w$, it follows that $u \leq wu = v$.

(ii) Let $\mathcal{M} = \{u_1, u_2, \dots\}$. By Dickson's lemma (Theorem 2.2.1), the set $\mathcal{M}$ has only finitely many minimal elements with respect to divisibility. Choose the first occurrence of each of them in the sequence and denote these occurrences by $u_{i_1}, \dots, u_{i_r}$, where $i_1 < \dots < i_r$. For $j \geq i_r$, the monomial u_j is divisible by some u_{i_k} . Indeed, among the elements of $\mathcal{M}$ which divide u_j , choose one of smallest total degree; it is a minimal element of $\mathcal{M}$. Part (i) and the fact that the sequence is decreasing now give

$$u_{i_k} \leq u_j \leq u_{i_r} \leq u_{i_k}.$$

Thus $u_j = u_{i_r}$ for every $j \geq i_r$, as desired. $\square$

The fact that a monomial order induces a well-ordering on $\text{Mon}(S)$ is assumed as a condition in the definition of monomial orders given in [3, Definition 1, page 55]. This assumption is *not* necessary as it follows from Dickson's lemma.

PROPOSITION 3.1.3. *Let $<$ be a monomial order. Then $<$ is a well-ordering on $\text{Mon}(S)$. This means that every nonempty subset of $\text{Mon}(S)$ has a smallest element under $<$. In other words, if $\mathcal{M} \subset \text{Mon}(S)$ is nonempty, then there is $w \in \mathcal{M}$ such that $v > w$ for every $v \neq w$ in $\mathcal{M}$.*

PROOF. Let $\mathcal{M} \subset \text{Mon}(S)$ be a nonempty subset of monomials. By Dickson's lemma (Theorem 2.2.1), we have $\mathcal{M}^{\min} = \{u_1, \dots, u_r\}$. Every $u \in \mathcal{M}$ is divisible by an element of $\mathcal{M}^{\min}$, and so Lemma 3.1.2 yields $u \geq u_i$ for some $1 \leq i \leq r$. Therefore, by taking u_0 to be the smallest element among $\{u_1, \dots, u_r\}$ with respect to the monomial order $<$, it follows that u_0 is the smallest element of $\mathcal{M}$. $\square$

Thus there is no infinite strictly decreasing sequence of monomials. This fact will guarantee the termination of every reduction procedure used later in this chapter.

3.1.1. The standard monomial orders. For $\mathbf{a} = (a_1, \dots, a_n) \in \mathbb{N}^n$, we continue to use the notation

$$\mathbf{x}^{\mathbf{a}} = x_1^{a_1} \cdots x_n^{a_n} \quad \text{and} \quad |\mathbf{a}| = a_1 + \cdots + a_n.$$

We shall always list the variables in the order $x_1 > x_2 > \cdots > x_n$. Changing the order of the variables changes each of the orders below.

DEFINITION 3.1.4 (Lexicographic order). Let $\mathbf{x}^{\mathbf{a}}, \mathbf{x}^{\mathbf{b}} \in \text{Mon}(S)$. We say that $\mathbf{x}^{\mathbf{a}} >_{\text{lex}} \mathbf{x}^{\mathbf{b}}$ if the first nonzero entry of

$$\mathbf{a} - \mathbf{b} = (a_1 - b_1, \dots, a_n - b_n)$$

is positive.

In lexicographic order, the exponent of x_1 is compared first; in case of equality, the exponent of x_2 is compared, and so on. For instance, in $\mathbb{k}[x, y, z]$ with $x > y > z$, one has

$$x^2y >_{\text{lex}} x^2z^{100} >_{\text{lex}} xy^{1000}.$$

In particular, $x >_{\text{lex}} y^d$ for every $d \geq 1$. Thus lexicographic order does not compare monomials by total degree.

DEFINITION 3.1.5 (Graded lexicographic order). We say that $\mathbf{x}^{\mathbf{a}} >_{\text{grlex}} \mathbf{x}^{\mathbf{b}}$ if either

$$|\mathbf{a}| > |\mathbf{b}|,$$

or $|\mathbf{a}| = |\mathbf{b}|$ and $\mathbf{x}^{\mathbf{a}} >_{\text{lex}} \mathbf{x}^{\mathbf{b}}$.

DEFINITION 3.1.6 (Graded reverse lexicographic order). We say that $\mathbf{x}^{\mathbf{a}} >_{\text{grevlex}} \mathbf{x}^{\mathbf{b}}$ if either $|\mathbf{a}| > |\mathbf{b}|$, or $|\mathbf{a}| = |\mathbf{b}|$ and the last nonzero entry of $\mathbf{a} - \mathbf{b}$ is negative.

The word “reverse” in the last definition refers to the fact that, after total degree has been compared, one inspects the exponent vectors from the last variable to the first. It does not mean that the variables are reversed: one still has $x_1 > x_2 > \cdots > x_n$.

EXAMPLE 3.1.7. In $\mathbb{k}[x, y, z]$ with $x > y > z$, the monomials of degree two occur in decreasing order as follows:

$$\begin{array}{l|l} \text{lex and grlex} & x^2 > xy > xz > y^2 > yz > z^2 \\ \text{grevlex} & x^2 > xy > y^2 > xz > yz > z^2. \end{array}$$

The agreement between lexicographic and graded lexicographic order in this display is due to the fact that all the monomials have the same degree. On the other hand,

$$x >_{\text{lex}} y^{100}, \quad y^{100} >_{\text{grlex}} x, \quad y^{100} >_{\text{grevlex}} x.$$

PROPOSITION 3.1.8. *Lexicographic, graded lexicographic, and graded reverse lexicographic order are monomial orders.*

PROOF. Each relation is visibly a total order and has 1 as its smallest monomial. Suppose $\mathbf{x}^{\mathbf{a}} > \mathbf{x}^{\mathbf{b}}$, and let $\mathbf{c} \in \mathbb{N}^n$. For lexicographic order, the first nonzero entry of $(\mathbf{a} + \mathbf{c}) - (\mathbf{b} + \mathbf{c})$ is the first nonzero entry of $\mathbf{a} - \mathbf{b}$. For the two graded orders, total degrees change by the same

amount $|c|$, and, when the original degrees are equal, the relevant first or last nonzero entry is again unchanged. Thus $\mathbf{x}^{\mathbf{a}+\mathbf{c}} > \mathbf{x}^{\mathbf{b}+\mathbf{c}}$ in all three cases. $\square$

There are many other useful monomial orders. The following construction both supplies examples and will be useful in the exercises.

PROPOSITION 3.1.9 (Weighted orders). *Let $\mathbf{w} = (w_1, \dots, w_n) \in \mathbb{N}^n$. Define $\mathbf{x}^{\mathbf{a}} >_{\mathbf{w}} \mathbf{x}^{\mathbf{b}}$ when either*

$$\mathbf{w} \cdot \mathbf{a} > \mathbf{w} \cdot \mathbf{b},$$

or the two weighted degrees are equal and $\mathbf{x}^{\mathbf{a}} >_{\text{lex}} \mathbf{x}^{\mathbf{b}}$. Then $>_{\mathbf{w}}$ is a monomial order.

PROOF. The relation is a total order and is compatible with multiplication. Since all the weights are nonnegative and the lexicographic tie breaker has 1 as its smallest monomial, 1 is also the smallest monomial for $>_{\mathbf{w}}$. $\square$

REMARK 3.1.10. If $w_i = 0$, weighted degree alone cannot distinguish $1, x_i, x_i^2, \dots$, but the lexicographic tie breaker in [Proposition 3.1.9](#) still produces a monomial order. Nonnegativity is essential for this construction: allowing a negative weight would put some positive power of a variable below 1 and would violate the definition.

An order such as graded lexicographic or graded reverse lexicographic order is called *degree compatible*: $\deg(u) < \deg(v)$ implies $u < v$. Lexicographic order is especially useful for elimination, whereas graded reverse lexicographic order often makes computations smaller. The mathematical results in this chapter are valid for every monomial order.

3.2. Basics of Gröbner bases

Fix a monomial order $<$ on S . Every nonzero polynomial $f \in S$ has a unique presentation

$$f = c_1 u_1 + \dots + c_r u_r,$$

where $c_i \in \mathbb{k} \setminus \{0\}$ and $u_1 > u_2 > \dots > u_r$ are distinct monomials. The first summand contains the initial data of f .

DEFINITION 3.2.1. Let $0 \neq f = c_1 u_1 + \dots + c_r u_r \in S$, where $u_1 > \dots > u_r$. We define:

(i) the *initial term* of f by

$$\text{in}_{<}(f) := c_1 u_1;$$

(ii) the *leading coefficient* of f by

$$\text{LC}_{<}(f) := c_1;$$

We also set $\text{in}_{<}(0) = 0$.

When the monomial order is clear, we omit it from the notation. As in Eisenbud's convention, the initial term includes its coefficient. For nonzero terms au and bv , with $a, b \in \mathbb{k} \setminus \{0\}$ and u, v

monomials, we compare au and bv by comparing u and v . We also write $au \mid bv$ when $u \mid v$; in that case the quotient is the term

$$\frac{bv}{au} = \frac{b}{a} \frac{v}{u}.$$

Thus nonzero scalar coefficients do not affect order comparisons or divisibility. We declare 0 to be smaller than every nonzero term.

EXAMPLE 3.2.2. Let $S = \mathbb{Q}[x, y, z]$ and let

$$f = 3x^2z - 5xy^3 + 7y^4 - 2z^{10}.$$

For lexicographic order with $x > y > z$, we have

$$\text{in}_{\text{lex}}(f) = 3x^2z, \quad \text{LC}_{\text{lex}}(f) = 3.$$

For either graded lexicographic or graded reverse lexicographic order,

$$\text{in}(f) = -2z^{10}, \quad \text{LC}(f) = -2.$$

The compatibility of a monomial order with multiplication gives the following fundamental identities.

LEMMA 3.2.3. Let $f, g \in S$ be nonzero polynomials, write $\text{in}(f) = au$ and $\text{in}(g) = bv$, and let $c \in \mathbb{k} \setminus \{0\}$. Then:

- (i) $\text{in}(cf) = c\text{in}(f)$;
- (ii) $\text{in}(fg) = \text{in}(f)\text{in}(g)$;
- (iii) if $f + g \neq 0$ and $\text{in}(f + g) = dw$, then

$$w \leq \max\{u, v\}.$$

If $u \neq v$, then the initial term with the larger monomial part is $\text{in}(f + g)$.

PROOF. Only part (ii) needs an explanation. The product $abuv$ occurs in fg . Every other monomial in the product has the form $u'v'$ with $u' \leq u$ and $v' \leq v$, with at least one strict inequality. Multiplicativity gives $u'v' < uv$. Thus $abuv$ is the initial term of fg . $\square$

3.2.1. Initial ideals and Gröbner bases.

DEFINITION 3.2.4. Let $I \subset S$ be an ideal. The *initial ideal* of I with respect to $<$ is

$$\text{in}_{<}(I) := (\text{in}_{<}(f) \mid f \in I \setminus \{0\}).$$

The initial ideal is a monomial ideal. It is important that its definition uses the initial terms of *all* polynomials in I , not only the initial terms of a chosen set of generators.

EXAMPLE 3.2.5. Let $S = \mathbb{k}[x, y]$ with lexicographic order $x > y$, and set

$$f_1 = x^2 - y, \quad f_2 = xy - 1, \quad I = (f_1, f_2).$$

The initial terms of the displayed generators are x^2 and xy . However,

$$yf_1 - xf_2 = x - y^2 \in I,$$

so $x \in \text{in}_<(I)$ even though $x \notin (x^2, xy)$. Similarly,

$$f_2 - y(x - y^2) = y^3 - 1 \in I,$$

and hence $y^3 \in \text{in}_<(I)$. We shall soon prove that

$$\text{in}_<(I) = (x, y^3).$$

This example illustrates why an arbitrary generating set need not reveal the initial ideal.

DEFINITION 3.2.6. Let $I \subset S$ be a nonzero ideal. A finite set $\mathcal{G} = \{g_1, \dots, g_s\} \subset I \setminus \{0\}$ is called a *Gröbner basis* of I with respect to $<$ if

$$\text{in}_<(I) = (\text{in}_<(g_1), \dots, \text{in}_<(g_s)).$$

We regard the empty set as the Gröbner basis of the zero ideal.

The following characterization is often the most convenient one.

PROPOSITION 3.2.7. Let $\mathcal{G} = \{g_1, \dots, g_s\} \subset I \setminus \{0\}$. The following conditions are equivalent:

- (i) $\mathcal{G}$ is a Gröbner basis of I .
- (ii) For every nonzero $f \in I$, there is an index i such that

$$\text{in}(g_i) \mid \text{in}(f).$$

PROOF. This is precisely the membership criterion for the monomial ideal

$$(\text{in}(g_1), \dots, \text{in}(g_s));$$

see [Proposition 2.1.6](#). □

A Gröbner basis is, in particular, a generating set of the original ideal. This useful fact is part of the reason for requiring the elements of a Gröbner basis to belong to I .

PROPOSITION 3.2.8. If $\mathcal{G} = \{g_1, \dots, g_s\}$ is a Gröbner basis of I , then

$$I = (g_1, \dots, g_s).$$

PROOF. The inclusion $(g_1, \dots, g_s) \subset I$ is clear. Let $0 \neq f \in I$. By [Proposition 3.2.7](#), some $\text{in}(g_i)$ divides $\text{in}(f)$. The quotient

$$q = \frac{\text{in}(f)}{\text{in}(g_i)}$$

is a term, and either $f - qg_i = 0$ or

$$\text{in}(f - qg_i) < \text{in}(f).$$

Moreover, $f - qg_i \in I$. If it is nonzero, repeat the same operation. The process cannot produce an infinite strictly decreasing sequence of initial terms by Proposition 3.1.3. It therefore ends at zero and expresses f as an S -linear combination of the elements of $\mathcal{G}$. $\square$

REMARK 3.2.9. The proof gives more than ideal generation. At each stage the initial term is cancelled, so the coefficient multiplying a basis element is explicitly determined once that basis element is chosen. The multivariate division algorithm in Section 3.6 organizes this reduction and also handles polynomials which do not belong to I .

3.2.2. Standard monomials and the vector space S/I .

DEFINITION 3.2.10. A monomial $u \in \text{Mon}(S)$ is called *standard modulo I* if

$$u \notin \text{in}_{<}(I).$$

We denote the set of standard monomials by

$$\mathcal{N}_{<}(I) := \text{Mon}(S) \setminus \text{in}_{<}(I).$$

THEOREM 3.2.11 (Macaulay's theorem). *The residue classes of the monomials in $\mathcal{N}_{<}(I)$ form a $\mathbb{k}$ -basis of S/I .*

PROOF. We first prove spanning. Let $f \in S$. If every monomial of f is standard, there is nothing to show. Otherwise, let u be the largest monomial of f which belongs to $\text{in}(I)$, and let cu be its term in f . By the definition of the initial ideal and the membership criterion for monomial ideals, there is a nonzero $h \in I$ such that $\text{in}(h) \mid u$. The quotient

$$q = \frac{cu}{\text{in}(h)}$$

is a term, and subtracting qh cancels the term cu of f . All the new monomials are smaller than u . Repeating this operation must terminate because $<$ is a well-order. We obtain a polynomial r , all of whose monomials are standard, such that $f - r \in I$. Thus the residue classes of the standard monomials span S/I .

For linear independence, suppose that a nonzero $\mathbb{k}$ -linear combination r of distinct standard monomials belongs to I . Then $\text{in}(r) \in \text{in}(I)$ by definition, but the monomial underlying $\text{in}(r)$ is one of the standard monomials occurring in r . This is a contradiction. $\square$

The representative obtained in the spanning argument is unique. Indeed, if r and r' are linear combinations of standard monomials and $r - r' \in I$, then the linear independence just proved gives $r = r'$. Later this representative will be denoted by $\text{NF}_{\mathcal{G}}(f)$ and computed by division by a Gröbner basis.

COROLLARY 3.2.12. *The following conditions are equivalent:*

- (i) S/I is a finite-dimensional $\mathbb{k}$ -vector space.

(ii) The set $\mathcal{N}_{<}(I)$ is finite.

(iii) For every $1 \leq i \leq n$, some power of x_i belongs to $\text{in}_{<}(I)$.

When these conditions hold,

$$\dim_{\mathbb{k}}(S/I) = |\mathcal{N}_{<}(I)|.$$

PROOF. The equivalence of (i) and (ii), as well as the displayed equality, follows from [Theorem 3.2.11](#). If $x_i^{a_i} \in \text{in}(I)$ for all i , then every standard monomial divides

$$x_1^{a_1-1} \cdots x_n^{a_n-1},$$

so there are only finitely many standard monomials. Conversely, if no power of some x_i belongs to $\text{in}(I)$, then $1, x_i, x_i^2, \dots$ are distinct standard monomials. $\square$

EXAMPLE 3.2.13. Return to

$$I = (x^2 - y, xy - 1) \subset \mathbb{k}[x, y]$$

with lexicographic order $x > y$. Once we verify that $\text{in}(I) = (x, y^3)$, the standard monomials are

$$1, y, y^2.$$

Consequently, S/I has dimension three over $\mathbb{k}$. The relations $x = y^2$ and $y^3 = 1$ show directly that every residue class is a unique $\mathbb{k}$ -linear combination of these three monomials.

PROPOSITION 3.2.14 (Finiteness of initial ideals). *For a fixed ideal $I \subset S$, there are only finitely many ideals of the form $\text{in}_{<}(I)$ as $<$ ranges over all monomial orders on S .*

PROOF. Suppose, to the contrary, that the collection of initial ideals of I is infinite. By [Lemma 2.6.3](#), it contains an infinite chain

$$J_1 \supsetneq J_2 \supsetneq J_3 \supsetneq \cdots,$$

where $J_i = \text{in}_{<_i}(I)$ for suitable monomial orders $<_i$. Taking complements in the set of all monomials gives strict inclusions

$$\mathcal{N}_{<_1}(I) \subsetneq \mathcal{N}_{<_2}(I) \subsetneq \cdots.$$

For every i , however, the residue classes of the monomials in $\mathcal{N}_{<_i}(I)$ form a $\mathbb{k}$ -basis of the same vector space S/I by [Theorem 3.2.11](#). This is impossible: a proper subset of a basis cannot span the vector space. Thus the collection of initial ideals is finite. $\square$

We shall also need a precise formulation of homogeneity.

DEFINITION 3.2.15. Write every polynomial uniquely as $f = \sum_{d \geq 0} f_d$, where f_d is homogeneous of degree d . An ideal $I \subset S$ is called *homogeneous* if it is generated by homogeneous polynomials.

PROPOSITION 3.2.16. *For an ideal $I \subset S$, the following conditions are equivalent:*

- (i) I is homogeneous;
- (ii) if $f = \sum_d f_d \in I$, then $f_d \in I$ for every d ;
- (iii) $I = \bigoplus_{d \geq 0} (I \cap S_d)$ as a $\mathbb{k}$ -vector space.

PROOF. Suppose that $I = (g_\lambda \mid \lambda \in \Lambda)$ with each g_λ homogeneous. Every particular element $f \in I$ is an S -linear combination of finitely many of the g_λ . It follows that every homogeneous component of f is a sum of products of homogeneous components of the coefficients with the g_λ , and hence belongs to I . Thus (i) implies (ii). Conditions (ii) and (iii) are equivalent by the uniqueness of the homogeneous decomposition. Finally, if (ii) holds, the homogeneous components of the elements of I are homogeneous elements of I and generate I . $\square$

There is also a graded consequence which will be useful whenever I is homogeneous.

COROLLARY 3.2.17. *Suppose that S has the standard grading and $I \subset S$ is a homogeneous ideal. Then $\text{in}_<(I)$ is homogeneous and*

$$\dim_{\mathbb{k}}(S/I)_m = \dim_{\mathbb{k}}(S/\text{in}_<(I))_m$$

for every $m \geq 0$. In particular, S/I and $S/\text{in}_<(I)$ have the same Hilbert function and the same Hilbert series.

PROOF. The initial term of a homogeneous polynomial has the same degree as the polynomial. Since a homogeneous ideal is generated, as a vector space, by its homogeneous elements, its initial ideal is generated by homogeneous monomials. In degree m , both quotients have as a $\mathbb{k}$ -basis the standard monomials of degree m . $\square$

3.3. Hilbert functions of homogeneous quotients

Let $I \subsetneq S = \mathbb{k}[x_1, \dots, x_n]$ be a homogeneous ideal and set $R = S/I$. The quotient R is a graded ring by Proposition 3.2.16. Its Hilbert function and Hilbert series are

$$H_R(m) := \dim_{\mathbb{k}}(R_m), \quad \text{Hilb}_R(t) := \sum_{m \geq 0} H_R(m)t^m.$$

If $J = \text{in}_<(I)$ for any monomial order, then J is homogeneous and Corollary 3.2.17 gives

$$\text{Hilb}_R(t) = \text{Hilb}_{S/J}(t).$$

Thus the monomial theory of Chapter 2 gives the following theorem.

THEOREM 3.3.1 (Hilbert theorem for homogeneous quotients). *There is a unique expression*

$$\text{Hilb}_R(t) = \frac{Q(t)}{(1-t)^d},$$

where $0 \leq d \leq n$, $Q(t) \in \mathbb{Z}[t]$, and $Q(1) > 0$. If $d \geq 1$, there is a unique polynomial $P_R(z) \in \mathbb{Q}[z]$ of degree $d - 1$ such that

$$P_R(m) = \dim_{\mathbb{k}} R_m$$

for all $m \gg 0$, and

$$P_R(z) = Q(1) \frac{z^{d-1}}{(d-1)!} + \text{lower degree terms.}$$

If $d = 0$, then R is finite dimensional over $\mathbb{k}$, its Hilbert function is eventually zero, and its Hilbert polynomial is the zero polynomial; in this case

$$Q(1) = \dim_{\mathbb{k}}(R).$$

PROOF. Let $J = \text{in}_{<}(I)$. By [Corollary 3.2.17](#), the quotients $R = S/I$ and S/J have the same Hilbert function and Hilbert series. Applying [Theorem 2.5.4](#) to the monomial ideal J gives the rational expression, the eventual polynomial, its leading coefficient, and the assertion when $d = 0$. Since $Q(1) \neq 0$, the denominator is reduced at $t = 1$; hence d is the exponent of $1 - t$ in the reduced denominator and is unique, after which $Q(t)$ is also unique. $\square$

DEFINITION 3.3.2. With the notation of [Theorem 3.3.1](#), the *dimension* and *degree* of the homogeneous quotient $R = S/I$ are

$$\dim(R) := d, \quad \deg(R) := Q(1).$$

Thus, when $d \geq 1$, the degree is the normalized leading coefficient of $P_R(z)$; when $d = 0$, it is $\dim_{\mathbb{k}}(R)$.

The condition $Q(1) \neq 0$ means that the denominator is reduced at $t = 1$; in fact, the theorem gives $Q(1) > 0$. Without it one could multiply numerator and denominator by the same power of $1 - t$, obscuring both dimension and degree.

EXAMPLE 3.3.3. For $I = (0)$, one has

$$\text{Hilb}_S(t) = \frac{1}{(1-t)^n}.$$

Thus $\dim(S) = n$ and $\deg(S) = 1$.

PROPOSITION 3.3.4 (Hilbert series of a homogeneous hypersurface quotient). *Let $0 \neq f \in S$ be homogeneous of positive degree e . Then*

$$\text{Hilb}_{S/(f)}(t) = \frac{1-t^e}{(1-t)^n} = \frac{1+t+\dots+t^{e-1}}{(1-t)^{n-1}}.$$

Consequently,

$$\dim(S/(f)) = n - 1, \quad \deg(S/(f)) = e.$$

PROOF. Since S is a domain and $f \neq 0$, multiplication by f gives an exact sequence of graded modules

$$0 \longrightarrow S(-e) \xrightarrow{f} S \longrightarrow S/(f) \longrightarrow 0.$$

Additivity of Hilbert series and the shift formula give

$$\text{Hilb}_{S/(f)}(t) = \text{Hilb}_S(t) - t^e \text{Hilb}_S(t).$$

The result follows from [Theorem 3.3.1](#). □

REMARK 3.3.5. The Hilbert function is more refined than dimension and degree. The latter record only the order of the pole at $t = 1$ and the value of the reduced numerator at 1. Different homogeneous quotient rings can have the same dimension and degree while having different Hilbert functions in small degrees.

The homogeneity of I is essential here: for a nonhomogeneous ideal, S/I does not inherit the standard grading. To treat an arbitrary ideal, we use the total-degree filtration instead.

3.4. Affine Hilbert functions, dimension, and degree

The homogeneous hypothesis can be replaced by a degree filtration, provided that the monomial order respects total degree.

THEOREM 3.4.1 (Degree-bounded Macaulay equality). *Let $I \subset S$ be any ideal, let $<$ be a degree-compatible monomial order, and set*

$$S_{\leq m} = \bigoplus_{j=0}^m S_j \quad \text{and} \quad F_m(S/I) = \frac{S_{\leq m} + I}{I}.$$

Then the residue classes of the standard monomials of degree at most m form a $\mathbb{k}$ -basis of $F_m(S/I)$. Consequently,

$$\dim_{\mathbb{k}} F_m(S/I) = \dim_{\mathbb{k}} (S/\text{in}_{<}(I))_{\leq m}$$

for every $m \geq 0$.

PROOF. Let $f \in S_{\leq m}$. If f has a nonstandard monomial, let u be the largest such monomial. By the definition of the initial ideal, there is a nonzero $h \in I$ such that $\text{in}(h)$ divides u . Multiply h by a suitable term so as to cancel the term of f supported at u . Every monomial introduced is smaller than u . Moreover, none has degree greater than $\deg(u)$, because $<$ is degree-compatible. Repeating this operation terminates by the well-ordering property in [Proposition 3.1.3](#) and produces a polynomial of degree at most m whose monomials are all standard and whose difference from f belongs to I . Thus the standard monomials of degree at most m span $F_m(S/I)$. They are linearly independent by [Theorem 3.2.11](#). The same monomials form the evident basis of $(S/\text{in}_{<}(I))_{\leq m}$, which proves the equality. □

Let $I \subsetneq S = \mathbb{k}[x_1, \dots, x_n]$, set $R = S/I$, and give R the total-degree filtration

$$F_m R = \frac{S_{\leq m} + I}{I} \quad (m \geq 0).$$

The *affine Hilbert function* and *affine Hilbert series* of this quotient are

$$H_R^{\text{aff}}(m) := \dim_{\mathbb{k}} F_m R, \quad \text{Hilb}_R^{\text{aff}}(t) := \sum_{m \geq 0} H_R^{\text{aff}}(m) t^m.$$

THEOREM 3.4.2 (Affine Hilbert theorem). *There is a unique integer d with $0 \leq d \leq n$ and a unique polynomial $Q(t) \in \mathbb{Z}[t]$ such that*

$$\text{Hilb}_R^{\text{aff}}(t) = \frac{Q(t)}{(1-t)^{d+1}} \quad \text{and} \quad Q(1) > 0.$$

Moreover, there is a unique polynomial $P_R^{\text{aff}}(z) \in \mathbb{Q}[z]$ of degree d such that

$$P_R^{\text{aff}}(m) = H_R^{\text{aff}}(m)$$

for all $m \gg 0$. If $e = Q(1)$, then

$$P_R^{\text{aff}}(z) = e \frac{z^d}{d!} + \text{lower degree terms}.$$

PROOF. Choose a degree-compatible monomial order and put $J = \text{in}_{<}(I)$. By the degree-bounded Macaulay equality in [Theorem 3.4.1](#),

$$H_R^{\text{aff}}(m) = \dim_{\mathbb{k}}(S/J)_{\leq m} = \sum_{j=0}^m \dim_{\mathbb{k}}(S/J)_j.$$

Therefore, as formal power series,

$$\begin{aligned} \text{Hilb}_R^{\text{aff}}(t) &= \sum_{m \geq 0} \sum_{j=0}^m \dim_{\mathbb{k}}(S/J)_j t^m \\ &= \frac{\text{Hilb}_{S/J}(t)}{1-t}. \end{aligned}$$

Applying [Theorem 3.3.1](#) to the homogeneous monomial quotient S/J gives

$$\text{Hilb}_{S/J}(t) = \frac{Q(t)}{(1-t)^d}, \quad Q(1) > 0,$$

for some $0 \leq d \leq n$ and $Q(t) \in \mathbb{Z}[t]$. This proves the asserted rational form. Since $Q(1) \neq 0$, the fraction is reduced at $t = 1$, so d and $Q(t)$ are unique. Finally, [Lemma 2.5.16](#), applied with denominator exponent $d + 1$, gives the eventual polynomial and its leading term. Its uniqueness follows because two polynomials which agree at all sufficiently large integers are equal. $\square$

DEFINITION 3.4.3. For an arbitrary quotient $R = S/I$, with the notation of Theorem 3.4.2, the *dimension* and *affine degree* are

$$\dim(R) := d, \quad \deg_{\text{aff}}(R) := e = Q(1).$$

Thus $\dim(R)$ is the degree of $P_R^{\text{aff}}(z)$, while $\deg_{\text{aff}}(R)$ is $d!$ times its leading coefficient. Equivalently, for every degree-compatible monomial order, these agree with the dimension and degree defined in Section 3.3 for the homogeneous monomial quotient $S/\text{in}_{<}(I)$.

REMARK 3.4.4 (Compatibility for homogeneous quotients). If I is homogeneous, then

$$F_m R = \bigoplus_{j=0}^m R_j,$$

and consequently

$$\text{Hilb}_R^{\text{aff}}(t) = \frac{\text{Hilb}_R(t)}{1-t}.$$

Thus the affine definition extends the homogeneous one: the two dimensions agree, and

$$\deg_{\text{aff}}(R) = \deg(R).$$

REMARK 3.4.5. If $\dim(R) = 0$, then $P_R^{\text{aff}}(z)$ is the nonzero constant polynomial $\dim_{\mathbb{k}}(R)$ and

$$\deg_{\text{aff}}(R) = \dim_{\mathbb{k}}(R).$$

For the zero ring, the affine Hilbert function and polynomial are both zero; we use the conventions

$$\dim(0) = -\infty, \quad \deg_{\text{aff}}(0) = 0.$$

REMARK 3.4.6. Over an arbitrary field, these are invariants of the filtered quotient ring $R = S/I$. No assertion about the set of $\mathbb{k}$ -rational zeros of I is intended. For example,

$$R = \mathbb{R}[x]/(x^2 + 1)$$

has dimension zero and affine degree two, although its set of real zeros $V(x^2 + 1) \subset \mathbb{A}_{\mathbb{R}}^1$ is empty. The geometric interpretation over an algebraically closed field will follow from Hilbert's Nullstellensatz in Chapter 4.

3.5. Hilbert basis theorem

We first prove that every ideal in S has a finite Gröbner basis. The argument uses only the finiteness theorem for monomial ideals from the previous chapter.

THEOREM 3.5.1 (Existence of finite Gröbner bases). *Let $I \subset S$ be an ideal and fix a monomial order $<$. Then I has a finite Gröbner basis with respect to $<$.*

PROOF. The assertion is clear for $I = (0)$, so assume $I \neq (0)$. The initial ideal $\text{in}_{<}(I)$ is a monomial ideal. By Hilbert basis theorem for monomial ideals, it has a finite set of monomial

generators, say

$$\text{in}_{<}(I) = (u_1, \dots, u_s).$$

We may take $u_1, \dots, u_s$ to be the minimal monomial generators of $\text{in}_{<}(I)$. By the definition of the initial ideal, each u_i is divisible by $\text{in}(f_i)$ for some nonzero $f_i \in I$. Since the monomial underlying $\text{in}(f_i)$ also belongs to $\text{in}_{<}(I)$ and u_i is a minimal generator, that monomial must be u_i . After rescaling f_i , we may therefore assume that

$$\text{in}(f_i) = u_i.$$

It follows that $\{f_1, \dots, f_s\}$ is a Gröbner basis of I . □

Combining [Theorem 3.5.1](#) and [Proposition 3.2.8](#) immediately gives the field case of Hilbert basis theorem.

COROLLARY 3.5.2 (Hilbert basis theorem over a field). *Every ideal of $\mathbb{k}[x_1, \dots, x_n]$ is finitely generated.*

REMARK 3.5.3. The proof of [Theorem 3.5.1](#) is an existence proof, not an algorithm: it starts with the unknown ideal $\text{in}(I)$ and chooses polynomials whose initial terms minimally generate it. Buchberger's algorithm, developed in the next section, begins with any finite generating set and constructs a Gröbner basis by detecting and resolving cancellations of initial terms.

We now give the usual general form of Hilbert basis theorem. Recall that a commutative ring R is called *Noetherian* if every ideal of R is finitely generated.

THEOREM 3.5.4 (Hilbert basis theorem). *If R is a Noetherian ring, then $R[t]$ is a Noetherian ring. Consequently, $R[t_1, \dots, t_n]$ is Noetherian for every $n \geq 1$.*

PROOF. Let $I \subset R[t]$ be an ideal. For every $d \geq 0$, define

$$A_d := \{a \in R \mid \text{there is an } f \in I \text{ with } \deg(f) \leq d \text{ and coefficient of } t^d \text{ equal to } a\}.$$

Equivalently, the nonzero elements of A_d are the leading coefficients of the elements of I of degree exactly d . It is straightforward to check that A_d is an ideal of R . Multiplication by t shows that

$$A_0 \subset A_1 \subset A_2 \subset \dots.$$

A Noetherian ring satisfies the ascending chain condition on ideals. Indeed, the union of an ascending chain is an ideal; a finite set of generators for the union is contained in one member of the chain, after which the chain is constant. Therefore there is an integer N such that

$$A_N = A_{N+1} = A_{N+2} = \dots.$$

Since R is Noetherian, for every $0 \leq d \leq N$ we may choose generators

$$A_d = (a_{d1}, \dots, a_{dr_d}).$$

For every nonzero a_{dj} , choose $f_{dj} \in I$ of degree d and leading coefficient a_{dj} ; zero generators may simply be omitted. We claim that the finite collection of all f_{dj} generates I .

Let $0 \neq f \in I$ have degree e and leading coefficient a . If $e \leq N$, write $a = \sum_j c_j a_{ej}$. Then

$$f - \sum_j c_j f_{ej}$$

has degree strictly smaller than e . If $e > N$, the equality $A_e = A_N$ gives a presentation $a = \sum_j c_j a_{Nj}$, and

$$f - \sum_j c_j t^{e-N} f_{Nj}$$

again has degree strictly smaller than e . Induction on the degree expresses f as an $R[t]$ -linear combination of the chosen polynomials. Thus I is finitely generated and $R[t]$ is Noetherian. The final assertion follows by induction on n . $\square$

COROLLARY 3.5.5. *For a commutative ring R , the following conditions are equivalent:*

- (i) *every ideal of R is finitely generated;*
- (ii) *every ascending chain of ideals of R eventually becomes constant.*

In particular, every ascending chain of ideals in S eventually becomes constant.

PROOF. We proved (i) $\Rightarrow$ (ii) inside the proof of [Theorem 3.5.4](#). Conversely, let $I \subset R$ and consider the set of ideals generated by finite subsets of I . If none of them equals I , one can successively choose

$$a_1 \in I, \quad a_2 \in I \setminus (a_1), \quad a_3 \in I \setminus (a_1, a_2), \quad \dots,$$

which gives a strictly ascending chain, contradicting (ii). $\square$

REMARK 3.5.6. For polynomial rings over a field, the Gröbner basis proof contains additional information that the general proof does not: it produces generators adapted to a chosen monomial order and identifies a monomial basis of the quotient.

3.6. Division algorithm and Buchberger algorithm

In one variable, division by a nonzero polynomial produces a unique quotient and a remainder of smaller degree. In several variables, the role of degree is played by a monomial order, and one generally divides by an ordered list of polynomials. The remainder can depend on the order of the divisors and on the choices made during division. The special feature of a Gröbner basis is that its remainder is canonical.

3.6.1. The multivariate division algorithm. Let $F = (f_1, \dots, f_s)$ be an ordered list of nonzero polynomials. We describe division of $f \in S$ by F . The symbol p denotes the part of f which remains to be processed, the polynomials q_i are the quotients, and r is the remainder.

ALGORITHM 3.6.1 (Multivariate division). Given $f \in S$ and an ordered list $F = (f_1, \dots, f_s)$ of nonzero polynomials, perform the following steps.

(1) Set

$$q_1 = \dots = q_s = 0, \quad r = 0, \quad p = f.$$

(2) While $p \neq 0$, choose the smallest index i for which

$$\text{in}(f_i) \mid \text{in}(p),$$

if such an index exists.

(a) If such an i exists, set

$$\alpha = \frac{\text{in}(p)}{\text{in}(f_i)}, \quad q_i := q_i + \alpha, \quad p := p - \alpha f_i.$$

(b) If no such i exists, set

$$r := r + \text{in}(p), \quad p := p - \text{in}(p).$$

(3) When $p = 0$, return $q_1, \dots, q_s$ and r .

The instruction to choose the smallest possible index makes the procedure deterministic for a fixed ordered list. One can instead permit any eligible divisor; all the conclusions of the next theorem remain valid.

THEOREM 3.6.2 (Division theorem). *Algorithm 3.6.1 terminates after finitely many steps and returns polynomials $q_1, \dots, q_s, r \in S$ such that*

$$f = q_1 f_1 + \dots + q_s f_s + r.$$

Moreover:

(i) *no monomial of r is divisible by any $\text{in}(f_i)$;*

(ii) *whenever $q_i \neq 0$, one has*

$$\text{in}(q_i f_i) \leq \text{in}(f).$$

(iii) *if $r \neq 0$, then $\text{in}(r) \leq \text{in}(f)$.*

PROOF. At every stage of the algorithm, the following identity is an invariant:

$$(*) \quad f = q_1 f_1 + \dots + q_s f_s + r + p.$$

It holds at the beginning. In case (a), the same polynomial αf_i is added to $q_i f_i$ and subtracted from p . In case (b), the initial term of p is moved from p to r . Thus the identity is preserved.

After either case, the new value of p is zero or has initial term strictly smaller than the preceding $\text{in}(p)$. In case (a), this follows because the initial term of p is cancelled; in case (b), it is simply removed. There is no infinite strictly decreasing sequence of monomials by

Proposition 3.1.3. Therefore the algorithm terminates. Putting $p = 0$ in $(*)$ gives the required presentation.

A term is moved to r only when its monomial is not divisible by any $\text{in}(f_i)$. Terms moved to r are never changed afterward, which proves (i). Finally, every term added to q_i satisfies

$$\text{in}(af_i) = \text{in}(p) \leq \text{in}(f).$$

The initial term of the sum $q_i f_i$ cannot exceed the largest of these terms, proving (ii). Every term moved to r is likewise at most $\text{in}(f)$, which proves (iii). $\square$

We call r a *remainder* of f on division by F and write

$$f \xrightarrow{F} r.$$

When the deterministic rule in [Algorithm 3.6.1](#) is used, we may denote it by $\text{rem}_F(f)$.

EXAMPLE 3.6.3 (Dependence on the order of the divisors). Let $S = \mathbb{k}[x, y]$ with lexicographic order $x > y$, and consider

$$f = xy^2, \quad f_1 = xy - 1, \quad f_2 = y^2 - 1.$$

Division first by f_1 gives

$$xy^2 = y(xy - 1) + y,$$

so the remainder is y . If the order of the divisors is reversed, division first by f_2 gives

$$xy^2 = x(y^2 - 1) + x,$$

so the remainder is x . Both remainders satisfy the condition in [Theorem 3.6.2](#). Thus uniqueness of the remainder fails for an arbitrary list of divisors.

DEFINITION 3.6.4. A polynomial $r \in S$ is *reduced with respect to* a set $F = \{f_1, \dots, f_s\}$ if no monomial of r is divisible by any $\text{in}(f_i)$. A single reduction step is the replacement

$$p \mapsto p - \frac{cu}{\text{in}(f_i)} f_i,$$

where cu is a term of p and $\text{in}(f_i) \mid u$.

The division algorithm always produces a reduced remainder. For arbitrary F , different sequences of reductions can lead to different reduced polynomials. For a Gröbner basis, this ambiguity disappears.

THEOREM 3.6.5 (Ideal membership and normal forms). *Let $\mathcal{G} = \{g_1, \dots, g_s\}$ be a Gröbner basis of I . Then:*

- (i) *a polynomial $f \in S$ belongs to I if and only if the remainder of f on division by $\mathcal{G}$ is zero;*
- (ii) *every sequence of reductions of f which ends in a polynomial reduced with respect to $\mathcal{G}$ gives the same result;*

(iii) there is a unique polynomial $\text{NF}_{\mathcal{G}}(f)$, called the normal form of f modulo I , such that

$$f - \text{NF}_{\mathcal{G}}(f) \in I$$

and every monomial of $\text{NF}_{\mathcal{G}}(f)$ is standard modulo I .

PROOF. Let division give

$$f = q_1 g_1 + \cdots + q_s g_s + r.$$

If $r = 0$, then $f \in I$. Conversely, suppose $f \in I$. Then $r = f - \sum_i q_i g_i \in I$. If $r \neq 0$, the Gröbner basis property implies that some $\text{in}(g_i)$ divides $\text{in}(r)$. This contradicts the defining property of a remainder. Thus $r = 0$, proving (i).

Suppose that two terminating sequences give reduced polynomials r and r' . Then $f - r \in I$ and $f - r' \in I$, so $r - r' \in I$. Every monomial occurring in $r - r'$ is standard, because both r and r' are reduced with respect to a Gröbner basis. If $r - r' \neq 0$, the monomial underlying $\text{in}(r - r')$ would simultaneously be standard and belong to $\text{in}(I)$, a contradiction. Hence $r = r'$, proving (ii) and (iii). $\square$

COROLLARY 3.6.6. The map

$$\text{NF}_{\mathcal{G}} : S \longrightarrow \text{Span}_{\mathbb{k}}(\mathcal{N}_{<}(I))$$

is $\mathbb{k}$ -linear, has kernel I , and induces the vector space isomorphism from [Theorem 3.2.11](#).

PROOF. Uniqueness of normal forms gives

$$\text{NF}_{\mathcal{G}}(af + bg) = a\text{NF}_{\mathcal{G}}(f) + b\text{NF}_{\mathcal{G}}(g)$$

for $a, b \in \mathbb{k}$. The assertion about the kernel is [Theorem 3.6.5\(i\)](#), and the remaining statement follows. $\square$

LEMMA 3.6.7. Let $I \subseteq J$ be ideals in S . If $\text{in}_{<}(I) = \text{in}_{<}(J)$, then $I = J$.

PROOF. Let $f \in J$, and let $r = \text{NF}_{\mathcal{G}}(f)$ for a Gröbner basis $\mathcal{G}$ of I . Since $f - r \in I \subseteq J$, we have $r \in J$. If $r \neq 0$, then $\text{in}(r) \in \text{in}_{<}(J) = \text{in}_{<}(I)$, so the monomial underlying $\text{in}(r)$ lies in $\text{in}_{<}(I)$. This contradicts the fact that every monomial of r is standard modulo I . Thus $r = 0$, so $f \in I$ by [Theorem 3.6.5](#). $\square$

EXAMPLE 3.6.8. In $\mathbb{k}[x, y]$ with lexicographic order $x > y$, let

$$\mathcal{G} = \{x - y^2, y^3 - 1\}.$$

We shall verify below that this is a Gröbner basis. Division gives

$$\begin{aligned} x^2 y &= xy(x - y^2) + xy^3 \\ &= xy(x - y^2) + y^3(x - y^2) + y^5 \\ &= xy(x - y^2) + y^3(x - y^2) + y^2(y^3 - 1) + y^2. \end{aligned}$$

Therefore

$$\text{NF}_G(x^2y) = y^2.$$

The computation can also be read in the quotient: the relations $x = y^2$ and $y^3 = 1$ give $x^2y = y^5 = y^2$.

3.6.2. S-polynomials and cancellation of initial terms. An arbitrary generating set fails to be a Gröbner basis precisely when initial terms can cancel and expose a new initial term. The S-polynomial is designed to produce the simplest such cancellation for a pair of polynomials.

DEFINITION 3.6.9. Let $f, g \in S$ be nonzero, and write

$$\text{in}(f) = \alpha u, \quad \text{in}(g) = \beta v,$$

where $\alpha, \beta \in \mathbb{k} \setminus \{0\}$ and u, v are monomials. Set

$$m = \text{lcm}(u, v).$$

The *S-polynomial* of f and g is

$$S(f, g) := \frac{m}{\text{in}(f)} f - \frac{m}{\text{in}(g)} g.$$

Thus $m/\text{in}(f) = (1/\alpha)(m/u)$, and similarly for g . The two summands in $S(f, g)$ both have initial term m , so those terms cancel. Consequently,

$$\text{in}(S(f, g)) < m$$

whenever $S(f, g) \neq 0$.

EXAMPLE 3.6.10. For

$$f_1 = x^2 - y, \quad f_2 = xy - 1$$

in lexicographic order $x > y$, one has

$$\text{lcm}(x^2, xy) = x^2y$$

and

$$S(f_1, f_2) = yf_1 - xf_2 = x - y^2.$$

This is exactly the polynomial which showed that $\{f_1, f_2\}$ is not a Gröbner basis.

We shall use the following terminology in the proof of Buchberger's criterion.

DEFINITION 3.6.11. Let $F = \{f_1, \dots, f_s\} \subset S \setminus \{0\}$. A presentation

$$h = h_1 f_1 + \dots + h_s f_s$$

is called a *standard representation* of h with respect to F if, for every i with $h_i \neq 0$,

$$\text{in}(h_i f_i) \leq \text{in}(h).$$

The empty sum is regarded as a standard representation of the zero polynomial.

By [Theorem 3.6.2](#), if a polynomial has remainder zero on division by F , then the presentation produced by division is a standard representation.

The next lemma explains why pairwise S -polynomials account for every possible cancellation among several initial terms.

LEMMA 3.6.12 (Cancellation lemma). *Let $g_1, \dots, g_t \in S \setminus \{0\}$, write*

$$\text{in}(g_i) = \alpha_i u_i$$

with $\alpha_i \in \mathbb{k} \setminus \{0\}$ and u_i a monomial, and let $c_i x^{\alpha_i}$ be nonzero terms such that

$$x^{\alpha_i} u_i = M$$

for every i . Suppose that the initial terms in

$$\sum_{i=1}^t c_i x^{\alpha_i} g_i$$

cancel. Then this sum is an S -linear combination of the S -polynomials $\{S(g_1, g_i) \mid 2 \leq i \leq t\}$, and every nonzero summand in this combination has initial term strictly smaller than M .

PROOF. For $i \geq 2$, put $L_i = \text{lcm}(u_1, u_i)$. The equality $x^{\alpha_i} u_i = M$ shows that $L_i \mid M$. The coefficient of M in the displayed sum is $\sum_i c_i \alpha_i$, and the cancellation hypothesis says

$$\sum_{i=1}^t c_i \alpha_i = 0.$$

For $i \geq 2$, the definition of an S -polynomial gives

$$\frac{M}{L_i} S(g_1, g_i) = \frac{M}{\alpha_1 u_1} g_1 - \frac{M}{\alpha_i u_i} g_i = \frac{x^{\alpha_1}}{\alpha_1} g_1 - \frac{x^{\alpha_i}}{\alpha_i} g_i.$$

It follows that

$$\begin{aligned} & - \sum_{i=2}^t c_i \alpha_i \frac{M}{L_i} S(g_1, g_i) \\ &= \sum_{i=2}^t c_i x^{\alpha_i} g_i - \frac{\sum_{i=2}^t c_i \alpha_i}{\alpha_1} x^{\alpha_1} g_1 \\ &= \sum_{i=1}^t c_i x^{\alpha_i} g_i. \end{aligned}$$

Finally,

$$\text{in} \left(\frac{M}{L_i} S(g_1, g_i) \right) < M,$$

because the two initial terms having monomial L_i cancel in $S(g_1, g_i)$. □

3.6.3. Buchberger's criterion.

THEOREM 3.6.13 (Buchberger's criterion). *Let $\mathcal{G} = \{g_1, \dots, g_s\} \subset S \setminus \{0\}$ be a finite generating set of a nonzero ideal $I \subset S$. The following conditions are equivalent:*

- (i) $\mathcal{G}$ is a Gröbner basis of I .
- (ii) For every $1 \leq i < j \leq s$, the polynomial $S(g_i, g_j)$ has a standard representation with respect to $\mathcal{G}$.
- (iii) For every $1 \leq i < j \leq s$, division of $S(g_i, g_j)$ by $\mathcal{G}$ gives remainder zero.

PROOF. Assume first that $\mathcal{G}$ is a Gröbner basis. Every S -polynomial belongs to I . By the ideal membership criterion in [Theorem 3.6.5](#), its remainder on division by $\mathcal{G}$ is zero. Thus (i) implies (iii). The division theorem shows that (iii) implies (ii).

It remains to prove that (ii) implies (i). Assume (ii), and let $0 \neq f \in I$. Among all presentations

$$f = h_1 g_1 + \dots + h_s g_s,$$

choose one for which the largest monomial underlying one of the terms $\text{in}(h_i g_i)$, with $h_i \neq 0$, is as small as possible; denote this monomial by M . Such a smallest monomial exists because $<$ is a well-order. By [Lemma 3.2.3](#), the monomial underlying $\text{in}(f)$ is at most M . Let A be the set of indices i for which the monomial underlying $\text{in}(h_i g_i)$ is M . We claim that the monomial underlying $\text{in}(f)$ is M .

Suppose, to the contrary, that $\text{in}(f) < M$. For $i \in A$, write $\text{in}(h_i) = c_i \mathbf{x}^{\alpha_i}$. Since the initial term of the sum has monomial smaller than M , the initial terms in

$$\sum_{i \in A} c_i \mathbf{x}^{\alpha_i} g_i$$

cancel. By [Lemma 3.6.12](#), this sum is a combination of S -polynomials multiplied by terms so that the initial term of every resulting product is smaller than M .

By hypothesis, each of these S -polynomials has a standard representation with respect to $\mathcal{G}$. After multiplying those representations by the terms appearing in the cancellation lemma, every product with a g_j still has initial term smaller than M . Also,

$$\text{in}((h_i - \text{in}(h_i))g_i) < M$$

for $i \in A$, and the summands indexed outside A already have initial term smaller than M . We have therefore constructed a new presentation of f for which the largest monomial underlying an initial term is smaller than M . This contradicts the choice of the original presentation. Thus the monomial underlying $\text{in}(f)$ is M .

For any $i \in A$, the monomial underlying $\text{in}(g_i)$ therefore divides the monomial underlying $\text{in}(f)$. Equivalently, $\text{in}(g_i) \mid \text{in}(f)$ under our convention for divisibility of terms. By [Proposition 3.2.7](#), $\mathcal{G}$ is a Gröbner basis. □

REMARK 3.6.14. Buchberger’s criterion reduces an infinite condition—checking the initial term of every element of I —to finitely many computations, one for every pair in a finite generating set. The remainders must be computed with respect to the whole set $\mathcal{G}$.

The following useful special case can often eliminate S -pairs without calculation.

PROPOSITION 3.6.15 (Product criterion). *Let $f, g \in S \setminus \{0\}$. If the monomials underlying $\text{in}(f)$ and $\text{in}(g)$ are relatively prime, then*

$$S(f, g) \xrightarrow{\{f, g\}} 0.$$

PROOF. Write $\text{in}(f) = \alpha u$ and $\text{in}(g) = \beta v$, where u, v are monomials and $\alpha, \beta \in \mathbb{k} \setminus \{0\}$. We first show that $\{f, g\}$ is a Gröbner basis of (f, g) . For a nonzero $h \in (f, g)$, choose a presentation

$$h = Af + Bg$$

for which the largest monomial underlying $\text{in}(Af)$ or $\text{in}(Bg)$ is as small as possible, omitting a zero summand when necessary; call this monomial M . If the monomial underlying $\text{in}(h)$ is M , then u or v divides it, as required. Suppose instead that $\text{in}(h) < M$. Then both summands are nonzero and their initial terms cancel. Write $\text{in}(A) = \alpha p$ and $\text{in}(B) = \beta q$, where p, q are monomials. Comparing monomial parts gives

$$pu = qv = M.$$

Since $\gcd(u, v) = 1$, it follows that $v \mid \text{in}(A)$ and $u \mid \text{in}(B)$. Choose the term C so that $\text{in}(A) = C\text{in}(g)$. The cancellation between Af and Bg then also gives

$$\text{in}(B) = -C\text{in}(f).$$

Consequently,

$$h = (A - Cg)f + (B + Cf)g,$$

and both products on the right have initial term strictly smaller than M . This contradicts the choice of the presentation. Thus $\{f, g\}$ is a Gröbner basis by [Proposition 3.2.7](#). Since $S(f, g) \in (f, g)$, [Theorem 3.6.5](#) shows that its remainder is zero. $\square$

The last proof can also be viewed as a direct cancellation argument. Care is still required when several pairs are present: relatively prime monomials underlying the initial terms allow that particular pair to be skipped, but do not by themselves say anything about the other pairs.

3.6.4. Buchberger’s algorithm. Buchberger’s criterion leads directly to an algorithm for constructing a Gröbner basis.

ALGORITHM 3.6.16 (Buchberger algorithm). Let $F = \{f_1, \dots, f_s\}$ be a finite generating set of a nonzero ideal I .

- (1) Delete any zero polynomial from F , set $\mathcal{G} := F$, and let P be the set of all unordered pairs of distinct elements of $\mathcal{G}$.
- (2) While P is nonempty, choose a pair $\{g, h\} \in P$ and remove it from P . Compute a remainder

$$r = \text{rem}_{\mathcal{G}}(S(g, h)).$$

If $r \neq 0$, add r to $\mathcal{G}$ and add to P every new pair $\{r, q\}$ with $q \in \mathcal{G} \setminus \{r\}$.

- (3) When P is empty, return $\mathcal{G}$.

Different choices of pairs, divisors, and nonzero scalar multiples of the remainders can lead to different intermediate sets. They do not affect termination or correctness.

THEOREM 3.6.17 (Correctness and termination of Buchberger's algorithm). *Algorithm 3.6.16 terminates after finitely many steps, and the set it returns is a Gröbner basis of I .*

PROOF. Every new polynomial r is an S -polynomial of elements of I minus an S -linear combination of elements of I . Thus $r \in I$, and adding it does not change the ideal generated by $\mathcal{G}$.

Suppose that $r \neq 0$ is added. Because r is reduced with respect to the current $\mathcal{G}$, no $\text{in}(g)$ with $g \in \mathcal{G}$ divides $\text{in}(r)$. Consequently,

$$(\text{in}(g) \mid g \in \mathcal{G}) \subsetneq (\text{in}(g) \mid g \in \mathcal{G} \cup \{r\}).$$

If infinitely many remainders were added, these monomial ideals would form an infinite strictly ascending chain. This is impossible by [Proposition 2.2.7](#), or equivalently by Hilbert basis theorem. Hence only finitely many new polynomials are added. There are then only finitely many pairs to process, so the algorithm terminates.

Every pair in the final set was placed in P and processed. If its S -polynomial was zero, it has the empty standard representation. Otherwise, at the moment the pair was processed, division gave

$$S(g, h) = \sum_i q_i g_i + r$$

with every nonzero product on the right having initial term at most $\text{in}(S(g, h))$. If $r = 0$, this was already a standard representation. If $r \neq 0$, the polynomial r was added to the basis, so the same equality is a standard representation with coefficient 1 on r . It remains a standard representation with respect to the final set. Buchberger's criterion now shows that the final $\mathcal{G}$ is a Gröbner basis of I . $\square$

REMARK 3.6.18. The proof of termination explains an important design feature of the algorithm. A nonzero remainder is not added merely because it is a new polynomial; it is added because its initial term is outside the monomial ideal generated by all initial terms found so far. Thus each addition makes irreversible progress.

3.6.5. Worked computations. We first complete the running example.

EXAMPLE 3.6.19. Let

$$I = (f_1, f_2) = (x^2 - y, xy - 1) \subset \mathbb{k}[x, y]$$

and use lexicographic order $x > y$. The first S-polynomial is

$$f_3 = S(f_1, f_2) = yf_1 - xf_2 = x - y^2.$$

Neither x^2 nor xy divides $\text{in}(f_3) = x$, so f_3 is a nonzero remainder and is added. Next,

$$S(f_2, f_3) = f_2 - yf_3 = y^3 - 1 =: f_4,$$

which is again reduced with respect to the initial terms already present.

At this point f_1 and f_2 are unnecessary. Indeed, division gives

$$f_2 = yf_3 + f_4,$$

$$f_1 = (x + y^2)f_3 + yf_4.$$

Thus

$$I = (f_3, f_4) = (x - y^2, y^3 - 1).$$

There is only one S-pair left to check:

$$\begin{aligned} S(f_3, f_4) &= y^3(x - y^2) - x(y^3 - 1) \\ &= x - y^5 \\ &= (x - y^2) - y^2(y^3 - 1). \end{aligned}$$

Its remainder is zero. By Buchberger's criterion,

$$\mathcal{G} = \{x - y^2, y^3 - 1\}$$

is a Gröbner basis and

$$\text{in}_{\text{lex}}(I) = (x, y^3).$$

This proves all the assertions made earlier about this example.

The next computation illustrates that several genuinely new initial terms can occur.

EXAMPLE 3.6.20. Let $S = \mathbb{k}[x, y, z]$ with lexicographic order $x > y > z$, and consider

$$I = (f_1, f_2) = (x^2 - y, xy - z).$$

We have

$$f_3 = S(f_1, f_2) = yf_1 - xf_2 = xz - y^2,$$

$$f_4 = S(f_2, f_3) = zf_2 - yf_3 = y^3 - z^2.$$

Both are nonzero remainders at the stages when they occur. We claim that

$$\mathcal{G} = \{f_1, f_2, f_3, f_4\} = \{x^2 - y, xy - z, xz - y^2, y^3 - z^2\}$$

is a Gröbner basis. The two S-pairs used to construct f_3 and f_4 now reduce to zero because $f_3, f_4 \in \mathcal{G}$. The remaining reductions are

$$\begin{aligned} S(f_1, f_3) &= zf_1 - xf_3 = yf_2, \\ S(f_1, f_4) &= y^3f_1 - x^2f_4 = z^2f_1 - yf_4, \\ S(f_2, f_4) &= y^2f_2 - xf_4 = zf_3, \\ S(f_3, f_4) &= y^3f_3 - xzf_4 = z^2f_3 - y^2f_4. \end{aligned}$$

Each right-hand side reduces immediately to zero. Therefore Buchberger's criterion proves the claim, and

$$\text{in}_{\text{lex}}(I) = (x^2, xy, xz, y^3).$$

The standard monomials form the set

$$\{x\} \cup \{y^a z^b \mid 0 \leq a \leq 2, b \geq 0\}.$$

They give a $\mathbb{k}$ -basis of S/I .

The identities in the preceding example are worth reading in two ways. They certify the S-pair reductions, and they also display explicitly how every relation created by the algorithm remains in the original ideal.

3.6.6. Minimal and reduced Gröbner bases. Buchberger's algorithm usually returns more polynomials than are needed. There are two standard ways to remove this redundancy.

DEFINITION 3.6.21. Let $\mathcal{G} = \{g_1, \dots, g_s\}$ be a Gröbner basis. It is called *minimal* if:

- (i) $\text{LC}(g_i) = 1$ for every i ;
- (ii) $\text{in}(g_i)$ does not divide $\text{in}(g_j)$ whenever $i \neq j$.

It is called *reduced* if:

- (i) $\text{LC}(g_i) = 1$ for every i ;
- (ii) no monomial of g_i is divisible by $\text{in}(g_j)$ for any $j \neq i$.

Because every other monomial of g_i is smaller than $\text{in}(g_i)$, it cannot be divisible by $\text{in}(g_i)$. Thus the second condition for a reduced basis says that the initial terms are pairwise nondivisible and that every other monomial of every g_i is standard modulo I . Every reduced Gröbner basis is minimal.

PROPOSITION 3.6.22. *Let $\mathcal{G}$ be a Gröbner basis of I . Then $\mathcal{G}$ is minimal if and only if its elements are monic, its initial terms are pairwise distinct, and*

$$\{\text{in}(g) \mid g \in \mathcal{G}\} = G(\text{in}(I)),$$

the unique minimal set of monomial generators of the initial ideal.

PROOF. The initial terms of a Gröbner basis generate $\text{in}(I)$. A finite set of monomial generators is minimal precisely when no member divides another. The assertion follows from the uniqueness of $G(\text{in}(I))$. $\square$

Minimal Gröbner bases need not be unique, since one may change the other terms without changing the initial terms. Reduced Gröbner bases remove this ambiguity.

THEOREM 3.6.23 (Existence and uniqueness of reduced Gröbner bases). *For every ideal $I \subset S$ and every monomial order $<$, there exists a unique reduced Gröbner basis of I .*

PROOF. We first prove existence. Starting with a finite Gröbner basis, remove any g_i whose initial term is divisible by the initial term of another member. The remaining initial terms still generate the same initial ideal. Continue until no such divisibility remains, and then divide each polynomial by its leading coefficient. This produces a minimal Gröbner basis

$$\mathcal{G} = \{g_1, \dots, g_s\}.$$

For each i , divide g_i by the ordered list $\mathcal{G} \setminus \{g_i\}$. Since none of the other initial terms divides $\text{in}(g_i)$, the resulting remainder $\tilde{g}_i$ is monic and has the same initial term as g_i . Moreover,

$$\tilde{g}_i - g_i \in (\mathcal{G} \setminus \{g_i\}) \subset I,$$

so $\tilde{g}_i \in I$. Replacing every g_i by $\tilde{g}_i$ preserves all initial terms and hence preserves the Gröbner basis property. The resulting basis is reduced.

For uniqueness, let $\mathcal{G}$ and $\mathcal{H}$ be two reduced Gröbner bases of I . By Proposition 3.6.22, their sets of initial terms both equal the unique set $G(\text{in}(I))$. Thus, for every $g \in \mathcal{G}$, there is a unique $h \in \mathcal{H}$ such that

$$\text{in}(g) = \text{in}(h).$$

Both polynomials are monic, so their initial terms cancel in $g - h$. Every remaining monomial of $g - h$ is standard modulo I , because both bases are reduced and their initial-term sets agree. But $g - h \in I$. By the linear independence in Theorem 3.2.11, this forces $g - h = 0$. Therefore $g = h$, and $\mathcal{G} = \mathcal{H}$. $\square$

REMARK 3.6.24. The existence argument is an effective post-processing procedure: remove elements with redundant initial terms, make the remaining elements monic, and reduce every other term by the other basis elements. Unlike an arbitrary output of Buchberger's algorithm, the final reduced basis depends only on I and the chosen monomial order.

COROLLARY 3.6.25. *Fix a monomial order $<$. Two ideals $I, J \subset S$ are equal if and only if their reduced Gröbner bases with respect to $<$ are equal.*

PROOF. Equal ideals have the same reduced Gröbner basis by the uniqueness in [Theorem 3.6.23](#). Conversely, a Gröbner basis generates its ideal by [Proposition 3.2.8](#), so equal reduced bases generate equal ideals. $\square$

EXAMPLE 3.6.26. In [Example 3.6.19](#), the set

$$\{x - y^2, y^3 - 1\}$$

is reduced: it is monic, x and y^3 minimally generate the initial ideal, and the other monomials y^2 and 1 are standard. The original Buchberger basis $\{f_1, f_2, f_3, f_4\}$ was not minimal because $x = \text{in}(f_3)$ divides both $x^2 = \text{in}(f_1)$ and $xy = \text{in}(f_2)$.

EXAMPLE 3.6.27. The basis

$$\{x^2 - y, xy - z, xz - y^2, y^3 - z^2\}$$

from [Example 3.6.20](#) is already reduced. Its initial terms x^2, xy, xz, y^3 are pairwise nondivisible, and its other monomials y, z, y^2, z^2 are all standard.

3.6.7. A glimpse of computational complexity. The algorithms in this chapter terminate, but termination alone says little about how long a computation will take. As the number of variables, the degrees, and the number of generators grow, a Gröbner basis computation may create many new polynomials and very large intermediate expressions. To formulate this warning precisely, we take a short detour into complexity theory. Only a small dictionary is needed here.

Want to earn a million dollars?

The playful question points toward the famous problem

$$P \stackrel{?}{=} NP.$$

It is one of the [Millennium Prize Problems](#); the [official problem description by Stephen Cook](#) gives precise definitions. Roughly speaking, it asks whether every decision problem whose yes answers have short, efficiently checkable certificates can also be decided efficiently. Here is a brief guide to the terminology.

DEFINITION 3.6.28 (A small complexity dictionary). A *decision problem* is a problem whose answer is yes or no.

- (i) *Polynomial time*: the running time grows at most like a fixed power of the input size.
- (ii) P : decision problems that can be solved in polynomial time.
- (iii) NP : decision problems whose yes answers have short proofs, called *certificates*, that can be checked in polynomial time.

- (iv) A *polynomial-time many-one reduction* translates an instance of one decision problem into an instance of another, in polynomial time, preserving the yes-or-no answer.
- (v) A problem is *NP-hard* if every problem in NP can be reduced to it in polynomial time.
- (vi) A decision problem is *NP-complete* if it is both in NP and NP-hard.

Thus $P \subseteq NP$: if a problem can be solved quickly, a claimed yes answer can be checked by solving the problem. Whether the reverse inclusion holds is unknown. The initials NP stand for “nondeterministic polynomial time,” not “non-polynomial time.” We shall use only one consequence of this language: if an NP-complete problem could be solved in polynomial time, then $P = NP$.

Our source problem is graph coloring. A finite simple graph $G = (V, E)$ is *3-colorable* if one can assign one of three colors to every vertex in such a way that adjacent vertices receive different colors. The decision problem 3-COLORABILITY asks whether such an assignment exists. A certificate lists one of the three colors for each vertex. Checking the labels and making sure that the endpoints of every edge have different colors takes polynomial time in the size of the graph. A fundamental result in complexity theory, which we use without proof, says that 3-COLORABILITY is NP-complete; see the [official problem description, Section 2](#). For the picture and the algebraic encoding below, fix a primitive cube root of unity $\omega \in \mathbb{C}$.

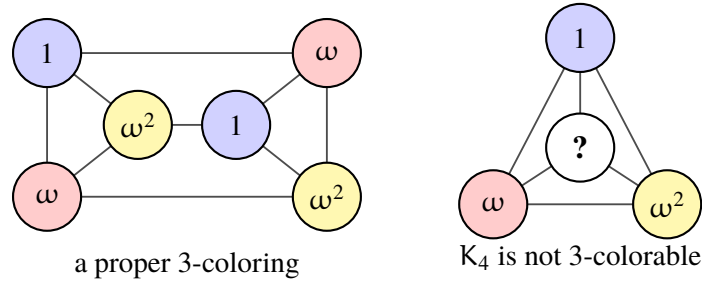


FIGURE 1. A yes-instance and a no-instance of 3-COLORABILITY. Labels $1, \omega, \omega^2$ keep the colors meaningful in grayscale. In K_4 , the outer triangle must use all three colors and leaves none for the central vertex.

We now translate this combinatorial problem into polynomial equations. Let $V = \{1, \dots, n\}$ and put

$$S_{\mathbb{Q}} = \mathbb{Q}[x_1, \dots, x_n], \quad S_{\mathbb{C}} = \mathbb{C}[x_1, \dots, x_n].$$

For a graph $G = (V, E)$, define the *coloring ideal*

$$J_G = (x_i^3 - 1 \mid i \in V) + (x_i^2 + x_i x_j + x_j^2 \mid \{i, j\} \in E) \subset S_{\mathbb{Q}},$$

and let $I_G = J_G S_{\mathbb{C}}$ be the same ideal after extending the coefficient field to $\mathbb{C}$. The use of $\mathbb{Q}$ records that the input has an exact finite encoding; the passage to $\mathbb{C}$ allows us to use the Nullstellensatz. This is the standard [graph-coloring ideal construction](#).

PROPOSITION 3.6.29 (The coloring ideal). *Let $\mathcal{G}_G$ be the reduced Gröbner basis of J_G for lexicographic order $x_1 > x_2 > \dots > x_n$. Then*

$$\begin{aligned} G \text{ is 3-colorable} &\iff V(I_G) \neq \emptyset \iff I_G \neq S_{\mathbb{C}} \\ &\iff J_G \neq S_{\mathbb{Q}} \iff \mathcal{G}_G \neq \{1\}. \end{aligned}$$

PROOF. Let

$$\mu_3 = \{a \in \mathbb{C} \mid a^3 = 1\} = \{1, \omega, \omega^2\}.$$

The equations $x_i^3 - 1 = 0$ require the coordinate a_i of every point $a \in V(I_G)$ to belong to μ_3 . These three values will be our colors. If $a_i, a_j \in \mu_3$, then

$$a_i^3 - a_j^3 = (a_i - a_j)(a_i^2 + a_i a_j + a_j^2) = 0.$$

If $a_i \neq a_j$, the first factor is nonzero, so the second factor is zero. If $a_i = a_j$, the second factor is $3a_i^2 \neq 0$. Therefore

$$a_i^2 + a_i a_j + a_j^2 = 0 \iff a_i \neq a_j.$$

It follows that the points of $V(I_G)$ are exactly the proper 3-colorings of G with color set μ_3 .

The weak Nullstellensatz in Theorem 4.1.6 gives $V(I_G) \neq \emptyset$ if and only if $I_G \neq S_{\mathbb{C}}$. A Gröbner basis of J_G over $\mathbb{Q}$ remains a Gröbner basis after extending coefficients to $\mathbb{C}$, with the same initial monomials. Thus extension of the coefficient field preserves whether the ideal is the unit ideal, so $I_G \neq S_{\mathbb{C}}$ if and only if $J_G \neq S_{\mathbb{Q}}$. Finally, the reduced Gröbner basis of the unit ideal is $\{1\}$, and uniqueness in Theorem 3.6.23 gives the last equivalence. $\square$

For the complexity statement, a sparse polynomial is encoded by listing its nonzero terms, with rational coefficients represented by numerators and denominators in binary and exponents also written in binary. The number of variables is part of the input.

COROLLARY 3.6.30 (A complexity consequence). *The decision problem of determining whether an ideal given by finitely many generators in $\mathbb{Q}[x_1, \dots, x_n]$ is proper is NP-hard under polynomial-time many-one reductions. As a consequence, if a deterministic algorithm computes the reduced lexicographic Gröbner basis over $\mathbb{Q}$ in time polynomial in the length of its input, then $P = NP$.*

PROOF. Encode a graph $G = (V, E)$ by explicitly listing its vertices and edges. The ideal J_G can be written down using $|V| + |E|$ sparse polynomials. Each has at most three terms, degree at most three, and coefficients in $\{-1, 0, 1\}$. Thus the construction of J_G takes polynomial time in the size of the encoded graph. By Proposition 3.6.29, G is 3-colorable if and only if J_G is proper. This is a polynomial-time many-one reduction from the NP-complete problem 3-COLORABILITY, proving the first assertion. The same proposition says that J_G is proper if and only if $\mathcal{G}_G \neq \{1\}$. A polynomial-time algorithm for computing $\mathcal{G}_G$ would therefore decide 3-COLORABILITY in polynomial time, forcing $P = NP$. $\square$

3.6.8. A practical summary. For a fixed monomial order, the main constructions of the chapter fit together as follows. Starting with generators of an ideal I , Buchberger's algorithm produces a Gröbner basis $\mathcal{G}$. Its initial terms generate $\text{in}(I)$. Division by $\mathcal{G}$ computes the unique normal form of any polynomial, and

$$f \in I \iff \text{NF}_{\mathcal{G}}(f) = 0.$$

The monomials not divisible by any $\text{in}(g)$, $g \in \mathcal{G}$, form a basis of S/I . Finally, interreducing $\mathcal{G}$ gives the unique reduced Gröbner basis.

The choice of monomial order matters. Different orders usually give different initial ideals, standard monomials, intermediate computations, and reduced Gröbner bases. What does not change is the ideal itself and, for homogeneous ideals, the Hilbert function recorded by every initial ideal.

3.7. Exercises

EXERCISE 3.1. Let $I, J \subset S$ be two ideals and $<$ a monomial order on S . Let $\mathcal{G}, \mathcal{G}'$ be Gröbner bases of I and J , respectively, with respect to $<$. Prove that if the monomials underlying $\text{in}_{<}(g)$ and $\text{in}_{<}(g')$ are relatively prime for every $g \in \mathcal{G}$ and $g' \in \mathcal{G}'$, then $\mathcal{G} \cup \mathcal{G}'$ is a Gröbner basis of $I + J$.

EXERCISE 3.2. Prove the following statements:

- (i) There is a unique monomial order on $\mathbb{k}[x_1]$.
- (ii) Let $n \geq 2$. Then there are infinitely many monomial orders on $S = \mathbb{k}[x_1, \dots, x_n]$.

EXERCISE 3.3. Let $I \subset S$ be an ideal and $<$ be a monomial order on S . Show that $\text{in}_{<}(\sqrt{I}) \subseteq \sqrt{\text{in}_{<}(I)}$.

EXERCISE 3.4. Using only the Gröbner-basis and normal-form results developed in this chapter, let $I \subset S$ be an ideal and let $<$ be a monomial order on S . Prove the following statements:

- (i) If $\text{in}_{<}(I)$ is radical, then I is radical.
- (ii) If $\text{in}_{<}(I)$ is prime, then I is prime.

EXERCISE 3.5. Let $f_1, \dots, f_m \in S$ be polynomials and $<$ be a monomial order on S . Assume that $\text{in}_{<}(f_1), \dots, \text{in}_{<}(f_m)$ is a regular sequence. Then prove the following statements:

- (i) $f_1, \dots, f_m$ is a regular sequence.
- (ii) $f_1, \dots, f_m$ is a Gröbner basis of $I = (f_1, \dots, f_m)$.

CHAPTER 4

The Algebra–Geometry Dictionary

Throughout this chapter, $\mathbb{k}$ is an algebraically closed field and $S = \mathbb{k}[x_1, \dots, x_n]$. Our goal is to make precise the correspondence between ideals of S and affine varieties in $\mathbb{A}_{\mathbb{k}}^n$. The fundamental result is Hilbert’s Nullstellensatz.

4.1. Hilbert’s Nullstellensatz (a first quick algebraic proof)

We first give the usual algebraic proof based on Zariski’s lemma; see, e.g., [7, §5]. We include all the needed details.

DEFINITION 4.1.1. Let A be a ring containing $\mathbb{k}$ as a subring. We say that A is a *finitely generated $\mathbb{k}$ -algebra* if there are elements $a_1, \dots, a_m \in A$ such that

$$A = \mathbb{k}[a_1, \dots, a_m].$$

An element $a \in A$ is *algebraic over $\mathbb{k}$* if it is a root of a nonzero polynomial in $\mathbb{k}[t]$.

We will also use the following slightly more general notion.

DEFINITION 4.1.2. Let $A \subset B$ be rings. An element $b \in B$ is *integral over A* if it satisfies an equation

$$b^r + c_{r-1}b^{r-1} + \dots + c_1b + c_0 = 0$$

with $r \geq 1$ and $c_0, \dots, c_{r-1} \in A$. The ring B is *integral over A* if every element of B is integral over A .

Thus, an element of a field extension of $\mathbb{k}$ is algebraic over $\mathbb{k}$ if and only if it is integral over $\mathbb{k}$.

LEMMA 4.1.3. Let $A \subset B$ be rings. If $B = A[b_1, \dots, b_s]$ and every b_i is integral over A , then B is a *finitely generated A -module*.

PROOF. For every $1 \leq i \leq s$, choose a monic equation for b_i of degree d_i . By successively replacing $b_i^{d_i}$ by an A -linear combination of lower powers, we see that the finitely many elements

$$b_1^{e_1} \dots b_s^{e_s}, \quad 0 \leq e_i < d_i,$$

generate B as an A -module. □

LEMMA 4.1.4. Let $A \subset L$ be rings such that L is a field and is a *finitely generated A -module*. Then A is a field.

PROOF. Since L is a finitely generated A -module, every element of L is integral over A by the determinant trick. Let $0 \neq a \in A$. Since $a^{-1} \in L$, there is an equation

$$(a^{-1})^r + c_{r-1}(a^{-1})^{r-1} + \cdots + c_1 a^{-1} + c_0 = 0$$

with $c_i \in A$. Multiplying by a^{r-1} gives

$$a^{-1} = -c_{r-1} - c_{r-2}a - \cdots - c_0 a^{r-1} \in A.$$

Therefore every nonzero element of A is a unit, and A is a field. $\square$

For completeness, the determinant trick used above is the following simple observation. If a ring B is generated by $v_1, \dots, v_s$ as an A -module and $b \in B$, write $bv_i = \sum_j a_{ij}v_j$ with $a_{ij} \in A$. Multiplication by the adjugate of the matrix $(b\delta_{ij} - a_{ij})$ shows that its determinant annihilates every v_i . Since these elements generate B and $1 \in B$, the monic polynomial $\det(t\delta_{ij} - a_{ij}) \in A[t]$ vanishes at b .

THEOREM 4.1.5 (Zariski's lemma). *Let L be a field which is a finitely generated $\mathbb{k}$ -algebra. Then L is a finite algebraic extension of $\mathbb{k}$.*

PROOF. Write $L = \mathbb{k}[a_1, \dots, a_m]$. Choose a maximal algebraically independent subset among the a_i and, after relabeling, suppose it is $a_1, \dots, a_r$. Set

$$F = \mathbb{k}(a_1, \dots, a_r).$$

By maximality, each of $a_{r+1}, \dots, a_m$ is algebraic over F . As a consequence, $L = F[a_{r+1}, \dots, a_m]$ is a finite-dimensional F -vector space.

The monic equations for $a_{r+1}, \dots, a_m$ have only finitely many coefficients in F . After clearing all their denominators at once, there is a nonzero polynomial $f \in \mathbb{k}[a_1, \dots, a_r]$ such that each a_i , $i > r$, is integral over

$$A = \mathbb{k}[a_1, \dots, a_r, 1/f].$$

Because L is a field and $0 \neq f \in L$, the element $1/f$ already belongs to L . As a consequence, $L = A[a_{r+1}, \dots, a_m]$. It follows from [Lemma 4.1.3](#) that L is a finitely generated A -module. Hence [Lemma 4.1.4](#) shows that A is a field.

We claim that this is impossible if $r > 0$. The algebraic independence of $a_1, \dots, a_r$ identifies $\mathbb{k}[a_1, \dots, a_r]$ with a polynomial ring $\mathbb{k}[t_1, \dots, t_r]$. View f as a polynomial in $t_2, \dots, t_r$ with coefficients in $\mathbb{k}[t_1]$ and choose one nonzero coefficient $c(t_1)$. There are infinitely many pairwise nonassociate irreducible polynomials in $\mathbb{k}[t_1]$: otherwise, if $p_1, \dots, p_v$ were all of them, an irreducible factor of $1 + p_1 \cdots p_v$ would give a contradiction. We may therefore choose an irreducible $q(t_1)$ which does not divide $c(t_1)$. Then q does not divide f in $\mathbb{k}[t_1, \dots, t_r]$. Moreover, q is a prime element of this polynomial ring, because the quotient by (q) is a polynomial ring over the field $\mathbb{k}[t_1]/(q)$. If q were a unit in A , then an equality $qg = f^N$ would hold for some

$g \in \mathbb{k}[t_1, \dots, t_r]$. Multiplying g by f if necessary, we may take $N \geq 1$. Since q is prime, this would force $q \mid f$, a contradiction. Thus A is not a field.

It follows that $r = 0$. Hence all the α_i are algebraic over $\mathbb{k}$, and adjoining them one at a time shows that $[L : \mathbb{k}] < \infty$. $\square$

For a point $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{A}_{\mathbb{k}}^n$, consider the evaluation map

$$\text{ev}_{\alpha} : S \longrightarrow \mathbb{k}, \quad f \longmapsto f(\alpha).$$

This map is surjective and

$$\text{Ker}(\text{ev}_{\alpha}) = (x_1 - \alpha_1, \dots, x_n - \alpha_n).$$

Indeed, repeated use of the one-variable division algorithm writes every $f \in S$ in the form

$$f = f(\alpha) + \sum_{i=1}^n (x_i - \alpha_i) g_i$$

for suitable $g_i \in S$. It follows that

$$\mathfrak{m}_{\alpha} := (x_1 - \alpha_1, \dots, x_n - \alpha_n)$$

is a maximal ideal of S .

THEOREM 4.1.6 (Weak Nullstellensatz). *Let $I \subset S$ be an ideal. Then*

$$V(I) = \emptyset \iff I = S.$$

Equivalently, every proper ideal of S has a common zero in $\mathbb{A}_{\mathbb{k}}^n$.

PROOF. It is clear that $V(S) = \emptyset$. Conversely, suppose that I is proper. Choose a maximal ideal $\mathfrak{m}$ containing I . The quotient

$$L = S/\mathfrak{m} = \mathbb{k}[\overline{x_1}, \dots, \overline{x_n}]$$

is both a field and a finitely generated $\mathbb{k}$ -algebra. By Zariski's lemma, L is algebraic over $\mathbb{k}$. Since $\mathbb{k}$ is algebraically closed, $L = \mathbb{k}$. Thus, for every i , there is an element $\alpha_i \in \mathbb{k}$ such that $\overline{x_i} = \alpha_i$. Therefore

$$(x_1 - \alpha_1, \dots, x_n - \alpha_n) \subset \mathfrak{m}.$$

The ideal on the left is maximal, so equality holds. Since $I \subset \mathfrak{m} = \mathfrak{m}_{\alpha}$, every polynomial in I vanishes at $\alpha = (\alpha_1, \dots, \alpha_n)$. Hence $\alpha \in V(I)$. $\square$

The proof yields the following important description.

COROLLARY 4.1.7. *The maximal ideals of $S = \mathbb{k}[x_1, \dots, x_n]$ are precisely the ideals*

$$\mathfrak{m}_{\alpha} = (x_1 - \alpha_1, \dots, x_n - \alpha_n), \quad \alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{A}_{\mathbb{k}}^n.$$

The weak Nullstellensatz detects whether a system has a solution. The strong form determines every polynomial consequence of that system.

THEOREM 4.1.8 (Hilbert’s Nullstellensatz). *Let $I \subset S$ be an ideal. Then*

$$I(V(I)) = \sqrt{I}.$$

PROOF. If $f \in \sqrt{I}$, then $f^N \in I$ for some $N \geq 1$. For every $\alpha \in V(I)$, we have $f(\alpha)^N = 0$, and thus $f(\alpha) = 0$. This proves $\sqrt{I} \subset I(V(I))$.

For the reverse inclusion, let $f \in I(V(I))$. If $f = 0$, there is nothing to prove, so assume that $f \neq 0$ and introduce one more variable y . Consider the ideal

$$J = (I, 1 - yf) \subset S[y].$$

The variety $V(J) \subset \mathbb{A}_k^{n+1}$ is empty. Indeed, if $(\alpha, b) \in V(J)$, then $\alpha \in V(I)$, so $f(\alpha) = 0$, whereas $1 - bf(\alpha) = 0$, which is impossible. By the weak Nullstellensatz, $J = S[y]$. As a consequence, there are $g_1, \dots, g_s \in I$ and $A_1, \dots, A_s, B \in S[y]$ such that

$$1 = \sum_{i=1}^s A_i(x, y) g_i(x) + B(x, y)(1 - yf(x)).$$

In the localization

$$S_f = \{\alpha/f^m \mid \alpha \in S, m \geq 0\},$$

substitute $y = f^{-1}$. We obtain

$$1 = \sum_{i=1}^s A_i(x, f^{-1}) g_i(x).$$

After multiplying by a sufficiently large power f^N , with $N \geq 1$, to clear all denominators, we get $f^N \in I$. Therefore $f \in \sqrt{I}$. $\square$

The substitution $y = f^{-1}$ used in this proof is commonly called the *Rabinowitsch trick*.

We now record the algebra–geometry dictionary furnished by the Nullstellensatz.

COROLLARY 4.1.9. *The maps*

$$\left\{ \text{radical ideals of } S \right\} \xrightleftharpoons[I(-)]{V(-)} \left\{ \text{affine varieties in } \mathbb{A}_k^n \right\}$$

are mutually inverse, inclusion-reversing bijections.

PROOF. If I is radical, then the Nullstellensatz gives $I(V(I)) = \sqrt{I} = I$. For every affine variety X , we already know that $V(I(X)) = X$. Both operations reverse inclusions by their definitions. $\square$

COROLLARY 4.1.10. *Let $I, J \subset S$ be ideals. Then the following statements hold:*

- (i) $V(I) = V(J)$ if and only if $\sqrt{I} = \sqrt{J}$;
- (ii) $V(I) \subset V(J)$ if and only if $\sqrt{I} \supset \sqrt{J}$;

- (iii) $V(I)$ is a single point if and only if $\sqrt{I}$ is a maximal ideal;
- (iv) $V(I) = \emptyset$ if and only if $\sqrt{I} = S$.

PROOF. These statements follow by applying $I(-)$ and using $I(V(I)) = \sqrt{I}$, together with Corollary 4.1.7. $\square$

EXAMPLE 4.1.11. In $\mathbb{k}[x, y]$, the ideals

$$I = (x^2, y^3) \quad \text{and} \quad J = (x, y)$$

define the same variety, namely the origin. The Nullstellensatz explains this equality algebraically: $\sqrt{I} = J$.

The dictionary also recognizes irreducible varieties.

DEFINITION 4.1.12. A nonempty affine variety X is *irreducible* if, whenever $X = X_1 \cup X_2$ with X_1 and X_2 affine varieties, one has $X = X_1$ or $X = X_2$. An *irreducible component* of an affine variety X is an irreducible affine variety contained in X which is maximal, under inclusion, among all such subvarieties of X .

THEOREM 4.1.13. Let $X \subset \mathbb{A}_{\mathbb{k}}^n$ be a nonempty affine variety. Then X is irreducible if and only if $I(X)$ is a prime ideal. As a consequence, for every proper ideal $I \subset S$, the variety $V(I)$ is irreducible if and only if $\sqrt{I}$ is prime.

PROOF. Suppose first that X is irreducible and $fg \in I(X)$. Then

$$X = (X \cap V(f)) \cup (X \cap V(g)).$$

Both sets on the right are affine varieties. Irreducibility gives $X \subset V(f)$ or $X \subset V(g)$. Therefore $f \in I(X)$ or $g \in I(X)$, and $I(X)$ is prime.

Conversely, suppose that $I(X)$ is prime and write $X = X_1 \cup X_2$. If both X_1 and X_2 were proper subsets of X , then $I(X_i) \neq I(X)$ for $i = 1, 2$: otherwise,

$$X_i = V(I(X_i)) = V(I(X)) = X.$$

Since $X_i \subset X$ gives $I(X) \subset I(X_i)$, we may therefore choose polynomials

$$f \in I(X_1) \setminus I(X) \quad \text{and} \quad g \in I(X_2) \setminus I(X).$$

The product fg vanishes on $X_1 \cup X_2 = X$, so $fg \in I(X)$, contradicting primality. Thus $X = X_1$ or $X = X_2$.

The last statement follows from $I(V(I)) = \sqrt{I}$. $\square$

EXAMPLE 4.1.14. The variety $V(xy) \subset \mathbb{A}_{\mathbb{k}}^2$ is the union

$$V(xy) = V(x) \cup V(y)$$

of the two coordinate axes. It is reducible, and correspondingly the radical ideal (xy) is not prime.

DEFINITION 4.1.15. Let $X \subset \mathbb{A}_{\mathbb{k}}^n$ be an affine variety. The *coordinate ring* of X is

$$\mathbb{k}[X] := S/I(X).$$

Two polynomials determine the same function on X precisely when their difference belongs to $I(X)$. Therefore $\mathbb{k}[X]$ can be identified with the ring of polynomial functions $X \rightarrow \mathbb{k}$. Under this identification, [Theorem 4.1.13](#) says that X is irreducible if and only if $\mathbb{k}[X]$ is an integral domain.

The affine Hilbert invariants of the preceding chapter now have their geometric interpretation.

DEFINITION 4.1.16. Let $X \subset \mathbb{A}_{\mathbb{k}}^n$ be a nonempty affine variety. We define its *dimension* and its *affine degree* by

$$\dim(X) := \dim(\mathbb{k}[X]), \quad \deg_{\text{aff}}(X) := \deg_{\text{aff}}(\mathbb{k}[X]).$$

These invariants use the filtration on $\mathbb{k}[X]$ induced by polynomials of total degree in the chosen affine coordinates. In particular, affine degree is attached to the embedding $X \subset \mathbb{A}_{\mathbb{k}}^n$.

THEOREM 4.1.17 (Affine Hilbert theorem for varieties). *Let $X \subset \mathbb{A}_{\mathbb{k}}^n$ be a nonempty affine variety, and give $\mathbb{k}[X]$ its induced total-degree filtration*

$$F_m \mathbb{k}[X] := \frac{S_{\leq m} + I(X)}{I(X)}.$$

Set

$$H_X^{\text{aff}}(m) := \dim_{\mathbb{k}} F_m \mathbb{k}[X], \quad \text{Hilb}_X^{\text{aff}}(t) := \sum_{m \geq 0} H_X^{\text{aff}}(m) t^m.$$

If $d = \dim(X)$, then there is a unique polynomial $Q_X(t) \in \mathbb{Z}[t]$ such that

$$\text{Hilb}_X^{\text{aff}}(t) = \frac{Q_X(t)}{(1-t)^{d+1}}, \quad Q_X(1) = \deg_{\text{aff}}(X) > 0.$$

There is a unique polynomial $P_X^{\text{aff}}(z) \in \mathbb{Q}[z]$ such that

$$P_X^{\text{aff}}(m) = H_X^{\text{aff}}(m)$$

for all $m \gg 0$. Moreover,

$$P_X^{\text{aff}}(z) = \deg_{\text{aff}}(X) \frac{z^d}{d!} + \text{lower degree terms}.$$

For $d = 0$, this says that P_X^{aff} is the positive constant $\deg_{\text{aff}}(X)$.

More explicitly, if $X = V(I)$ for an ideal $I \subset S$, then

$$I(X) = \sqrt{I} \quad \text{and hence} \quad \mathbb{k}[X] \cong S/\sqrt{I}.$$

Thus P_X^{aff} , $\dim(X)$, and $\deg_{\text{aff}}(X)$ are respectively the affine Hilbert polynomial, dimension, and affine degree of $S/\sqrt{I}$.

PROOF. The displayed identification of the coordinate ring is [Theorem 4.1.8](#). The remaining assertions follow by applying [Theorem 3.4.2](#) to $\mathbb{k}[X] = S/I(X)$. $\square$

REMARK 4.1.18. Let $I \subset S$ be an ideal. The quotient S/I should not be confused with the coordinate ring of the variety $V(I)$. By Hilbert’s Nullstellensatz,

$$\mathbb{k}[V(I)] \cong S/\sqrt{I}.$$

The rings S/I and $S/\sqrt{I}$ may be different: the former may contain nonzero nilpotent elements, while the latter does not. Their affine degrees may also be different. For example,

$$V(x^2) = V(x) = \{0\} \subset \mathbb{A}_{\mathbb{k}}^1,$$

so $\mathbb{k}[V(x^2)] \cong \mathbb{k}[x]/(x)$ has affine degree 1, whereas $\mathbb{k}[x]/(x^2)$ has affine degree 2.

More generally, for an arbitrary subset $Y \subset \mathbb{A}_{\mathbb{k}}^n$, define $I(Y)$ to be the ideal of all polynomials vanishing on Y . The smallest affine variety containing Y , i.e., its closure in the Zariski topology, is

$$\overline{Y} = V(I(Y)).$$

Indeed, $V(I(Y))$ contains Y , and every variety containing Y is of the form $V(J)$ with $J \subset I(Y)$, hence contains $V(I(Y))$.

4.2. Hilbert’s Nullstellensatz (a second proof)

We now give the elementary proof of the weak Nullstellensatz from [\[1\]](#). The proof is geometric in spirit: after a linear change of coordinates, we project a system of equations from $\mathbb{A}_{\mathbb{k}}^n$ to $\mathbb{A}_{\mathbb{k}}^{n-1}$. A resultant ensures that a point obtained by induction lifts to a common zero of the original system.

We begin with the coordinate change that puts one polynomial in a suitable position.

LEMMA 4.2.1. *Let F be an infinite field and let $0 \neq f \in F[x_1, \dots, x_n]$ be a nonconstant polynomial, where $n \geq 2$. Set $d = \deg(f)$. There are $\lambda_1, \dots, \lambda_{n-1} \in F$ such that the coefficient of x_n^d in*

$$f(x_1 + \lambda_1 x_n, \dots, x_{n-1} + \lambda_{n-1} x_n, x_n)$$

is nonzero.

PROOF. Let f_d be the homogeneous component of f of degree d . The coefficient of x_n^d after the indicated substitution is

$$f_d(\lambda_1, \dots, \lambda_{n-1}, 1).$$

The polynomial $f_d(t_1, \dots, t_{n-1}, 1)$ is nonzero: two different degree- d monomials remain different after setting the last variable equal to 1. Since F is infinite, [Proposition 1.1.7](#) gives a point $(\lambda_1, \dots, \lambda_{n-1}) \in F^{n-1}$ at which this polynomial does not vanish. $\square$

The substitution in [Lemma 4.2.1](#) is an invertible linear change of coordinates. After dividing by its nonzero leading coefficient, the transformed polynomial is monic when regarded as an element of $F[x_1, \dots, x_{n-1}][x_n]$.

We next define the resultant while keeping track of degree bounds. This is important here because specialization may lower the degree of a polynomial.

DEFINITION (Sylvester matrix and resultant). Let A be a commutative ring with identity, and fix integers $d, e \geq 0$ with $d + e > 0$. Write

$$p = a_0 + a_1 t + \dots + a_d t^d, \quad q = b_0 + b_1 t + \dots + b_e t^e$$

in $A[t]$. The bounds d and e are part of the data; in particular, the coefficients a_d and b_e are allowed to vanish. For $r \geq 0$, put $A[t]_{<r} = \bigoplus_{i=0}^{r-1} A t^i$, with $A[t]_{<0} = 0$. For $h \in A[t]_{<d+e}$, let $[h]$ denote its coefficient row vector in the ordered basis $1, t, \dots, t^{d+e-1}$.

The *Sylvester matrix* of p and q , with these degree bounds, is the $(d+e) \times (d+e)$ matrix whose rows, in order, are

$$[p], [tp], \dots, [t^{e-1}p], [q], [tq], \dots, [t^{d-1}q].$$

The first list is omitted if $e = 0$, and the second is omitted if $d = 0$. When $d, e > 0$, it has the schematic form

$$\text{Syl}_t^{d,e}(p, q) = \begin{pmatrix} a_0 & a_1 & \dots & a_d & 0 & \dots & 0 \\ 0 & a_0 & a_1 & \dots & a_d & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & & \ddots & 0 \\ 0 & \dots & 0 & a_0 & a_1 & \dots & a_d \\ b_0 & b_1 & \dots & b_e & 0 & \dots & 0 \\ 0 & b_0 & b_1 & \dots & b_e & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & & \ddots & 0 \\ 0 & \dots & 0 & b_0 & b_1 & \dots & b_e \end{pmatrix}.$$

The first block has e rows, the second block has d rows, and all blank entries are zero.

The (d, e) -*resultant* of p and q with respect to t is

$$\text{Res}_t^{d,e}(p, q) := \det(\text{Syl}_t^{d,e}(p, q)) \in A.$$

When p and q have degrees exactly d and e , respectively, we abbreviate this element as $\text{Res}_t(p, q)$.

LEMMA 4.2.2 (Basic properties of the resultant). *With the notation of the preceding definition, the resultant has the following properties:*

(i) *There are $u \in A[t]_{<e}$ and $v \in A[t]_{<d}$ such that*

$$\text{Res}_t^{d,e}(p, q) = up + vq.$$

In particular, $\text{Res}_t^{d,e}(p, q) \in (p, q) \cap A$.

(ii) *If $\varphi : A \rightarrow B$ is a ring homomorphism and $\varphi(p), \varphi(q) \in B[t]$ are presented with the same degree bounds d, e , then*

$$\varphi\left(\text{Res}_t^{d,e}(p, q)\right) = \text{Res}_t^{d,e}(\varphi(p), \varphi(q)).$$

(iii) *If $p = 1$ (presented with the same degree bound d) and q is monic of degree e , then $\text{Res}_t^{d,e}(p, q) = 1$.*

PROOF. Set $M = \text{Syl}_t^{d,e}(p, q)$ and let $e_0 = (1, 0, \dots, 0)$. Write

$$e_0 \text{adj}(M) = (u_0, \dots, u_{e-1}, v_0, \dots, v_{d-1})$$

and set

$$u = \sum_{i=0}^{e-1} u_i t^i, \quad v = \sum_{j=0}^{d-1} v_j t^j,$$

where an empty sum is understood to be zero. Since

$$\text{adj}(M)M = \det(M) \text{Id},$$

the first row of this matrix identity says that the coefficient row of $up + vq$ is $(\det(M), 0, \dots, 0)$. Hence $up + vq = \det(M) = \text{Res}_t^{d,e}(p, q)$, proving (i).

For (ii), applying φ to the coefficients of p and q applies φ entrywise to M , and determinants commute with ring homomorphisms. Finally, if $p = 1$ and q is monic of degree e , then the displayed Sylvester matrix has block form

$$\begin{pmatrix} \text{Id}_e & 0 \\ * & L \end{pmatrix},$$

where L is lower triangular with all diagonal entries equal to the leading coefficient $b_e = 1$. Therefore $\det(M) = 1$, proving (iii). $\square$

EXAMPLE 4.2.3. Let

$$f = xy - 1 \quad \text{and} \quad g = x^2 + y^2 - 4$$

be regarded as polynomials in x with coefficients in $\mathbb{k}[y]$. Here $d = 1$ and $e = 2$, so the rows of the Sylvester matrix are the coefficient vectors of f, xf, g in the ordered basis $1, x, x^2$. Thus

$$M = \begin{pmatrix} -1 & y & 0 \\ 0 & -1 & y \\ y^2 - 4 & 0 & 1 \end{pmatrix}, \quad \text{Res}_x(f, g) = \det(M) = y^4 - 4y^2 + 1.$$

In particular, the y -coordinate of every common zero of f and g must satisfy $y^4 - 4y^2 + 1 = 0$.

We can now prove the weak Nullstellensatz without Zariski's lemma.

SECOND PROOF OF THEOREM 4.1.6. Let $I \subsetneq \mathbb{k}[x_1, \dots, x_n]$. We prove that $V(I) \neq \emptyset$ by induction on n . If $I = 0$, then every point is a common zero, so assume that $I \neq 0$.

When $n = 1$, the ideal I is generated by a nonconstant polynomial $h \in \mathbb{k}[x_1]$. Since $\mathbb{k}$ is algebraically closed, h has a root $a_1 \in \mathbb{k}$, and this root is a common zero of I .

Suppose now that $n \geq 2$ and that the result holds for polynomial rings in $n - 1$ variables. Choose a nonzero polynomial in I . By Lemma 4.2.1, after an invertible linear change of coordinates and multiplication by a nonzero scalar, we may assume that I contains a polynomial

$$g = g_0 + g_1 x_n + \cdots + g_{e-1} x_n^{e-1} + x_n^e, \quad g_i \in A := \mathbb{k}[x_1, \dots, x_{n-1}].$$

An invertible change of coordinates preserves properness of an ideal and transports common zeros bijectively, so this reduction loses no generality.

Consider the elimination ideal

$$I_0 = I \cap A.$$

It is proper because I is proper. By the induction hypothesis, there is a point $\alpha = (a_1, \dots, a_{n-1}) \in \mathbb{k}^{n-1}$ such that $h(\alpha) = 0$ for every $h \in I_0$. Let

$$\varepsilon_\alpha : A[x_n] \longrightarrow \mathbb{k}[x_n]$$

be evaluation of the first $n - 1$ variables at α , and set $J = \varepsilon_\alpha(I)$. Since ε_α is surjective, J is an ideal of $\mathbb{k}[x_n]$.

We claim that J is proper. Suppose otherwise. Then there is $f \in I$ such that

$$f(a_1, \dots, a_{n-1}, x_n) = 1.$$

Set $d = \deg_{x_n}(f)$ and recall that $e = \deg_{x_n}(g)$. Using these fixed degree bounds, take the resultant

$$R = \text{Res}_{x_n}^{d,e}(f, g) \in A.$$

By Lemma 4.2.2(i), $R \in (f, g) \subset I$. Hence $R \in I \cap A = I_0$, so $R(\alpha) = 0$. On the other hand, parts (ii) and (iii) of the same lemma give

$$R(\alpha) = \text{Res}_{x_n}^{d,e}(1, g(\alpha, x_n)) = 1,$$

because $g(\alpha, x_n)$ is monic of degree e . This contradiction proves the claim.

The ideal J is nonzero because it contains the monic polynomial $g(\alpha, x_n)$. Since J is proper, we have $J = (h)$ for a nonconstant polynomial $h \in \mathbb{k}[x_n]$. Choose a root $\alpha_n \in \mathbb{k}$ of h . Then every polynomial of J vanishes at α_n . Therefore

$$f(\alpha_1, \dots, \alpha_{n-1}, \alpha_n) = 0 \quad \text{for every } f \in I.$$

Thus $(\alpha_1, \dots, \alpha_n) \in V(I)$, completing the induction. $\square$

The two proofs emphasize complementary ideas. The first proof classifies maximal ideals by studying finitely generated algebras which happen to be fields. The second proof constructs a common zero inductively, using a coordinate change to make projection possible and a resultant to guarantee that the point downstairs lifts. Together with the Rabinowitsch trick, either proof of the weak form yields the strong Nullstellensatz and hence the full algebra–geometry dictionary above.

4.3. Primary decomposition and associated primes

The primary decomposition of monomial ideals in Chapter 2 is particularly explicit and its irredundant decomposition into pure-power ideals (equivalently, irreducible monomial ideals) is unique. We now prove the existence of primary decompositions for arbitrary ideals. In this generality the primary components need not be unique, but their radicals are controlled by the associated primes.

DEFINITION 4.3.1. A proper ideal Q of a ring R is *irreducible* if an equality

$$Q = I \cap J$$

between ideals of R implies that $Q = I$ or $Q = J$. A *primary decomposition* of a proper ideal I is an expression

$$I = Q_1 \cap \dots \cap Q_s$$

in which every Q_i is primary. The decomposition is *minimal* if it is irredundant and the prime ideals $\sqrt{Q_1}, \dots, \sqrt{Q_s}$ are pairwise distinct.

Thus the first condition is the usual notion of an irreducible ideal; when all the ideals involved are monomial, it specializes to the definition used in Chapter 2.

LEMMA 4.3.2. *Every proper ideal in a Noetherian ring is a finite intersection of irreducible ideals.*

PROOF. Suppose otherwise, and let $\mathcal{C}$ be the collection of proper ideals which cannot be written as finite intersections of irreducible ideals. The ascending chain condition gives a maximal member Q of $\mathcal{C}$. The ideal Q is not irreducible, so

$$Q = Q_1 \cap Q_2$$

for ideals Q_1, Q_2 which properly contain Q . By the maximality of Q , each Q_i is a finite intersection of irreducible ideals. Intersecting these two expressions gives such an expression for Q , a contradiction. $\square$

LEMMA 4.3.3. *Every irreducible ideal in a Noetherian ring is primary.*

PROOF. Let Q be irreducible and suppose that $fg \in Q$ but $f \notin Q$. We must show that some power of g belongs to Q . Suppose, to the contrary, that $g^r \notin Q$ for every $r \geq 1$. The ascending chain

$$Q : g \subset Q : g^2 \subset Q : g^3 \subset \dots$$

stabilizes. Choose $m \geq 1$ such that

$$Q : g^m = Q : g^{m+1}.$$

We claim that

$$Q = (Q + (f)) \cap (Q + (g^m)).$$

Only the inclusion from right to left requires proof. Let

$$h = q_1 + af = q_2 + bg^m$$

belong to the intersection, where $q_1, q_2 \in Q$. Since $fg \in Q$, the first expression gives $gh \in Q$. The second expression then gives $bg^{m+1} \in Q$. Thus

$$b \in Q : g^{m+1} = Q : g^m,$$

and so $bg^m \in Q$ and $h \in Q$. The displayed intersection is therefore correct. But $f \notin Q$ and $g^m \notin Q$, so both ideals on the right properly contain Q , contradicting irreducibility. $\square$

THEOREM 4.3.4 (Lasker–Noether). *Every proper ideal of $S = \mathbb{k}[x_1, \dots, x_n]$ has a minimal primary decomposition.*

PROOF. The ring S is Noetherian by the Hilbert basis theorem. Hence Lemma 4.3.2 writes a proper ideal I as a finite intersection of irreducible ideals, and Lemma 4.3.3 shows that all these ideals are primary.

If several components are primary with respect to the same prime P , we replace them by their intersection. This intersection is again P -primary. Indeed, if ab belongs to all the components and a fails to belong to one of them, then $b \in P$. A sufficiently large power of b belongs simultaneously to all the finitely many components. Moreover, the radical of the intersection is

the intersection of the radicals of its components, and hence is P . Finally, remove every redundant component. The resulting primary decomposition is minimal. $\square$

The following intrinsic set records the prime ideals which occur in a minimal primary decomposition.

DEFINITION 4.3.5. Let R be a ring and let M be an R -module. The set of *associated primes* of M is

$$\text{Ass}_R(M) = \{P \subset R \mid P = \text{Ann}_R(m) \text{ for some } m \in M \text{ and } P \text{ is prime}\}.$$

For $M = S/I$, this can be written as

$$\text{Ass}_S(S/I) = \{I : f \mid f \notin I \text{ and } I : f \text{ is prime}\}.$$

LEMMA 4.3.6. If $Q \subset S$ is P -primary, then

$$\text{Ass}_S(S/Q) = \{P\}.$$

PROOF. First suppose that $Q : f$ is prime and $f \notin Q$. If $a \in Q : f$, then $af \in Q$ and primaryness gives $a \in P$. Thus $Q : f \subset P$. Conversely, if $a \in P = \sqrt{Q}$, then $a^N \in Q \subset Q : f$ for some N . Since $Q : f$ is prime, $a \in Q : f$, and hence $Q : f = P$.

It remains to see that P actually occurs. Because S is Noetherian, P is finitely generated; the equality $P = \sqrt{Q}$ therefore gives an integer $N \geq 1$ such that $P^N \subset Q$. Indeed, if $P = (p_1, \dots, p_r)$ and $p_i^{n_i} \in Q$, one may take $N = 1 + \sum_i (n_i - 1)$: every product of N generators contains some $p_i^{n_i}$. Choose the least such N and an element $u \in P^{N-1} \setminus Q$. Then $P \subset Q : u$. If $au \in Q$, primaryness and $u \notin Q$ give $a \in P$. Consequently $Q : u = P$, as required. $\square$

We use two elementary facts about associated primes. If M is a submodule of N , then

$$\text{Ass}_R(M) \subset \text{Ass}_R(N).$$

Also, for a finite direct sum,

$$\text{Ass}_R(M_1 \oplus \dots \oplus M_s) = \text{Ass}_R(M_1) \cup \dots \cup \text{Ass}_R(M_s).$$

For the second assertion, one inclusion follows by placing an element in one summand. For the other, suppose that $P = \text{Ann}(m_1, \dots, m_s) = \bigcap_i \text{Ann}(m_i)$ is prime. If every $\text{Ann}(m_i)$ were strictly larger than P , choose $a_i \in \text{Ann}(m_i) \setminus P$. The product $a_1 \dots a_s$ would belong to P , contradicting primality. Thus $P = \text{Ann}(m_i)$ for some i .

THEOREM 4.3.7. Let

$$I = Q_1 \cap \dots \cap Q_s$$

be a minimal primary decomposition, and set $P_i = \sqrt{Q_i}$. Then

$$\text{Ass}_S(S/I) = \{P_1, \dots, P_s\}.$$

In particular, the number of components and their radicals are independent of the chosen minimal primary decomposition.

PROOF. The diagonal map

$$S/I \longrightarrow \bigoplus_{i=1}^s S/Q_i, \quad \bar{f} \longmapsto (\bar{f}, \dots, \bar{f}),$$

is injective. The preceding facts and [Lemma 4.3.6](#) give

$$\text{Ass}_S(S/I) \subset \{P_1, \dots, P_s\}.$$

Fix i . Irredundancy provides an element

$$g_i \in \bigcap_{j \neq i} Q_j \setminus Q_i.$$

Choose the least $N \geq 1$ for which $P_i^N g_i \subset Q_i$, and choose $h \in P_i^{N-1}$ such that $h g_i \notin Q_i$. Set $u = h g_i$. Then $u \notin I$, while $P_i u \subset I$. If $au \in I$, then $au \in Q_i$ and $u \notin Q_i$; since Q_i is P_i -primary, we obtain $a \in P_i$. It follows that

$$I : u = P_i,$$

so $P_i \in \text{Ass}_S(S/I)$. □

EXAMPLE 4.3.8. Primary components need not be unique. In $\mathbb{k}[x, y]$, let

$$I = (x^2, xy).$$

For every $m \geq 1$,

$$I = (x) \cap (x^2, xy, y^m).$$

The first component is (x) -primary, and the second is (x, y) -primary because it is an Artinian monomial ideal; see [Exercise 2.7](#). These are minimal primary decompositions, and

$$\text{Ass}_S(S/I) = \{(x), (x, y)\}.$$

Thus the component belonging to (x, y) varies with m , whereas its radical does not.

4.4. Primary decomposition and irreducible components

DEFINITION 4.4.1. Let $I \subset S$ be a proper ideal. The primes which are minimal under inclusion in $\text{Ass}_S(S/I)$ are called the *minimal primes* of I . The remaining associated primes are called *embedded primes*.

The terminology is justified by the next result. It also explains precisely which parts of a primary decomposition are visible in the zero set $V(I)$.

THEOREM 4.4.2. *Let $P_1, \dots, P_r$ be the minimal primes of a proper ideal $I \subset S$. Then*

$$\sqrt{I} = P_1 \cap \dots \cap P_r,$$

and

$$V(I) = V(P_1) \cup \cdots \cup V(P_r)$$

is the decomposition of $V(I)$ into its irreducible components.

PROOF. Choose a minimal primary decomposition $I = Q_1 \cap \cdots \cap Q_s$ and write $P_i = \sqrt{Q_i}$. After relabeling, we may assume that $P_1, \dots, P_r$ are the minimal primes. By [Theorem 4.3.7](#), the primes P_i are exactly the associated primes of S/I . Taking radicals gives

$$\sqrt{I} = P_1 \cap \cdots \cap P_s.$$

Every P_i contains a prime which is minimal among the P_j . The nonminimal primes may therefore be omitted from this intersection, which proves the first displayed equality. As a consequence,

$$V(I) = V(\sqrt{I}) = V(P_1) \cup \cdots \cup V(P_r).$$

Each $V(P_i)$ is irreducible by [Theorem 4.1.13](#).

It remains to prove maximality. Let $Y \subset V(I)$ be an irreducible affine variety. Then $I(Y)$ is a prime ideal containing $\sqrt{I}$. Since

$$P_1 P_2 \cdots P_r \subset P_1 \cap \cdots \cap P_r = \sqrt{I} \subset I(Y)$$

and $I(Y)$ is prime, some P_i is contained in $I(Y)$. It follows that $Y \subset V(P_i)$. The distinct minimal primes $P_1, \dots, P_r$ are pairwise incomparable, so the inclusion-reversing dictionary shows that none of the varieties $V(P_i)$ is properly contained in another one. Thus the displayed varieties are precisely the maximal irreducible subvarieties of $V(I)$. $\square$

EXAMPLE 4.4.3. For $I = (x^2, xy) \subset \mathbb{k}[x, y]$, the prime (x) is minimal and (x, y) is embedded. Accordingly,

$$V(I) = V(x)$$

has only one irreducible component. The embedded prime records algebra which the set of points alone cannot see.

EXAMPLE 4.4.4. Here is an elementary instance of the same principle. Let

$$C = V(x^2 + y^2 - 1) \subset \mathbb{A}_{\mathbb{k}}^2$$

and let $L = V(x - 1)$. The quotient by their combined equations is

$$\mathbb{k}[x, y]/(x^2 + y^2 - 1, x - 1) \cong \mathbb{k}[y]/(y^2).$$

The residue class of y is a nonzero nilpotent, and the residue classes of 1 and y form a $\mathbb{k}$ -basis. On the other hand,

$$C \cap L = V(x - 1, y) = \{(1, 0)\},$$

whose coordinate ring is $\mathbb{k}[x, y]/(x - 1, y) \cong \mathbb{k}$. Thus the quotient by the given equations can contain algebraic information not visible in their common zero set.

4.5. Exercises

EXERCISE 4.1. Let $\mathbf{a} = (a_1, \dots, a_n) \in \mathbb{A}_{\mathbb{k}}^n$. Prove directly, by induction on n , that every $f \in S$ can be written as

$$f = f(\mathbf{a}) + \sum_{i=1}^n (x_i - a_i)g_i$$

for suitable $g_1, \dots, g_n \in S$.

EXERCISE 4.2. Let $I \subset S$ be an ideal and $f \in S$. Use the Rabinowitsch trick to prove that

$$f \in \sqrt{I} \iff 1 \in (I, 1 - yf) \subset S[y].$$

EXERCISE 4.3. Determine the varieties of the following ideals and compute their radicals:

- (i) $(x^3, x^2y, y^4) \subset \mathbb{k}[x, y]$;
- (ii) $(xy, xz) \subset \mathbb{k}[x, y, z]$;
- (iii) $(x^2 - 1, y - x) \subset \mathbb{k}[x, y]$, assuming $\text{char}(\mathbb{k}) \neq 2$.

EXERCISE 4.4. Let $X, Y \subset \mathbb{A}_{\mathbb{k}}^n$ be affine varieties. Prove that

$$I(X \cup Y) = I(X) \cap I(Y)$$

and use the Nullstellensatz to recover the identity $V(I \cap J) = V(I) \cup V(J)$ for ideals $I, J \subset S$.

EXERCISE 4.5. Prove that the Zariski closure of a subset $Y \subset \mathbb{A}_{\mathbb{k}}^n$ is $V(I(Y))$. Show in particular that Y is Zariski dense in $\mathbb{A}_{\mathbb{k}}^n$ if and only if $I(Y) = 0$.

EXERCISE 4.6. Let $I \subset S$ be a radical ideal. Suppose that

$$I = \mathfrak{p}_1 \cap \dots \cap \mathfrak{p}_s$$

is an irredundant intersection of prime ideals such that no $\mathfrak{p}_i$ contains another $\mathfrak{p}_j$. Prove that

$$V(I) = V(\mathfrak{p}_1) \cup \dots \cup V(\mathfrak{p}_s)$$

is the decomposition of $V(I)$ into its irreducible components.

EXERCISE 4.7. Let $Q \subset S$ be a P -primary ideal and let $f \notin Q$. Show that $Q : f$ is P -primary. Deduce that if $Q : f$ is prime, then $Q : f = P$.

EXERCISE 4.8. Verify directly that, for every $m \geq 1$,

$$(x^2, xy) = (x) \cap (x^2, xy, y^m) \subset \mathbb{k}[x, y].$$

Show that the decomposition is minimal and determine the associated primes.

CHAPTER 5

Elimination and projective methods

Gröbner bases become especially useful when the variables are divided into groups. With a suitable monomial order, one can remove one group of variables and retain all polynomial consequences involving the others. This simple idea is called *elimination*. It gives algorithms for projection, implicitization, intersections, radical membership, and saturations.

Projective geometry is a natural companion to elimination. It records directions at infinity and treats homogeneous equations without choosing a preferred affine chart. For homogeneous quotients, the Hilbert series records algebraic dimension and degree; over an algebraically closed field, these invariants give the dimension and degree of projective varieties through their homogeneous coordinate rings.

5.1. Projective space and homogeneous equations

We begin with the set-theoretic definition.

DEFINITION 5.1.1. Let $\mathbb{k}$ be a field. On $\mathbb{k}^{n+1} \setminus \{0\}$, consider the equivalence relation

$$(a_0, \dots, a_n) \sim (b_0, \dots, b_n)$$

if there exists $\lambda \in \mathbb{k}^\times$ such that $b_i = \lambda a_i$ for every i . The set of equivalence classes is the *projective n -space over $\mathbb{k}$* , denoted by $\mathbb{P}_{\mathbb{k}}^n$. The class of $(a_0, \dots, a_n)$ is written

$$[a_0 : \dots : a_n],$$

and these are called *homogeneous coordinates*.

At least one homogeneous coordinate is nonzero, but there is usually no preferred coordinate to normalize. For $0 \leq i \leq n$, set

$$U_i = \{[a_0 : \dots : a_n] \in \mathbb{P}_{\mathbb{k}}^n \mid a_i \neq 0\}.$$

There is a bijection

$$U_i \longrightarrow \mathbb{A}_{\mathbb{k}}^n, \quad [a_0 : \dots : a_n] \longmapsto \left(\frac{a_0}{a_i}, \dots, \frac{\widehat{a_i}}{a_i}, \dots, \frac{a_n}{a_i} \right).$$

Its inverse inserts 1 in the i -th position. Thus projective space is covered by $n + 1$ affine charts.

The chart U_0 identifies $\mathbb{A}_{\mathbb{k}}^n$ with the subset

$$\{[1 : a_1 : \dots : a_n]\} \subset \mathbb{P}_{\mathbb{k}}^n.$$

For $n \geq 1$, its complement

$$H_\infty = V_+(x_0) = \{[0 : a_1 : \cdots : a_n]\} \cong \mathbb{P}_{\mathbb{k}}^{n-1}$$

is called the *hyperplane at infinity* relative to this chart. No hyperplane is intrinsically the hyperplane at infinity; this terminology depends on the chosen affine chart.

EXAMPLE 5.1.2. The projective line is

$$\mathbb{P}_{\mathbb{k}}^1 = U_0 \cup \{[0 : 1]\}.$$

The chart U_0 consists of the points $[1 : t]$, with $t \in \mathbb{k}$, and the remaining point is often denoted by ∞ . This notation describes a set of points; it does not by itself identify the projective line with an affine algebraic variety.

There is a useful topological picture over the real numbers.

EXAMPLE 5.1.3. The formula

$$\Phi : \mathbb{P}_{\mathbb{R}}^1 \longrightarrow \{(a, b) \in \mathbb{R}^2 \mid a^2 + b^2 = 1\}, \quad [u : v] \longmapsto \left(\frac{u^2 - v^2}{u^2 + v^2}, \frac{2uv}{u^2 + v^2} \right)$$

is well-defined on real points because $u^2 + v^2 \neq 0$ for $(u, v) \neq (0, 0)$. Writing $[u : v] = [\cos \theta : \sin \theta]$, it sends the class of θ modulo π to the point with angle 2θ modulo 2π . It is therefore a homeomorphism between the real projective line and a circle. The double angle is essential: choosing a normalized representative (u, v) would leave a sign ambiguity. This is a statement about real points and their topology, not an assertion that the circle is an affine algebraic model of $\mathbb{P}_{\mathbb{R}}^1$.

Let

$$S = \mathbb{k}[x_0, \dots, x_n]$$

with its standard grading. If $f \in S$ is homogeneous of degree e , then

$$f(\lambda a_0, \dots, \lambda a_n) = \lambda^e f(a_0, \dots, a_n).$$

Consequently, the vanishing of f depends only on the projective point $[a_0 : \cdots : a_n]$.

DEFINITION 5.1.4. For a homogeneous ideal $I \subset S$, its *projective zero locus* is

$$V_+(I) = \{[a_0 : \cdots : a_n] \in \mathbb{P}_{\mathbb{k}}^n \mid f(a_0, \dots, a_n) = 0 \text{ for every } f \in I\}.$$

For homogeneous polynomials $f_1, \dots, f_s$, we also write $V_+(f_1, \dots, f_s)$ for $V_+((f_1, \dots, f_s))$.

Homogeneity guarantees that the vanishing condition is independent of the chosen representative. For example, $x_0 - 1$ vanishes at the representative $(1, a_1, \dots, a_n)$ but generally not at a nontrivial scalar multiple of that representative. Thus a nonhomogeneous polynomial need not give a well-defined equation on projective space.

PROPOSITION 5.1.5. *Let $I, J \subset S$ be homogeneous ideals. Then:*

- (i) $V_+(I + J) = V_+(I) \cap V_+(J)$;
- (ii) $V_+(I \cap J) = V_+(I) \cup V_+(J)$;
- (iii) *if $I \subseteq J$, then $V_+(I) \supseteq V_+(J)$.*

PROOF. The first and third assertions follow directly from the definition. For the second, the inclusion from right to left is clear. If $p \notin V_+(I) \cup V_+(J)$, choose homogeneous elements $f \in I$ and $g \in J$ which do not vanish at p . Then $fg \in IJ \subseteq I \cap J$ and $(fg)(p) \neq 0$. Hence $p \notin V_+(I \cap J)$. $\square$

For a subset $X \subset \mathbb{P}_{\mathbb{k}}^n$, define its *projective vanishing ideal* by

$$I_+(X) = \bigoplus_{d \geq 0} \{f \in S_d \mid f(a_0, \dots, a_n) = 0 \text{ for every } [a_0 : \dots : a_n] \in X\}.$$

This is a homogeneous ideal. The definition is independent of the chosen homogeneous coordinates because a form of degree d satisfies $f(\lambda a) = \lambda^d f(a)$.

THEOREM 5.1.6 (Projective Nullstellensatz). *Assume that $\mathbb{k}$ is algebraically closed and let $I \subset S$ be a homogeneous ideal. Write*

$$\mathfrak{m} = (x_0, \dots, x_n) \subset S,$$

for the irrelevant ideal. Then:

- (i) $V_+(I) = \emptyset$ *if and only if* $\mathfrak{m} \subseteq \sqrt{I}$. *Equivalently, $V_+(I) = \emptyset$ if and only if $\mathfrak{m}^N \subseteq I$ for some $N \geq 1$.*
- (ii) *If $V_+(I) \neq \emptyset$, then*

$$I_+(V_+(I)) = \sqrt{I}.$$

PROOF. First suppose that I is proper. Since I is homogeneous, its affine zero locus is the cone

$$V(I) = \{0\} \cup \{a \in \mathbb{k}^{n+1} \setminus \{0\} \mid [a] \in V_+(I)\}.$$

Consequently, $V_+(I) = \emptyset$ if and only if $V(I) = \{0\}$. Theorem 4.1.8 gives

$$V(I) = \{0\} \iff \sqrt{I} = I(\{0\}) = (x_0, \dots, x_n) = \mathfrak{m}.$$

If $I = S$, both $V_+(I) = \emptyset$ and $\mathfrak{m} \subseteq \sqrt{I}$ hold, so the first equivalence follows in all cases.

If $\mathfrak{m} \subseteq \sqrt{I}$, then for every i some power $x_i^{e_i}$ belongs to I . For

$$N = 1 + \sum_{i=0}^n (e_i - 1),$$

every monomial of degree N is divisible by one of these powers, and hence $\mathfrak{m}^N \subseteq I$. The converse is immediate: if $\mathfrak{m}^N \subseteq I$, then $x_i^N \in I$ for every i , so $\mathfrak{m} \subseteq \sqrt{I}$. This proves (i).

Now assume that $V_+(I)$ is nonempty. Every element of $\sqrt{I}$ vanishes on $V(I)$. Writing such an element as a sum of homogeneous components and evaluating at λa , with $\lambda \in \mathbb{k}^\times$, shows that

each component vanishes on every nonzero point of $V(I)$: the field $\mathbb{k}$ is infinite, so a polynomial in λ which vanishes for every $\lambda \neq 0$ has all coefficients zero. Thus

$$\sqrt{I} \subseteq I_+(V_+(I)).$$

Conversely, an element of $I_+(V_+(I))$ vanishes on every nonzero point of $V(I)$. Its constant homogeneous component is zero because $V_+(I)$ is nonempty, so it also vanishes at the origin. It therefore vanishes on all of $V(I)$, and [Theorem 4.1.8](#) yields

$$I_+(V_+(I)) \subseteq I(V(I)) = \sqrt{I}.$$

This proves (ii). □

The nonempty hypothesis in part (ii) is necessary. Indeed,

$$V_+(\mathfrak{m}) = \emptyset, \quad I_+(\emptyset) = S, \quad \sqrt{\mathfrak{m}} = \mathfrak{m}.$$

Thus the irrelevant ideal $\mathfrak{m}$, rather than only the unit ideal, accounts for an empty projective zero locus.

REMARK 5.1.7. Over a field that is not algebraically closed, $V_+(I)$ records only the $\mathbb{k}$ -rational projective zeros, so its point set need not reflect the Hilbert series of S/I . We therefore interpret dimension and degree geometrically only for projective varieties over algebraically closed fields.

EXAMPLE 5.1.8. A nonzero homogeneous polynomial f of positive degree e defines a projective zero locus

$$V_+(f) \subset \mathbb{P}_{\mathbb{k}}^n.$$

Its intersection with U_0 is obtained by setting $x_0 = 1$:

$$f(1, x_1, \dots, x_n) = 0.$$

Its points at infinity relative to this chart are obtained by setting $x_0 = 0$.

5.2. Dimension and degree of projective varieties

The dimension and degree of a homogeneous quotient were defined algebraically in [Section 3.3](#). We now apply those invariants to projective varieties through their homogeneous coordinate rings.

DEFINITION 5.2.1. Assume that $\mathbb{k}$ is algebraically closed. A nonempty subset $X \subset \mathbb{P}_{\mathbb{k}}^n$ is a *projective variety* if $X = V_+(I)$ for some homogeneous ideal $I \subset S = \mathbb{k}[x_0, \dots, x_n]$. Its *homogeneous coordinate ring*, relative to this embedding, is

$$R_X := \mathbb{k}[X] := S/I_+(X).$$

By [Theorem 5.1.6](#), the ideal $I_+(X)$ is proper, homogeneous, and radical. Moreover, $\dim(R_X) \geq 1$: otherwise R_X would be finite dimensional over $\mathbb{k}$, so $\mathfrak{m}^N \subseteq I_+(X)$ for some N , and part (i) of

[Theorem 5.1.6](#) would force X to be empty. We define the *dimension* and *degree* of X by

$$\dim(X) := \dim(R_X) - 1, \quad \deg(X) := \deg(R_X).$$

Equivalently, if

$$\text{Hilb}_{R_X}(t) = \frac{Q_X(t)}{(1-t)^d}, \quad Q_X(1) > 0,$$

then

$$\dim(X) = d - 1, \quad \deg(X) = Q_X(1).$$

The degree is the degree for the given embedding $X \subset \mathbb{P}_{\mathbb{k}}^n$. If $X = V_+(I)$ for a homogeneous ideal I , then [Theorem 5.1.6](#) gives

$$I_+(X) = \sqrt{I}, \quad R_X \cong S/\sqrt{I}.$$

REMARK 5.2.2. [Theorem 3.3.1](#) is an algebraic statement about a graded quotient S/I . Over an arbitrary field, it makes no assertion about the dimension or degree of the set of rational points $V_+(I)$. When $\mathbb{k}$ is algebraically closed and $V_+(I)$ is nonempty, the projective variety uses the homogeneous coordinate ring $S/\sqrt{I}$. The quotient S/I may contain nilpotent elements, so its degree can differ from the degree of the projective variety defined by the same equations.

EXAMPLE 5.2.3. Assume that $\mathbb{k}$ is algebraically closed. The homogeneous coordinate ring of $\mathbb{P}_{\mathbb{k}}^n$ is $S = \mathbb{k}[x_0, \dots, x_n]$, and

$$\text{Hilb}_S(t) = \frac{1}{(1-t)^{n+1}}.$$

Therefore $\dim(\mathbb{P}_{\mathbb{k}}^n) = n$ and $\deg(\mathbb{P}_{\mathbb{k}}^n) = 1$.

COROLLARY 5.2.4 (Projective hypersurfaces). *Assume that $\mathbb{k}$ is algebraically closed. Let $0 \neq f \in S$ be homogeneous of positive degree e . If $V_+(f)$ is nonempty and f is squarefree, meaning that it has no repeated irreducible factor, then the projective variety $V_+(f)$ has dimension $n - 1$ and degree e .*

PROOF. Since f is squarefree, (f) is radical, so $S/(f)$ is the homogeneous coordinate ring of $V_+(f)$. Applying [Proposition 3.3.4](#) to the polynomial ring in $n + 1$ variables gives $\dim(S/(f)) = n$ and $\deg(S/(f)) = e$. The result follows from the definition of the invariants of a projective variety. $\square$

REMARK 5.2.5. If $\mathbb{k}$ is algebraically closed, $V_+(f)$ is nonempty, and f has repeated irreducible factors, the homogeneous coordinate ring of $V_+(f)$ is instead $S/\sqrt{(f)}$. Thus the degree of the variety is the degree of the product of the distinct irreducible factors of f , while the degree of the quotient $S/(f)$ is $\deg(f)$.

EXAMPLE 5.2.6 (A real conic with no real points). Set $S = \mathbb{R}[x, y, z]$ and let

$$I = (x^2 + y^2 + z^2) \subset S.$$

There is no nonzero real triple (a, b, c) satisfying $a^2 + b^2 + c^2 = 0$. Hence

$$V_+(I) = \emptyset \subset \mathbb{P}_{\mathbb{R}}^2.$$

This does not contradict [Theorem 5.1.6](#), because $\mathbb{R}$ is not algebraically closed. By [Proposition 3.3.4](#),

$$\text{Hilb}_{S/I}(t) = \frac{1-t^2}{(1-t)^3} = \frac{1+t}{(1-t)^2}.$$

Therefore the homogeneous quotient S/I has dimension 2 and degree 2. These invariants do not belong to the empty real point set. After extending scalars to $\mathbb{C}$, the projective zero locus contains, for example, $[1 : i : 0]$. Since $x^2 + y^2 + z^2$ is squarefree over $\mathbb{C}$, the resulting complex projective variety has dimension 1 and degree 2.

5.3. Elimination ideals

We return to the affine polynomial ring

$$S = \mathbb{k}[x_1, \dots, x_n].$$

For $0 \leq \ell \leq n$, set

$$S_\ell = \mathbb{k}[x_{\ell+1}, \dots, x_n],$$

with the conventions $S_0 = S$ and $S_n = \mathbb{k}$.

DEFINITION 5.3.1. For an ideal $I \subset S$, the ideal

$$I_\ell = I \cap S_\ell$$

is called the ℓ -th *elimination ideal* of I . It consists of all polynomial consequences of I which do not involve $x_1, \dots, x_\ell$.

The lexicographic order is particularly effective because any monomial involving an eliminated variable is larger than every monomial involving only the remaining variables.

THEOREM 5.3.2 (Elimination theorem). *Use lexicographic order*

$$x_1 > x_2 > \dots > x_n.$$

If $\mathcal{G}$ is a Gröbner basis of $I \subset S$, then

$$\mathcal{G}_\ell = \mathcal{G} \cap S_\ell$$

is a Gröbner basis of the elimination ideal $I_\ell = I \cap S_\ell$ with respect to the induced lexicographic order on S_ℓ .

PROOF. Let $0 \neq f \in I_\ell$. Since $\mathcal{G}$ is a Gröbner basis of I , there is $g \in \mathcal{G}$ such that

$$\text{in}(g) \mid \text{in}(f).$$

The initial term of f belongs to S_ℓ , so $\text{in}(g)$ involves none of the variables $x_1, \dots, x_\ell$. We claim that $g \in S_\ell$. Indeed, in lexicographic order every monomial which involves one of $x_1, \dots, x_\ell$ is larger than every monomial in S_ℓ whose earlier exponents are all zero. If such a monomial occurred in g , it would be larger than the monomial underlying $\text{in}(g)$, a contradiction. Thus $g \in \mathcal{G}_\ell$.

We have proved that the initial term of every nonzero element of I_ℓ is divisible by the initial term of an element of $\mathcal{G}_\ell$. The divisibility criterion in [Proposition 3.2.7](#) now shows that $\mathcal{G}_\ell$ is a Gröbner basis of I_ℓ . $\square$

REMARK 5.3.3. The proof uses only the following property of the order: every monomial involving an eliminated variable is larger than every monomial involving only the retained variables. Orders with this property are called *elimination orders*. Lexicographic order is the basic example.

EXAMPLE 5.3.4 (A complete elimination computation). Let

$$I = (x + y - z, x^2 - y) \subset \mathbb{k}[x, y, z]$$

and use lexicographic order $x > y > z$. Set

$$\begin{aligned} g_1 &= x + y - z, \\ g_2 &= y^2 - 2yz + z^2 - y. \end{aligned}$$

The identity

$$g_2 = (x^2 - y) - (x - y + z)(x + y - z)$$

shows that $g_2 \in I$, while the same identity solved for $x^2 - y$ shows that $I = (g_1, g_2)$. Their initial terms are

$$\text{in}(g_1) = x, \quad \text{in}(g_2) = y^2.$$

They are relatively prime, so the relatively prime criterion proves that $\mathcal{G} = \{g_1, g_2\}$ is a Gröbner basis. The elimination theorem gives

$$I \cap \mathbb{k}[y, z] = (y^2 - 2yz + z^2 - y).$$

This calculation also has a geometric interpretation. The equations of I give $x = z - y$ and $x^2 = y$. Eliminating x therefore produces

$$(z - y)^2 - y = 0.$$

Conversely, every pair (b, c) satisfying this equation extends to the point $(c - b, b, c) \in V(I)$. In this example, the projection of $V(I)$ to the (y, z) -plane is exactly the hypersurface defined by the first elimination ideal.

For an arbitrary ideal, projection always gives one inclusion.

PROPOSITION 5.3.5. *Let $\pi : \mathbb{A}_{\mathbb{k}}^n \rightarrow \mathbb{A}_{\mathbb{k}}^{n-\ell}$ forget the first ℓ coordinates. Then*

$$\pi(V(I)) \subseteq V(I_\ell).$$

PROOF. If $\alpha \in V(I)$ and $f \in I_\ell$, then f involves only the last $n - \ell$ coordinates and

$$f(\pi(\alpha)) = f(\alpha) = 0.$$

Thus $\pi(\alpha) \in V(I_\ell)$. □

Equality need not hold: a projection of an algebraic set need not be closed.

PROPOSITION 5.3.6 (Closure of a projection). *Assume that $\mathbb{k}$ is algebraically closed. Then*

$$\overline{\pi(V(I))} = V(I_\ell),$$

where the bar denotes Zariski closure in $\mathbb{A}_{\mathbb{k}}^{n-\ell}$.

PROOF. A polynomial in S_ℓ vanishes on $\pi(V(I))$ precisely when, regarded as an element of S , it vanishes on $V(I)$. Therefore

$$\begin{aligned} I(\pi(V(I))) &= I(V(I)) \cap S_\ell \\ &= \sqrt{I} \cap S_\ell \\ &= \sqrt{I \cap S_\ell} = \sqrt{I_\ell}. \end{aligned}$$

The second equality is Hilbert's Nullstellensatz. For the third, an element $f \in S_\ell$ has a power in I if and only if it has a power in $I \cap S_\ell$. A set and its Zariski closure have the same vanishing ideal, so the algebra–geometry dictionary gives

$$\overline{\pi(V(I))} = V(I(\pi(V(I)))) = V(\sqrt{I_\ell}) = V(I_\ell).$$

□

For parametrized sets, the same closure statement also follows directly from the kernel computation in the next section.

5.4. Implicitization and rational parametrization

A parametrization describes points by auxiliary parameters. Implicitization removes those parameters and finds all polynomial equations satisfied by the image. The elimination theorem turns this passage into a Gröbner basis computation.

5.4.1. Polynomial parametrizations. Let

$$p_1, \dots, p_n \in \mathbb{k}[t_1, \dots, t_m]$$

and consider the polynomial map

$$\varphi : \mathbb{A}_{\mathbb{k}}^m \longrightarrow \mathbb{A}_{\mathbb{k}}^n, \quad \alpha \longmapsto (p_1(\alpha), \dots, p_n(\alpha)).$$

It induces the substitution homomorphism

$$\varphi^* : \mathbb{k}[x_1, \dots, x_n] \longrightarrow \mathbb{k}[t_1, \dots, t_m], \quad x_i \longmapsto p_i.$$

PROPOSITION 5.4.1 (Implicitization by elimination). *In the polynomial ring*

$$T = \mathbb{k}[t_1, \dots, t_m, x_1, \dots, x_n],$$

let

$$J = (x_1 - p_1, \dots, x_n - p_n).$$

Then

$$J \cap \mathbb{k}[x_1, \dots, x_n] = \text{Ker}(\varphi^*).$$

Thus a Gröbner basis of J for a lexicographic order which eliminates $t_1, \dots, t_m$ gives generators for all polynomial relations among $p_1, \dots, p_n$.

PROOF. Define

$$\Psi : T \longrightarrow \mathbb{k}[t_1, \dots, t_m]$$

by fixing every t_j and sending x_i to p_i . Repeatedly replacing x_i by p_i shows that every polynomial $F \in T$ has an expression

$$F = \sum_{i=1}^n A_i(x_i - p_i) + r(t_1, \dots, t_m).$$

The coefficients A_i need not be unique, but the remainder is unique and equals $\Psi(F)$. Hence $\text{Ker}(\Psi) = J$. The restriction of Ψ to $\mathbb{k}[x_1, \dots, x_n]$ is φ^* , so taking the contraction to this subring gives the claimed equality. $\square$

If $\mathbb{k}$ is infinite, a polynomial in the parameters which vanishes at every point of $\mathbb{k}^m$ is the zero polynomial. It follows that $\text{Ker}(\varphi^*)$ is the ideal of all polynomials which vanish on the image of φ . Over an algebraically closed field, its zero locus is the Zariski closure of that image. The word *closure* is important: the image of a polynomial map need not itself be closed.

EXAMPLE 5.4.2 (The parabola). Consider

$$\varphi : \mathbb{A}_{\mathbb{k}}^1 \longrightarrow \mathbb{A}_{\mathbb{k}}^2, \quad t \longmapsto (t, t^2).$$

In $\mathbb{k}[t, x, y]$, the graph ideal is

$$J = (x - t, y - t^2).$$

For lexicographic order $t > x > y$, the set

$$\mathcal{G} = \{t - x, x^2 - y\}$$

generates J and is a Gröbner basis: its initial terms t and x^2 are relatively prime. The elimination theorem gives

$$J \cap \mathbb{k}[x, y] = (x^2 - y).$$

Thus the implicit equation is $y = x^2$.

5.4.2. Rational parametrizations. Suppose now that

$$p_1, \dots, p_n, q \in \mathbb{k}[t_1, \dots, t_m], \quad q \neq 0,$$

and consider the rational map on the set where q does not vanish:

$$\alpha \mapsto \left(\frac{p_1(\alpha)}{q(\alpha)}, \dots, \frac{p_n(\alpha)}{q(\alpha)} \right).$$

Clearing denominators alone may introduce points coming from the locus $q = 0$. An auxiliary variable prevents this by forcing q to be invertible.

PROPOSITION 5.4.3 (Rational implicitization). *In*

$$T = \mathbb{k}[u, t_1, \dots, t_m, x_1, \dots, x_n],$$

let

$$J = (qx_1 - p_1, \dots, qx_n - p_n, 1 - uq).$$

Then

$$J \cap \mathbb{k}[x_1, \dots, x_n] = \text{Ker}(\mathbb{k}[x_1, \dots, x_n] \longrightarrow \mathbb{k}[t_1, \dots, t_m, 1/q]),$$

where $x_i \mapsto p_i/q$.

PROOF. The relation $1 - uq$ makes q invertible, with inverse u . Therefore

$$T/J \cong \mathbb{k}[t_1, \dots, t_m, 1/q],$$

under the map sending x_i to p_i/q and u to $1/q$. The kernel of the quotient map is J . Restricting that map to $\mathbb{k}[x_1, \dots, x_n]$ proves the contraction formula. $\square$

To compute the contraction, one uses a lexicographic order with

$$u > t_1 > \dots > t_m > x_1 > \dots > x_n$$

and keeps the basis elements involving only the x -variables. The equation $1 - uq$ is the same localization device which will reappear in the computation of radical membership and saturations.

EXAMPLE 5.4.4 (A shifted circle). Consider the real circle

$$C' = V(x^2 + (y - 1)^2 - 1) = V(x^2 + y^2 - 2y) \subset \mathbb{R}^2.$$

The formulas

$$\varphi : \mathbb{R} \longrightarrow C', \quad t \mapsto \left(\frac{4t}{t^2 + 4}, \frac{2t^2}{t^2 + 4} \right)$$

define a rational parametrization. Indeed, if $D = t^2 + 4$, then

$$\left(\frac{4t}{D} \right)^2 + \left(\frac{2t^2}{D} \right)^2 - 2 \left(\frac{2t^2}{D} \right) = 0.$$

Since $D > 0$ for every real t , the formulas are defined on all of $\mathbb{R}$.

The parametrization covers every real point of C' except $(0, 2)$. To see this, let $(a, b) \in C'$ with $b \neq 2$ and put

$$t = \frac{2a}{2-b}.$$

The relation $a^2 = b(2-b)$ gives

$$t^2 + 4 = \frac{8}{2-b},$$

and substitution recovers a and b from the displayed parametrization. If $b = 2$, the circle equation forces $a = 0$. For finite t one has

$$2 - \frac{2t^2}{t^2 + 4} = \frac{8}{t^2 + 4} > 0,$$

so $(0, 2)$ is never attained, although

$$\lim_{|t| \rightarrow \infty} \varphi(t) = (0, 2).$$

Thus the missing point is recovered as a topological limit of real points.

For implicitization, one may use the ideal

$$((t^2 + 4)x - 4t, (t^2 + 4)y - 2t^2, 1 - u(t^2 + 4)) \subset \mathbb{R}[u, t, x, y].$$

Eliminating u and t yields the relation $x^2 + y^2 - 2y$.

5.4.3. A filtered Hilbert function. Hilbert functions of homogeneous quotients count their graded pieces. For a nonhomogeneous ideal, one must instead use the total-degree filtration.

EXAMPLE 5.4.5. Let

$$A = \mathbb{R}[x, y]/(x^2 + y^2 - 1)$$

and let $F_m A$ be the image of the polynomials of total degree at most m . Use a degree-compatible order for which $x^2 > y^2$, for instance graded lexicographic order with $x > y$. The single polynomial $x^2 + y^2 - 1$ is a Gröbner basis of its principal ideal and

$$\text{in}((x^2 + y^2 - 1)) = (x^2).$$

By the degree-bounded Macaulay equality in [Theorem 3.4.1](#), the standard monomials of degree at most m form a basis of $F_m A$. They are

$$1, y, \dots, y^m, \quad x, xy, \dots, xy^{m-1}.$$

Here the second list is empty when $m = 0$. Hence

$$H(m) := \dim_{\mathbb{R}} F_m A = 2m + 1$$

for every $m \geq 0$. This is precisely why the nonhomogeneous equation must be treated through a filtration rather than by writing $(S/I)_m$.

5.5. Ideal operations by elimination

We now use one auxiliary variable to compute operations on ideals in $S = \mathbb{k}[x_1, \dots, x_n]$. In each construction, a lexicographic order with the auxiliary variable larger than all the x_i turns the required contraction into a Gröbner basis computation.

5.5.1. Intersections.

THEOREM 5.5.1 (Intersection by elimination). *Let $I, J \subset S$ be ideals and let t be a new variable. Then*

$$I \cap J = (tI + (1-t)J) \cap S$$

as ideals of $S[t]$ contracted to S .

PROOF. If $f \in I \cap J$, then

$$f = tf + (1-t)f \in tI + (1-t)J.$$

This proves the inclusion from left to right.

Conversely, let $f \in (tI + (1-t)J) \cap S$. Evaluate an expression for f at $t = 1$. The terms in $(1-t)J$ vanish and the remaining terms belong to I , so $f \in I$. Evaluating at $t = 0$ similarly gives $f \in J$. Hence $f \in I \cap J$. $\square$

EXAMPLE 5.5.2. Let $I = (x)$ and $J = (y)$ in $\mathbb{k}[x, y]$. Then

$$tI + (1-t)J = (tx, (1-t)y) \subset \mathbb{k}[t, x, y].$$

The theorem gives

$$(tx, (1-t)y) \cap \mathbb{k}[x, y] = (xy),$$

in agreement with $(x) \cap (y) = (xy)$. The point of the formula is not this easy example, but that the same computation works for arbitrary finitely generated ideals.

For more than two ideals, one may intersect successively. After each step, a reduced Gröbner basis removes generators which arose only from the auxiliary computation.

5.5.2. Radical membership. Recall from the Rabinowitsch argument in the proof of [Theorem 4.1.8](#) that, for $f \in S$,

$$f \in \sqrt{I} \iff (I, 1 - tf) = S[t].$$

This is also an effective test: compute a Gröbner basis of $(I, 1 - tf)$ and check whether its reduced basis is $\{1\}$. We do not repeat the proof here. The forward implication there is witnessed by the finite geometric-series identity

$$1 = (1 - tf)(1 + tf + \dots + (tf)^{N-1}) + t^N f^N$$

whenever $f^N \in I$; the reverse implication is the localization step $t = f^{-1}$ from the same proof.

EXAMPLE 5.5.3. Let $I = (x^2, y) \subset \mathbb{k}[x, y]$ and take $f = x$. Since $x^2 \in I$, the identity with $N = 2$ reads

$$1 = (1 - tx)(1 + tx) + t^2 x^2.$$

Thus $(I, 1 - tx) = \mathbb{k}[t, x, y]$, which detects $x \in \sqrt{I}$ even though $x \notin I$.

5.5.3. Saturation. Radical membership asks whether some power of one element belongs to an ideal. Saturation keeps all elements which enter the ideal after multiplication by some power of a fixed element.

DEFINITION 5.5.4. For an ideal $I \subset S$ and $f \in S$, the *saturation of I by f* is

$$I : f^\infty = \bigcup_{N \geq 0} (I : f^N) = \{g \in S \mid f^N g \in I \text{ for some } N \geq 0\}.$$

The union is an ideal because the ascending chain

$$I = I : f^0 \subseteq I : f \subseteq I : f^2 \subseteq I : f^3 \subseteq \dots$$

eventually stabilizes by Hilbert basis theorem.

THEOREM 5.5.5 (Saturation by elimination). *Let t be a new variable. Then*

$$I : f^\infty = (I, 1 - tf) \cap S.$$

PROOF. Suppose that $g \in I : f^\infty$, and choose $N \geq 0$ such that $f^N g \in I$. For $N = 0$ the assertion is clear. For $N \geq 1$, the finite geometric-series identity gives

$$g = (1 - tf)g(1 + tf + \dots + (tf)^{N-1}) + t^N f^N g.$$

The first summand belongs to $(1 - tf)$ and the second belongs to $IS[t]$. Thus $g \in (I, 1 - tf) \cap S$.

For the reverse inclusion, pass to the localization S_f and substitute $t = f^{-1}$. The relation $1 - tf$ becomes zero. If $g \in (I, 1 - tf) \cap S$, its image in S_f therefore belongs to IS_f . By the defining property of localization, there exists $N \geq 0$ such that $f^N g \in I$. Hence $g \in I : f^\infty$. $\square$

Equivalently, the quotient map

$$S[t] \longrightarrow S_f, \quad t \longmapsto f^{-1},$$

has kernel $(1 - tf)$, and contraction from the localized ideal IS_f gives $I : f^\infty$. Thus rational implicitization and saturation use exactly the same algebraic device.

EXAMPLE 5.5.6. Let

$$I = (x^2, xy) \subset \mathbb{k}[x, y]$$

and saturate by y . Since $yx \in I$, we have $x \in I : y^\infty$, and hence $(x) \subseteq I : y^\infty$. Conversely, if $y^N g \in I$, monomial membership in $I = (x^2, xy)$ shows that every monomial of g is divisible by x . Thus $g \in (x)$, and

$$I : y^\infty = (x).$$

The elimination formula computes the same answer as

$$(x^2, xy, 1 - ty) \cap \mathbb{k}[x, y] = (x).$$

REMARK 5.5.7 (Radical membership versus saturation). The two questions are related but different:

$$f \in \sqrt{I} \iff f^N \in I \text{ for some } N,$$

whereas

$$g \in I : f^\infty \iff f^N g \in I \text{ for some } N.$$

In particular,

$$f \in \sqrt{I} \iff 1 \in I : f^\infty \iff I : f^\infty = S.$$

A single auxiliary ideal $(I, 1 - tf)$ detects all three statements, but the element whose membership is tested changes.

5.5.4. The computational pattern. The constructions above can be summarized as follows. Introduce one auxiliary variable, form an ideal in a larger polynomial ring, compute a Gröbner basis in an elimination order, and retain the polynomials involving only the original variables. The relevant extensions are

operation in S	ideal in $S[t]$
$I \cap J$	$tI + (1 - t)J$
$I : f^\infty$	$(I, 1 - tf).$

For radical membership, one asks whether the second extended ideal is the unit ideal. For rational implicitization, the same relation $1 - tf$ forces the denominator to be invertible before the parameters are eliminated.

This pattern also explains why ordinary substitution is not always enough. Substitution describes a quotient or a localization; elimination computes the kernel seen by the variables that remain.

5.6. Exercises

EXERCISE 5.1. Prove that the sets U_i cover $\mathbb{P}_{\mathbb{k}}^n$. Write explicitly the change of coordinates from the chart U_i to the chart U_j on their intersection.

EXERCISE 5.2. Assume that $n \geq 1$. Describe $V_+(x_0 x_1) \subset \mathbb{P}_{\mathbb{k}}^n$ as a union of two projective hyperplanes. Compute its intersection with the affine chart U_0 and with the hyperplane at infinity $V_+(x_0)$.

EXERCISE 5.3. Let $n \geq 1$ and let $f \in \mathbb{k}[x_0, \dots, x_n]$ be homogeneous of positive degree e . Use Proposition 3.3.4 to prove that the Hilbert polynomial of $S/(f)$ has degree $n - 1$ and leading

term

$$e \frac{z^{n-1}}{(n-1)!}.$$

EXERCISE 5.4. Assume that $n \geq 1$. Set $S = \mathbb{k}[x_0, \dots, x_n]$. Compute the Hilbert series, dimension, and degree of the following homogeneous quotients:

- (i) $S/(x_0)$;
- (ii) $\mathbb{k}[x_0, x_1, x_2]/(f)$, where f is a nonzero cubic form;
- (iii) $S/(x_0^2)$.

If $\mathbb{k}$ is algebraically closed, compare the common projective variety $V_+(x_0) = V_+(x_0^2)$ with the corresponding quotient degrees.

EXERCISE 5.5. Verify directly that $V_+(x^2 + y^2 + z^2) \subset \mathbb{P}_{\mathbb{R}}^2$ is empty. Show that $[1 : i : 0]$ is a complex point after extending scalars to $\mathbb{C}$. Compute the dimension and degree of the homogeneous quotient $\mathbb{R}[x, y, z]/(x^2 + y^2 + z^2)$. Explain why the empty real zero locus does not contradict this Hilbert-series computation, and why the complex point is consistent with it after extending scalars.

EXERCISE 5.6. Let $I \subset \mathbb{k}[x, y, z]$ and use lexicographic order $x > y > z$. If a Gröbner basis of I is

$$\{x - y^2, y^3 - z, z^2 - 1\},$$

determine Gröbner bases for $I \cap \mathbb{k}[y, z]$ and $I \cap \mathbb{k}[z]$. Describe the possible z -coordinates of points of $V(I)$.

EXERCISE 5.7. Use elimination to find the equation of the projection to the (y, z) -plane of

$$V(x + y - z, x^2 - y) \subset \mathbb{A}_{\mathbb{k}}^3.$$

Then solve for the omitted coordinate and prove directly that the projection is closed in this example.

EXERCISE 5.8. Implicitize each polynomial parametrization:

- (i) $t \mapsto (t^2, t^3)$;
- (ii) $t \mapsto (t + t^2, t^2)$;
- (iii) $(s, t) \mapsto (s, t, st)$.

For each map, write the graph ideal and specify an elimination order.

EXERCISE 5.9. Let $p_1, \dots, p_n \in \mathbb{k}[t_1, \dots, t_m]$. Prove directly, without using Gröbner bases, that the graph ideal

$$(x_1 - p_1, \dots, x_n - p_n)$$

is prime. Deduce that the ideal of polynomial relations among the p_i is prime.

EXERCISE 5.10. For the shifted-circle parametrization in [Example 5.4.4](#), verify all of the following directly:

- (i) the image lies on C' ;
- (ii) $t = 2x/(2-y)$ recovers the parameter whenever $y \neq 2$;
- (iii) the point $(0,2)$ is not in the image;
- (iv) the image is Zariski dense in C' .

EXERCISE 5.11. Let $A = \mathbb{k}[x,y]/(x^2 + y^2 - 1)$ and assume that $\text{char}(\mathbb{k}) \neq 2$. With the total-degree filtration, compute $\dim_{\mathbb{k}} F_m A$ by reducing powers of x . Compare the answer with the standard-monomial proof in [Example 5.4.5](#).

EXERCISE 5.12. Let $I = (x^2, y^2)$ and $J = (x - y)$ in $\mathbb{k}[x,y]$. Form the ideal $tI + (1 - t)J$ in $\mathbb{k}[t, x, y]$ and use an elimination order to compute $I \cap J$. Verify your answer by direct ideal containment in both directions.

EXERCISE 5.13. Prove by induction that for every $N \geq 1$,

$$g = (1 - tf)g \sum_{j=0}^{N-1} (tf)^j + t^N f^N g.$$

Use this identity to prove the forward inclusion in [Theorem 5.5.5](#) without referring to localization.

EXERCISE 5.14. Compute the following saturations by elimination:

- (i) $(x^3, x^2y) : x^\infty$ in $\mathbb{k}[x,y]$;
- (ii) $(x^2, xy) : y^\infty$ in $\mathbb{k}[x,y]$;
- (iii) $(xz, yz) : z^\infty$ in $\mathbb{k}[x,y,z]$.

In each case, also find the smallest N for which $I : f^N = I : f^\infty$.

EXERCISE 5.15. Let $I \subset S$ and $f, g \in S$. Prove

$$(I : f^\infty) : g^\infty = I : (fg)^\infty.$$

Give both a proof from the definition and a computational interpretation using two inverse-denominator variables.

EXERCISE 5.16. Show that

$$f \in \sqrt{I} \iff I : f^\infty = S.$$

Explain carefully why this does not say that $\sqrt{I} = I : f^\infty$. Give an explicit example where these two ideals are different.

EXERCISE 5.17. Let $I, J, K \subset S$. Use the elimination formula for intersections to design one computation of $I \cap J \cap K$. Prove that the result is independent of the order in which the pairwise intersections are taken, even though the intermediate Gröbner bases may differ.

CHAPTER 6

Selected Applications: Symmetry and Gröbner degenerations

Throughout this chapter, $S = \mathbb{k}[x_1, \dots, x_n]$ is a polynomial ring over a field $\mathbb{k}$. We first use lexicographic order to understand the polynomials which are unchanged by permutations of the variables. We shall then use Gröbner bases to turn this structural result into an explicit normal-form algorithm. In the second half of the chapter, we study a different use of initial ideals: every ideal can be placed in a one-parameter family whose special fiber is a monomial initial ideal. This construction is called a *Gröbner degeneration*.

6.1. Symmetric polynomials

Let $\mathfrak{S}_n$ be the symmetric group on $\{1, \dots, n\}$. Every permutation $\sigma \in \mathfrak{S}_n$ acts on S by permuting the variables:

$$\sigma(f)(x_1, \dots, x_n) = f(x_{\sigma(1)}, \dots, x_{\sigma(n)}).$$

DEFINITION 6.1.1. A polynomial $f \in S$ is called *symmetric* if

$$\sigma(f) = f \quad \text{for every } \sigma \in \mathfrak{S}_n.$$

The set of symmetric polynomials is denoted by

$$S^{\mathfrak{S}_n} = \{f \in S \mid \sigma(f) = f \text{ for every } \sigma \in \mathfrak{S}_n\}.$$

It is clear that $S^{\mathfrak{S}_n}$ is a subring of S . For instance, the polynomial $x_1x_2 + x_1x_3$ is not symmetric in $\mathbb{k}[x_1, x_2, x_3]$, whereas

$$x_1x_2 + x_1x_3 + x_2x_3$$

is symmetric.

There is a distinguished family of symmetric polynomials.

DEFINITION 6.1.2. For $0 \leq r \leq n$, the r -th *elementary symmetric polynomial* is

$$e_r(x_1, \dots, x_n) = \sum_{1 \leq i_1 < \dots < i_r \leq n} x_{i_1} \cdots x_{i_r}.$$

We use the convention $e_0 = 1$.

Thus, when $n = 3$, we have

$$e_1 = x_1 + x_2 + x_3, \quad e_2 = x_1x_2 + x_1x_3 + x_2x_3, \quad e_3 = x_1x_2x_3.$$

These polynomials occur naturally as the coefficients in

$$\prod_{i=1}^n (t + x_i) = t^n + e_1 t^{n-1} + e_2 t^{n-2} + \cdots + e_n.$$

We now prove that the e_i account for all symmetric polynomials. The proof is a particularly clean application of lexicographic order.

THEOREM 6.1.3 (Fundamental theorem on symmetric polynomials). *Every symmetric polynomial $f \in \mathbb{k}[x_1, \dots, x_n]$ can be written uniquely in the form*

$$f = P(e_1, \dots, e_n)$$

for a polynomial $P \in \mathbb{k}[y_1, \dots, y_n]$. Equivalently,

$$S^{\mathfrak{S}_n} = \mathbb{k}[e_1, \dots, e_n],$$

and $e_1, \dots, e_n$ are algebraically independent over $\mathbb{k}$.

PROOF. Consider lexicographic order with $x_1 > x_2 > \cdots > x_n$. Let $0 \neq f \in S^{\mathfrak{S}_n}$ and write

$$\text{in}_{\text{lex}}(f) = c x_1^{a_1} \cdots x_n^{a_n}, \quad c \in \mathbb{k} \setminus \{0\}.$$

We claim that

$$a_1 \geq a_2 \geq \cdots \geq a_n.$$

Indeed, suppose that $a_i < a_j$ for some $i < j$. Interchanging x_i and x_j produces another term of f , with coefficient c , which is lexicographically larger than $\text{in}_{\text{lex}}(f)$. This is impossible.

Set $a_{n+1} = 0$. Since

$$\text{in}_{\text{lex}}(e_r) = x_1 \cdots x_r,$$

we obtain

$$\text{in}_{\text{lex}} \left(e_1^{a_1 - a_2} e_2^{a_2 - a_3} \cdots e_{n-1}^{a_{n-1} - a_n} e_n^{a_n} \right) = x_1^{a_1} \cdots x_n^{a_n}.$$

Subtracting

$$c e_1^{a_1 - a_2} e_2^{a_2 - a_3} \cdots e_{n-1}^{a_{n-1} - a_n} e_n^{a_n}$$

from f gives either zero or a symmetric polynomial with strictly smaller initial term. Lexicographic order has no infinite strictly decreasing sequence of monomials. Repeating the procedure therefore terminates and expresses f as a polynomial in $e_1, \dots, e_n$.

It remains to prove uniqueness. For $\mathbf{a} = (a_1, \dots, a_n) \in \mathbb{N}^n$, multiplicativity of initial terms gives

$$\text{in}_{\text{lex}}(e_1^{a_1} \cdots e_n^{a_n}) = x_1^{a_1 + \cdots + a_n} x_2^{a_2 + \cdots + a_n} \cdots x_n^{a_n}.$$

The map

$$(\mathbf{a}_1, \dots, \mathbf{a}_n) \longmapsto (\mathbf{a}_1 + \cdots + \mathbf{a}_n, \mathbf{a}_2 + \cdots + \mathbf{a}_n, \dots, \mathbf{a}_n)$$

is injective. Consequently, distinct monomials in $e_1, \dots, e_n$ have distinct initial terms in S . If $0 \neq P \in \mathbb{k}[y_1, \dots, y_n]$, the largest of the initial terms supplied by its terms cannot be canceled in $P(e_1, \dots, e_n)$. Thus $P(e_1, \dots, e_n) \neq 0$. This proves algebraic independence and uniqueness. $\square$

EXAMPLE 6.1.4. In two variables,

$$x_1^2 + x_1x_2 + x_2^2 = e_1^2 - e_2.$$

In three variables, the same initial-term procedure gives

$$x_1^2 + x_2^2 + x_3^2 = e_1^2 - 2e_2$$

and

$$x_1^3 + x_2^3 + x_3^3 = e_1^3 - 3e_1e_2 + 3e_3.$$

6.1.1. Complete homogeneous symmetric polynomials and power sums. The elementary symmetric polynomials are not the only useful symmetric family.

DEFINITION 6.1.5. For $r \geq 0$, the r -th *complete homogeneous symmetric polynomial* is

$$h_r(x_1, \dots, x_n) = \sum_{1 \leq i_1 \leq \dots \leq i_r \leq n} x_{i_1} \cdots x_{i_r}.$$

We again set $h_0 = 1$.

Thus h_r is the sum of all monomials of total degree r in $x_1, \dots, x_n$. For example,

$$h_2(x_1, x_2, x_3) = x_1^2 + x_2^2 + x_3^2 + x_1x_2 + x_1x_3 + x_2x_3.$$

Introduce the formal power series

$$E(t) = \prod_{i=1}^n (1 - x_i t) = \sum_{r=0}^n (-1)^r e_r t^r$$

and

$$H(t) = \prod_{i=1}^n \frac{1}{1 - x_i t} = \sum_{r \geq 0} h_r t^r.$$

The identity $E(t)H(t) = 1$ gives, for every $r \geq 1$,

$$h_r - e_1 h_{r-1} + e_2 h_{r-2} - \dots + (-1)^r e_r = 0,$$

where $e_r = 0$ for $r > n$. It follows inductively, without any restriction on the characteristic, that

$$\mathbb{k}[e_1, \dots, e_n] = \mathbb{k}[h_1, \dots, h_n].$$

In particular, $h_1, \dots, h_n$ are also algebraically independent.

For $r \geq 1$, the r -th *power sum* is

$$p_r = x_1^r + \dots + x_n^r.$$

Logarithmically differentiating $E(t)$ means taking its formal derivative with respect to t and dividing by $E(t)$. This division is valid in $\mathbb{k}[x_1, \dots, x_n][[t]]$ because $E(0) = 1$, so $E(t)$ is invertible. The product rule gives

$$\frac{E'(t)}{E(t)} = \sum_{i=1}^n \frac{-x_i}{1 - x_i t}.$$

In characteristic zero this is the formal identity $(\log E(t))' = E'(t)/E(t)$; the quotient itself requires no logarithm and is defined in every characteristic. Multiplying by $-t$ and expanding each geometric series yields

$$-t \frac{E'(t)}{E(t)} = \sum_{i=1}^n \frac{x_i t}{1 - x_i t} = \sum_{r \geq 1} \left(\sum_{i=1}^n x_i^r \right) t^r = \sum_{r \geq 1} p_r t^r.$$

Thus $-tE'(t) = E(t) \sum_{r \geq 1} p_r t^r$. Comparing coefficients of t^r gives Newton's identities

$$p_r - e_1 p_{r-1} + e_2 p_{r-2} - \dots + (-1)^{r-1} e_{r-1} p_1 + (-1)^r r e_r = 0$$

for $1 \leq r \leq n$. Thus $p_1, \dots, p_n$ belong to $\mathbb{k}[e_1, \dots, e_n]$. In fact every p_r belongs to this ring by [Theorem 6.1.3](#), since every power sum is symmetric. If $\text{char}(\mathbb{k}) = 0$, or more generally if $\text{char}(\mathbb{k}) > n$, the integers $1, \dots, n$ are invertible in $\mathbb{k}$. Newton's identities can then be solved successively for $e_1, \dots, e_n$, and

$$S^{\mathfrak{S}_n} = \mathbb{k}[p_1, \dots, p_n].$$

The characteristic hypothesis cannot be omitted. For instance, when $n = 2$ and $\text{char}(\mathbb{k}) = 2$, one has

$$p_1 = e_1, \quad p_2 = (x_1 + x_2)^2 = e_1^2.$$

Thus $\mathbb{k}[p_1, p_2] = \mathbb{k}[e_1]$ does not contain $e_2 = x_1 x_2$.

There are two further classical families. A *partition with at most n parts* is a tuple of integers

$$\lambda = (\lambda_1, \dots, \lambda_n), \quad \lambda_1 \geq \dots \geq \lambda_n \geq 0.$$

The corresponding *monomial symmetric polynomial* m_λ is the sum of the distinct monomials obtained by permuting the exponents in $x_1^{\lambda_1} \dots x_n^{\lambda_n}$. The corresponding *Schur polynomial* may be defined over $\mathbb{Z}$ by

$$s_\lambda = \frac{\det(x_i^{\lambda_j + n - j})_{1 \leq i, j \leq n}}{\det(x_i^{n - j})_{1 \leq i, j \leq n}}.$$

The quotient is a polynomial with integer coefficients. The families $\{m_\lambda\}$ and $\{s_\lambda\}$ form $\mathbb{k}$ -bases of $S^{\mathfrak{S}_n}$ over every field. They should be distinguished from the algebraically independent generating sets $e_1, \dots, e_n$ and $h_1, \dots, h_n$.

6.2. A Gröbner basis for symmetric polynomials

The proof of [Theorem 6.1.3](#) gives an algorithm which repeatedly cancels initial terms. We now encode the whole procedure in one reduced Gröbner basis.

Consider the polynomial ring

$$T = \mathbb{k}[x_1, \dots, x_n, y_1, \dots, y_n]$$

and the ideal

$$I_{\text{sym}} = (e_1(\mathbf{x}) - y_1, \dots, e_n(\mathbf{x}) - y_n) \subset T.$$

We use lexicographic order with

$$x_1 > \dots > x_n > y_1 > \dots > y_n.$$

For $1 \leq k \leq n$ and $r \geq 0$, write

$$h_r^{(k)} = h_r(x_k, \dots, x_n),$$

and set $h_0^{(k)} = 1$.

The following identity will supply the desired relations.

LEMMA 6.2.1. *For every $1 \leq k \leq n$, one has*

$$h_k^{(k)} + \sum_{i=1}^k (-1)^i h_{k-i}^{(k)} e_i(x_1, \dots, x_n) = 0.$$

PROOF. We have

$$\begin{aligned} \left(\sum_{i=0}^n (-1)^i e_i(\mathbf{x}) t^i \right) \left(\sum_{j \geq 0} h_j^{(k)} t^j \right) &= \prod_{i=1}^n (1 - x_i t) \prod_{i=k}^n \frac{1}{1 - x_i t} \\ &= \prod_{i=1}^{k-1} (1 - x_i t). \end{aligned}$$

The polynomial on the right has degree $k-1$. Its coefficient of t^k is therefore zero, and comparison of the coefficient of t^k on the left gives the result. $\square$

THEOREM 6.2.2. *For $1 \leq k \leq n$, set*

$$g_k = h_k^{(k)} + \sum_{i=1}^k (-1)^i h_{k-i}^{(k)} y_i.$$

Then

$$\mathcal{G}_{\text{sym}} = \{g_1, \dots, g_n\}$$

is the reduced Gröbner basis of I_{sym} . Moreover,

$$\text{in}_{\text{lex}}(g_k) = x_k^k$$

and

$$\text{in}_{\text{lex}}(I_{\text{sym}}) = (x_1, x_2^2, \dots, x_n^n).$$

PROOF. By Lemma 6.2.1, substituting $y_i = e_i(\mathbf{x})$ in g_k gives zero. The kernel of the substitution map

$$T \longrightarrow S, \quad x_i \longmapsto x_i, \quad y_i \longmapsto e_i(\mathbf{x}),$$

is I_{sym} . Thus $g_k \in I_{\text{sym}}$ for every k .

The largest monomial in $h_k^{(k)}$ is x_k^k . Every other term in g_k either has smaller exponent in x_k or first involves a variable which occurs later in the chosen lexicographic order. Hence

$$\text{in}_{\text{lex}}(g_k) = x_k^k.$$

Set

$$K = (x_1, x_2^2, \dots, x_n^n) \subset T.$$

We have obtained

$$K \subseteq \text{in}_{\text{lex}}(I_{\text{sym}}).$$

Give T the positive grading

$$\deg(x_i) = 1, \quad \deg(y_i) = i.$$

The ideal I_{sym} is homogeneous, and the substitution map above gives a graded isomorphism

$$T/I_{\text{sym}} \cong S.$$

Consequently,

$$\text{Hilb}_{T/I_{\text{sym}}}(t) = \frac{1}{(1-t)^n}.$$

On the other hand, the standard monomials of K give

$$\begin{aligned} \text{Hilb}_{T/K}(t) &= \left(\prod_{i=1}^n (1 + t + \dots + t^{i-1}) \right) \left(\prod_{i=1}^n \frac{1}{1-t^i} \right) \\ &= \prod_{i=1}^n \left(\frac{1-t^i}{1-t} \right) \left(\prod_{i=1}^n \frac{1}{1-t^i} \right) \\ &= \frac{1}{(1-t)^n}. \end{aligned}$$

For an ideal which is homogeneous with respect to a positive grading, the standard-monomial proof that the ideal and its initial ideal have the same Hilbert series works without change. Thus

$$\text{Hilb}_{T/\text{in}_{\text{lex}}(I_{\text{sym}})}(t) = \text{Hilb}_{T/I_{\text{sym}}}(t) = \text{Hilb}_{T/K}(t).$$

The inclusion $K \subseteq \text{in}_{\text{lex}}(I_{\text{sym}})$, compared degree by degree, now gives equality. Therefore $\mathcal{G}_{\text{sym}}$ is a Gröbner basis of I_{sym} .

Each g_k is monic. Every other monomial in $h_k^{(k)}$ has exponent less than k in x_k , and exponent at most $k < j$ in every x_j with $j > k$. A monomial in $h_{k-i}^{(k)} y_i$, with $i \geq 1$, has even smaller total degree in the x -variables. Thus no other monomial of any g_k is divisible by one of $x_1, x_2^2, \dots, x_n^n$. The Gröbner basis is reduced. $\square$

EXAMPLE 6.2.3. When $n = 3$, Theorem 6.2.2 gives

$$\begin{aligned} g_1 &= x_1 + x_2 + x_3 - y_1, \\ g_2 &= x_2^2 + x_2 x_3 + x_3^2 - (x_2 + x_3) y_1 + y_2, \\ g_3 &= x_3^3 - x_3^2 y_1 + x_3 y_2 - y_3. \end{aligned}$$

Their initial terms are x_1, x_2^2, x_3^3 .

6.2.1. Normal forms of symmetric polynomials. Let $f \in S^{\mathfrak{S}_n}$ and regard f as an element of T which does not involve the y_i . Divide f by the reduced Gröbner basis $\mathcal{G}_{\text{sym}}$ and let $R(\mathbf{x}, \mathbf{y})$ be its normal form. By Theorem 6.1.3, there is a unique polynomial $P \in \mathbb{k}[y_1, \dots, y_n]$ such that

$$f = P(e_1, \dots, e_n).$$

Hence

$$f - P(y_1, \dots, y_n) \in I_{\text{sym}}.$$

The polynomial $P(\mathbf{y})$ is already reduced because it contains none of the initial terms $x_1, x_2^2, \dots, x_n^n$. Uniqueness of normal forms gives

$$R(\mathbf{x}, \mathbf{y}) = P(\mathbf{y}).$$

Thus division by $\mathcal{G}_{\text{sym}}$ is an explicit algorithm for writing a symmetric polynomial in the elementary symmetric polynomials.

EXAMPLE 6.2.4. For $n = 3$, reduction by the displayed Gröbner basis gives

$$\text{NF}_{\mathcal{G}_{\text{sym}}}(x_1^2 + x_2^2 + x_3^2) = y_1^2 - 2y_2$$

and

$$\text{NF}_{\mathcal{G}_{\text{sym}}}(x_1^3 + x_2^3 + x_3^3) = y_1^3 - 3y_1 y_2 + 3y_3.$$

Replacing y_i by e_i recovers the identities above.

6.2.2. The coinvariant algebra. The quotient by the elementary symmetric polynomials of positive degree is called the *coinvariant algebra*:

$$C_n = \frac{\mathbb{k}[x_1, \dots, x_n]}{(e_1, \dots, e_n)}.$$

Specializing $y_1 = \dots = y_n = 0$ in the preceding Gröbner basis gives the polynomials

$$q_k = h_k(x_k, \dots, x_n), \quad 1 \leq k \leq n.$$

The identity in [Lemma 6.2.1](#) shows inductively that

$$(q_1, \dots, q_k) = (e_1, \dots, e_k)$$

for every k . Moreover, their initial terms

$$x_1, x_2^2, \dots, x_n^n$$

are pairwise relatively prime. It follows from Buchberger's criterion that $q_1, \dots, q_n$ form a Gröbner basis of $(e_1, \dots, e_n)$.

We obtain the following concrete description.

COROLLARY 6.2.5. *The residue classes of the monomials*

$$x_1^{a_1} \cdots x_n^{a_n}, \quad 0 \leq a_i < i,$$

form a $\mathbb{k}$ -basis of C_n . In particular,

$$\dim_{\mathbb{k}}(C_n) = 1 \cdot 2 \cdots n = n!$$

and

$$\text{Hilb}_{C_n}(t) = \prod_{i=1}^n (1 + t + \cdots + t^{i-1}).$$

PROOF. The initial ideal is $(x_1, x_2^2, \dots, x_n^n)$. The stated monomials are precisely its standard monomials. The dimension and Hilbert series follow immediately. $\square$

The description also gives an elementary explanation for the good behavior of the generators. Indeed,

$$q_n = x_n^n,$$

and, successively, $q_{n-1}, q_{n-2}, \dots, q_1$ are monic polynomials in $x_{n-1}, x_{n-2}, \dots, x_1$, respectively. Since a monic polynomial in $A[z]$ is a nonzerodivisor for every coefficient ring A , the reversed sequence

$$q_n, q_{n-1}, \dots, q_1$$

is a regular sequence. This conclusion required no depth or homological machinery.

The elementary symmetric polynomials themselves form a regular sequence as well. The first element $q_1 = e_1$ is nonzero, and S is a domain. For $i \geq 2$, divide a polynomial f by $q_1, \dots, q_{i-1}$. If $f q_i \in (q_1, \dots, q_{i-1})$ and the remainder r were nonzero, then

$$\text{in}(r q_i) = \text{in}(r) \text{in}(q_i) = \text{in}(r) x_i^i$$

would be divisible by one of $x_1, x_2^2, \dots, x_{i-1}^{i-1}$. Each of these generators is relatively prime to x_i^i , so the same generator would divide $\text{in}(r)$, contrary to the definition of a remainder. Thus $f \in (q_1, \dots, q_{i-1})$, and $q_1, \dots, q_n$ is a regular sequence in this order. Finally, [Lemma 6.2.1](#) gives

$$q_i \equiv (-1)^{i+1} e_i \pmod{(e_1, \dots, e_{i-1})},$$

and we already know that $(q_1, \dots, q_{i-1}) = (e_1, \dots, e_{i-1})$. It follows that

$$e_1, \dots, e_n$$

is a regular sequence.

REMARK 6.2.6 (The flag variety). There is a classical geometric interpretation of the coinvariant algebra. Let $\text{Fl}_n(\mathbb{C})$ be the variety of complete flags

$$0 = V_0 \subset V_1 \subset \dots \subset V_n = \mathbb{C}^n, \quad \dim_{\mathbb{C}}(V_i) = i.$$

The Borel presentation is the graded-ring isomorphism

$$H^*(\text{Fl}_n(\mathbb{C}); \mathbb{Z}) \cong \frac{\mathbb{Z}[x_1, \dots, x_n]}{(e_1, \dots, e_n)},$$

where the cohomological degree of each x_i is 2. After tensoring with $\mathbb{Q}$, this is the coinvariant algebra over $\mathbb{Q}$, with its ordinary polynomial degrees doubled. The Schubert classes, indexed by the permutations in $\mathfrak{S}_n$, give another integral basis. Their number is $n!$, in agreement with [Corollary 6.2.5](#). This topological interpretation is not needed for the algebraic arguments below.

6.3. One-parameter families

An initial ideal is simpler than the original ideal, but it retains a surprising amount of information. The reason is that the two ideals occur as fibers of one algebraic family. We begin by making this statement precise.

Let t be a new variable and set $T = S[t]$. For an ideal $\mathcal{J} \subset T$, put

$$A = T/\mathcal{J}.$$

For every $a \in \mathbb{k}$, the *fiber at a* is

$$A_a = A \otimes_{\mathbb{k}[t]} \frac{\mathbb{k}[t]}{(t-a)} \cong \frac{T}{(\mathcal{J}, t-a)}.$$

Since $T/(t-a) \cong S$, there is an ideal $I_a \subset S$ such that $A_a \cong S/I_a$.

DEFINITION 6.3.1. A $\mathbb{k}[t]$ -module M is *flat* if tensoring with M preserves injections: whenever $N' \rightarrow N$ is an injective homomorphism of $\mathbb{k}[t]$ -modules, the induced map

$$N' \otimes_{\mathbb{k}[t]} M \longrightarrow N \otimes_{\mathbb{k}[t]} M$$

is injective. A family $T/\mathcal{J}$ is called *flat* if it is flat as a $\mathbb{k}[t]$ -module, and it is called *free* if it is free as a $\mathbb{k}[t]$ -module.

Every free module is flat. Moreover, because $\mathbb{k}[t]$ is a principal ideal domain, a $\mathbb{k}[t]$ -module is flat if and only if it is torsion-free. We will not need this general characterization: the families constructed below will come with an explicit free basis.

DEFINITION 6.3.2. Let $I, I_0 \subset S$ be ideals. A *one-parameter flat degeneration of I to I_0* is an ideal $\mathcal{J} \subset S[t]$ such that:

- (i) $S[t]/\mathcal{J}$ is flat over $\mathbb{k}[t]$;
- (ii) for every $\alpha \neq 0$, the fiber $T/(\mathcal{J}, t - \alpha)$ is isomorphic as a $\mathbb{k}$ -algebra to S/I ;
- (iii) the special fiber $T/(\mathcal{J}, t)$ is isomorphic to S/I_0 .

The ideals obtained after specializing at two nonzero values of t need not be literally equal. The definition asks for isomorphic quotient rings; in our construction the isomorphisms will be diagonal changes of coordinates.

6.3.1. An algebraic family whose special quotient has nilpotents. Before giving the general construction, we consider an example over $\mathbb{R}$. Let

$$F = x^2 + y^2 t^2 - t^4 \in \mathbb{R}[x, y, t]$$

and set

$$\mathcal{J} = (F).$$

Since F is monic as a polynomial in x of degree two, the residue classes of

$$y^b t^c \quad \text{and} \quad xy^b t^c, \quad b, c \geq 0,$$

form an $\mathbb{R}$ -basis of $\mathbb{R}[x, y, t]/(F)$. Equivalently,

$$\mathbb{R}[x, y, t]/(F) = \bigoplus_{b \geq 0} \mathbb{R}[t] y^b \oplus \bigoplus_{b \geq 0} \mathbb{R}[t] x y^b$$

is a free $\mathbb{R}[t]$ -module.

At $t = \alpha \neq 0$, the fiber is

$$\frac{\mathbb{R}[x, y]}{(x^2 + \alpha^2 y^2 - \alpha^4)}.$$

The change of coordinates

$$x = \alpha^2 X, \quad y = \alpha Y$$

transforms the equation into

$$\alpha^4 (X^2 + Y^2 - 1) = 0.$$

Thus every nonzero fiber is isomorphic to

$$\mathbb{R}[X, Y]/(X^2 + Y^2 - 1),$$

and the equation $X^2 + Y^2 - 1 = 0$ has the unit circle as its real zero set. At $t = 0$, however, the fiber is

$$\frac{\mathbb{R}[x, y]}{(x^2)},$$

and the residue class of x is nonzero and nilpotent. The ideals (x^2) and (x) have the same real zero set, but the two quotient rings are different.

This example is a weighted homogenization of the polynomial $x^2 + y^2 - 1$ for the weights

$$\deg(x) = 2, \quad \deg(y) = 1.$$

We now develop the general construction.

6.4. Weights and initial forms

Let $\mathbf{w} = (w_1, \dots, w_n) \in \mathbb{Z}_{>0}^n$. For a monomial $\mathbf{x}^{\mathbf{a}} = x_1^{a_1} \cdots x_n^{a_n}$, define its $\mathbf{w}$ -degree by

$$\deg_{\mathbf{w}}(\mathbf{x}^{\mathbf{a}}) = \mathbf{w} \cdot \mathbf{a} = w_1 a_1 + \cdots + w_n a_n.$$

If

$$0 \neq f = \sum_{\mathbf{a}} c_{\mathbf{a}} \mathbf{x}^{\mathbf{a}},$$

set

$$\deg_{\mathbf{w}}(f) = \max\{\mathbf{w} \cdot \mathbf{a} \mid c_{\mathbf{a}} \neq 0\}$$

and define the $\mathbf{w}$ -initial form of f by

$$\text{in}_{\mathbf{w}}(f) = \sum_{\mathbf{w} \cdot \mathbf{a} = \deg_{\mathbf{w}}(f)} c_{\mathbf{a}} \mathbf{x}^{\mathbf{a}}.$$

For an ideal $I \subset S$, its $\mathbf{w}$ -initial ideal is

$$\text{in}_{\mathbf{w}}(I) = (\text{in}_{\mathbf{w}}(f) \mid 0 \neq f \in I).$$

Unlike an initial ideal determined by a monomial order, $\text{in}_{\mathbf{w}}(I)$ need not be a monomial ideal: several terms of a polynomial can have the same maximal $\mathbf{w}$ -degree.

The key point is that a positive weight vector can reproduce any prescribed finite collection of comparisons made by a monomial order. We first prove the elementary theorem of alternatives needed for this fact.

LEMMA 6.4.1 (Gordan's alternative). *Let $\mathbf{d}_1, \dots, \mathbf{d}_m \in \mathbb{Q}^n$. Exactly one of the following conditions holds:*

- (i) *There is a vector $\mathbf{v} \in \mathbb{Q}^n$ such that $\mathbf{d}_i \cdot \mathbf{v} > 0$ for every i .*
- (ii) *There are nonnegative rational numbers $\lambda_1, \dots, \lambda_m$, not all zero, such that*

$$\lambda_1 \mathbf{d}_1 + \cdots + \lambda_m \mathbf{d}_m = 0.$$

PROOF. The two conditions cannot hold simultaneously: taking the scalar product of the equality in (ii) with the vector in (i) would give a strictly positive number equal to zero.

Consider the convex hull

$$C = \left\{ \lambda_1 \mathbf{d}_1 + \cdots + \lambda_m \mathbf{d}_m \mid \lambda_i \geq 0, \sum_i \lambda_i = 1 \right\} \subset \mathbb{R}^n.$$

If $0 \in C$, condition (ii) holds with real coefficients. Restrict to the support of such a solution. The equations on that support have rational coefficients, and their real solution has all coordinates positive. Gaussian elimination shows that the affine solution space is defined over $\mathbb{Q}$; rational points are dense in it. A sufficiently close rational solution still has all coordinates positive, so the λ_i may be chosen rational.

Suppose now that $0 \notin C$. The set C is compact, so choose $\mathbf{q} \in C$ of smallest Euclidean norm. For every i and every $0 \leq s \leq 1$, the point

$$\mathbf{q} + s(\mathbf{d}_i - \mathbf{q})$$

belongs to C . The function

$$s \mapsto \|\mathbf{q} + s(\mathbf{d}_i - \mathbf{q})\|^2$$

has a minimum at $s = 0$ on the interval $[0, 1]$. Its right derivative at zero is nonnegative, and therefore

$$2\mathbf{q} \cdot (\mathbf{d}_i - \mathbf{q}) \geq 0.$$

Hence

$$\mathbf{q} \cdot \mathbf{d}_i \geq \|\mathbf{q}\|^2 > 0$$

for all i . The inequalities are strict, so a rational vector sufficiently close to $\mathbf{q}$ satisfies the same inequalities. This gives condition (i). $\square$

LEMMA 6.4.2 (Finite-weight lemma). *Let $<$ be a monomial order on S and let*

$$u_1 < v_1, \dots, u_m < v_m$$

be finitely many strict comparisons of monomials. Then there is a vector $\mathbf{w} \in \mathbb{Z}_{>0}^n$ such that

$$\deg_{\mathbf{w}}(u_i) < \deg_{\mathbf{w}}(v_i)$$

for every i .

PROOF. Write

$$u_i = \mathbf{x}^{\mathbf{a}_i}, \quad v_i = \mathbf{x}^{\mathbf{b}_i}.$$

Apply Lemma 6.4.1 to the vectors

$$\mathbf{b}_i - \mathbf{a}_i \quad (1 \leq i \leq m), \quad \mathbf{e}_1, \dots, \mathbf{e}_n,$$

where $\mathbf{e}_j$ is the j -th standard basis vector. If the first alternative holds, the resulting vector has positive coordinates and gives all the desired inequalities. After multiplying by a common denominator, it belongs to $\mathbb{Z}_{>0}^n$.

Suppose, toward a contradiction, that the second alternative holds. After clearing denominators, there are nonnegative integers $c_1, \dots, c_m, d_1, \dots, d_n$, not all zero, such that

$$\sum_{i=1}^m c_i(\mathbf{b}_i - \mathbf{a}_i) + (d_1, \dots, d_n) = 0.$$

At least one c_i is positive, since otherwise all the d_j would be zero. Set

$$U = \prod_{i=1}^m u_i^{c_i}, \quad V = \prod_{i=1}^m v_i^{c_i}, \quad \mathbf{x}^{\mathbf{d}} = x_1^{d_1} \cdots x_n^{d_n}.$$

The displayed vector equality says that

$$U = V\mathbf{x}^{\mathbf{d}}.$$

On the one hand, multiplicativity of the monomial order and the inequalities $u_i < v_i$ give $U < V$. On the other hand, $1 \leq \mathbf{x}^{\mathbf{d}}$ gives

$$U = V\mathbf{x}^{\mathbf{d}} \geq V,$$

a contradiction. □

REMARK 6.4.3. The lemma concerns only finitely many comparisons. A single weight vector usually cannot reproduce a monomial order on all monomials. For example, no fixed positive weight can reproduce the inequalities $x_1 > x_2^d$ for every $d \geq 1$ which hold in lexicographic order.

6.4.1. Homogenization. Let $0 \neq f \in S$ and put $d = \deg_{\mathbf{w}}(f)$. The $\mathbf{w}$ -homogenization of f is

$$\text{hom}_{\mathbf{w}}(f) = t^d f\left(\frac{x_1}{t^{w_1}}, \dots, \frac{x_n}{t^{w_n}}\right) = \sum_{\mathbf{a}} c_{\mathbf{a}} \mathbf{x}^{\mathbf{a}} t^{d - \mathbf{w} \cdot \mathbf{a}} \in S[t].$$

All exponents of t are nonnegative by the definition of d . If we give x_i degree w_i and t degree one, then $\text{hom}_{\mathbf{w}}(f)$ is homogeneous of degree d . Moreover,

$$\text{hom}_{\mathbf{w}}(f)|_{t=1} = f$$

and

$$\text{hom}_{\mathbf{w}}(f)|_{t=0} = \text{in}_{\mathbf{w}}(f).$$

For an ideal $I \subset S$, set

$$\text{hom}_{\mathbf{w}}(I) = (\text{hom}_{\mathbf{w}}(f) \mid 0 \neq f \in I) \subset S[t].$$

The next theorem identifies a weight for which the special fiber is a prescribed monomial initial ideal and, at the same time, proves that the family is free.

6.5. Gröbner degenerations

THEOREM 6.5.1 (Gröbner degeneration). *Let $I \subset S$ be an ideal and let $<$ be a monomial order. There is a vector $\mathbf{w} \in \mathbb{Z}_{>0}^n$ such that*

$$\text{in}_{\mathbf{w}}(I) = \text{in}_{<}(I).$$

For such a vector, the ideal

$$\mathcal{J} = \text{hom}_{\mathbf{w}}(I) \subset T = S[t]$$

has the following properties:

- (i) $T/\mathcal{J}$ is a free $\mathbb{k}[t]$ -module.
- (ii) For every $\alpha \in \mathbb{k} \setminus \{0\}$,

$$\frac{T}{(\mathcal{J}, t - \alpha)} \cong \frac{S}{I}.$$

- (iii) *The special fiber is*

$$\frac{T}{(\mathcal{J}, t)} \cong \frac{S}{\text{in}_{<}(I)}.$$

More precisely, if $\mathcal{B}$ is the set of standard monomials of $\text{in}_{<}(I)$, then

$$T/\mathcal{J} \cong \bigoplus_{\mathbf{u} \in \mathcal{B}} \mathbb{k}[t]\mathbf{u}$$

as a $\mathbb{k}[t]$ -module.

PROOF. Choose a finite monic Gröbner basis

$$\mathcal{G} = \{g_1, \dots, g_s\}$$

of I with respect to $<$. For every pair g_i, g_j , fix a finite sequence of initial-term reductions of $S(g_i, g_j)$ to zero by $\mathcal{G}$. Record the following finite collection of comparisons: in each g_i , its initial term is larger than every other monomial; in each S -polynomial, the canceled least common multiple is larger than every monomial which survives the initial cancellation; and, at every later reduction step, the monomial being canceled is larger than every monomial introduced by that step. Apply [Lemma 6.4.2](#) to all these comparisons. We obtain $\mathbf{w} \in \mathbb{Z}_{>0}^n$ such that:

- (a) $\text{in}_{<}(g_i)$ has strictly larger $\mathbf{w}$ -degree than every other monomial of g_i ;
- (b) in each chosen reduction sequence, the initial cancellation and every later reduction step strictly lowers the $\mathbf{w}$ -degree from the monomial being canceled to each monomial introduced by that step.

Put

$$\widehat{g}_i = \text{hom}_{\mathbf{w}}(g_i).$$

On T , order monomials first by the positive degree

$$\deg(x_i) = w_i, \quad \deg(t) = 1,$$

and break ties by comparing their x -parts with $<$. This is a monomial order. Every term of $\widehat{g}_i$ has the same weighted degree. Condition (a) implies that

$$\text{in}(\widehat{g}_i) = \text{in}_<(g_i),$$

with no factor of t .

The chosen reduction sequences of the S -polynomials lift after homogenization. For a fixed pair, let D be the $\mathbf{w}$ -degree of the least common multiple of the two initial terms. Condition (b) shows first that every monomial surviving the initial cancellation has $\mathbf{w}$ -degree smaller than D . If a later monomial m is canceled by a multiple of g_k , induction along the reduction sequence gives $\deg_{\mathbf{w}}(m) \leq D$, and the corresponding term in the homogenized polynomial is

$$mt^{D-\deg_{\mathbf{w}}(m)}$$

of total weighted degree D . Multiplying $\widehat{g}_k$ by the appropriate monomial and the appropriate nonnegative power of t performs the same cancellation. Condition (b) also guarantees inductively that every monomial introduced later has $\mathbf{w}$ -degree at most D , so every required exponent of t is nonnegative. Thus every $S(\widehat{g}_i, \widehat{g}_j)$ reduces to zero by

$$\widehat{\mathcal{G}} = \{\widehat{g}_1, \dots, \widehat{g}_s\}.$$

Buchberger's criterion shows that $\widehat{\mathcal{G}}$ is a Gröbner basis of the ideal

$$H = (\widehat{g}_1, \dots, \widehat{g}_s) \subset T.$$

Its initial ideal is

$$(\text{in}_<(g_1), \dots, \text{in}_<(g_s)) = \text{in}_<(I),$$

and none of these generators involves t . Consequently, the standard monomials of H are precisely

$$ut^q, \quad u \in \mathcal{B}, \quad q \geq 0.$$

Macaulay's theorem therefore gives

$$T/H \cong \bigoplus_{u \in \mathcal{B}} \mathbb{k}[t]u.$$

In particular, T/H is free over $\mathbb{k}[t]$, and multiplication by t on this quotient is injective.

We claim that

$$H = \text{hom}_{\mathbf{w}}(I).$$

One inclusion is clear because every g_i belongs to I . Conversely, let $0 \neq f \in I$. A representation $f = \sum_i a_i g_i$ can be homogenized after multiplying by a sufficiently large power of t , and it gives

$$t^N \text{hom}_{\mathbf{w}}(f) \in H$$

for some $N \geq 0$. Since multiplication by t is injective on T/H , the ideal H is saturated with respect to t . It follows that $\text{hom}_{\mathbf{w}}(f) \in H$. This proves the claim and also proves (i).

At $t = 0$, the Gröbner basis $\widehat{\mathcal{G}}$ specializes to

$$\text{in}_{<}(g_1), \dots, \text{in}_{<}(g_s).$$

Therefore

$$\text{in}_{\mathbf{w}}(I) = \frac{(\mathcal{J}, t)}{(t)} = \text{in}_{<}(I),$$

and (iii) follows.

Finally, let $\alpha \neq 0$. For $0 \neq f \in I$ of $\mathbf{w}$ -degree d , we have

$$\text{hom}_{\mathbf{w}}(f)(\mathbf{x}, \alpha) = \alpha^d f\left(\frac{x_1}{\alpha^{w_1}}, \dots, \frac{x_n}{\alpha^{w_n}}\right).$$

The diagonal automorphism

$$x_i \longmapsto \alpha^{w_i} x_i$$

identifies the specialized ideal with I . This proves (ii). $\square$

COROLLARY 6.5.2 (Prime and radical initial ideals). *Let $I \subset S$ be an ideal and let $<$ be a monomial order. If $\text{in}_{<}(I)$ is prime, then I is prime. If $\text{in}_{<}(I)$ is radical, then I is radical.*

PROOF. Choose the weight vector $\mathbf{w}$ in [Theorem 6.5.1](#), and set $\mathcal{J} = \text{hom}_{\mathbf{w}}(I)$ and

$$A = \frac{S[t]}{\mathcal{J}}.$$

Let $\mathcal{B}$ be the set of standard monomials of $\text{in}_{<}(I)$. The theorem gives

$$A \cong \bigoplus_{u \in \mathcal{B}} \mathbb{k}[t]u, \quad A/tA \cong S/\text{in}_{<}(I).$$

We first record what the free basis implies. Every nonzero $\alpha \in A$ can be written as

$$\alpha = t^q \alpha_0, \quad q \geq 0, \quad \alpha_0 \notin tA :$$

in the basis $\mathcal{B}$, factor from all the coefficients of α the largest common power of t . Moreover, multiplication by t is injective on A .

If A/tA is a domain and $\alpha b = 0$ with $\alpha, b \neq 0$, write $\alpha = t^q \alpha_0$ and $b = t^r b_0$ as above. Canceling t^{q+r} gives $\alpha_0 b_0 = 0$. Modulo t , this is a product of two nonzero elements of A/tA , a contradiction. Thus A is a domain. Similarly, if A/tA is reduced and $\alpha^m = 0$ with $\alpha \neq 0$, write $\alpha = t^q \alpha_0$. Canceling t^{qm} and reducing modulo t shows that the nonzero class of α_0 is nilpotent in A/tA , a contradiction. Thus A is reduced.

Put $K = \mathbb{k}(t)$. After extending scalars to K , the diagonal change of variables $x_i \mapsto t^{w_i} x_i$ identifies the generic fiber with

$$A \otimes_{\mathbb{k}[t]} K \cong (S/I) \otimes_{\mathbb{k}} K.$$

Localization preserves domains and reduced rings. Hence, according as $\text{in}_<(I)$ is prime or radical, $(S/I) \otimes_{\mathbb{k}} K$ is a domain or is reduced. Finally, the natural map

$$S/I \longrightarrow (S/I) \otimes_{\mathbb{k}} K$$

is injective, so the same property holds for S/I . Therefore I is respectively prime or radical. $\square$

REMARK 6.5.3. The freeness in [Theorem 6.5.1](#) is stronger than the flatness required in the definition of a degeneration. It also explains directly why the standard monomials of the initial ideal provide compatible bases in every fiber.

6.6. Exercises

EXERCISE 6.1. Work in three variables. Use the initial-term procedure in the proof of [Theorem 6.1.3](#) to express the following symmetric polynomials in e_1, e_2, e_3 :

- (i) $\sum_{i < j} x_i^2 x_j^2$;
- (ii) $\sum_{i \neq j} x_i^3 x_j$;
- (iii) $(x_1 - x_2)^2 (x_1 - x_3)^2 (x_2 - x_3)^2$.

EXERCISE 6.2. Let $\text{char}(\mathbb{k}) = p > 0$ and $n \geq p$. Show that $p_p = p_1^p$. Give another example, besides the case $p = n = 2$ in the text, in which $\mathbb{k}[p_1, \dots, p_n]$ is strictly smaller than $S^{\mathbb{S}_n}$.

EXERCISE 6.3. Assume that $n \geq 4$. Use the identity $E(t)H(t) = 1$ to express h_1, h_2, h_3, h_4 in terms of the elementary symmetric polynomials. Then solve the same identities successively for e_1, e_2, e_3, e_4 in terms of h_1, h_2, h_3, h_4 .

EXERCISE 6.4. For $n = 2$ and $n = 3$, verify [Theorem 6.2.2](#) directly using Buchberger's criterion. In each case, compute every S -polynomial and exhibit its reduction to zero.

EXERCISE 6.5. Use division by $\mathcal{G}_{\text{sym}}$ to express the following polynomials in the elementary symmetric polynomials:

- (i) $x_1^4 + x_2^4 + x_3^4$ in three variables;
- (ii) $\sum_{i \neq j} x_i^2 x_j$ in three variables;
- (iii) $\prod_{1 \leq i < j \leq n} (x_i + x_j)$ for $n = 3$ and $n = 4$.

EXERCISE 6.6. Let $f \in \mathbb{k}[x_1, \dots, x_n]$ be arbitrary, not necessarily symmetric, and let $R(\mathbf{x}, \mathbf{y})$ be its normal form modulo $\mathcal{G}_{\text{sym}}$. Prove that

$$f(\mathbf{x}) = R(\mathbf{x}, e_1(\mathbf{x}), \dots, e_n(\mathbf{x})).$$

Show that the monomials $x_1^{a_1} \cdots x_n^{a_n}$ with $0 \leq a_i < i$ form a basis of S as a free module over $S^{\mathbb{S}_n}$.

CHAPTER 6. SELECTED APPLICATIONS: SYMMETRY AND GRÖBNER DEGENERATIONS

EXERCISE 6.7. Let C_n be the coinvariant algebra. Prove that its highest nonzero graded component has degree

$$1 + 2 + \cdots + (n-1) = \binom{n}{2}$$

and is one-dimensional, generated by the residue class of

$$x_2 x_3^2 \cdots x_n^{n-1}.$$

Deduce directly from Macaulay's theorem that

$$t^{\binom{n}{2}} \text{Hilb}_{C_n}(t^{-1}) = \text{Hilb}_{C_n}(t).$$

EXERCISE 6.8. For the family

$$\mathbb{R}[x, y, t]/(x^2 + y^2 t^2 - t^4),$$

write down explicitly the isomorphism between the fibers at two nonzero parameters a and b . Show directly that the special fiber has nilradical generated by the residue class of x .

EXERCISE 6.9. Let $<$ be lexicographic order on $\mathbb{k}[x, y, z]$ with $x > y > z$. Find a positive integral weight vector which simultaneously realizes the comparisons

$$x^2 y > x y^4 z, \quad x z^{10} > y^{20}, \quad y^3 > z^{100}.$$

Explain why no one positive weight vector can realize all lexicographic comparisons.

EXERCISE 6.10. Prove the following rational form of the separation step used in [Lemma 6.4.1](#). If a system of linear equations and weak inequalities with rational coefficients has a real solution, and the inequalities which are prescribed to be strict are strict at that solution, then it has a rational solution satisfying the same strict inequalities.

EXERCISE 6.11. Let $I = (x^2 - y^3, x^3 - y^2) \subset \mathbb{k}[x, y]$ and choose a monomial order. Compute a Gröbner basis, choose a weight vector as in [Theorem 6.5.1](#), and write down the corresponding homogenized ideal in $\mathbb{k}[x, y, t]$. Determine the general and special fibers and exhibit a $\mathbb{k}[t]$ -basis of the total family.

EXERCISE 6.12. Let $I \subset S$ be homogeneous for the usual grading. Use the free basis in [Theorem 6.5.1](#) to recover, degree by degree, the equality

$$\text{Hilb}_{S/I}(t) = \text{Hilb}_{S/\text{in}_{<}(I)}(t).$$

Explain precisely where homogeneity for the usual grading is used.

EXERCISE 6.13. Let $I \subset S$ and suppose that $S/\text{in}_{<}(I)$ is finite-dimensional over $\mathbb{k}$. Use the Gröbner degeneration to prove that every fiber has the same $\mathbb{k}$ -dimension. Apply this statement to the family

$$\frac{\mathbb{k}[x, t]}{(x^3 - t^2 x)}.$$

CHAPTER 6. SELECTED APPLICATIONS: SYMMETRY AND GRÖBNER DEGENERATIONS

Show that its fiber at every $\alpha \neq 0$ is isomorphic to $\mathbb{k}[x]/(x^3 - \alpha)$, whereas its special fiber is $\mathbb{k}[x]/(x^3)$. Thus every fiber has $\mathbb{k}$ -dimension three. If $\mathbb{k}$ is algebraically closed of characteristic different from 2, compare the zeros of $x^3 - \alpha$ and x^3 .

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